

Hospitality OS Phase 1 Master Blueprint v2.0.9

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****Generated review edition****

- Package: 'Hospitality_OS_Phase_1_Clean_Build_Package_v2.0.9'
- Canonical source root: 'cd6b8cfc2175004cc2057deb7d9de1ad0d6264a23c82f00e15e96faa526c0c96'
- Status: ****Package M0 candidate - M0R and implementation are not authorized****

Hospitality OS Phase 1 Source of Truth

****Normative human-readable projection****

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Authority

The canonical authority is the pinned reconciled source ZIP in '01_CANONICAL_SOURCE'. The JSON files in '02_MACHINE_READABLE' and all Markdown, Excel, Word and PDF files are generated projections. When projections conflict, the pinned canonical JSON controls.

Package status

- Package M0: candidate for independent Codex review.
- M0R: prohibited until Codex approves or a recorded founder adjudication satisfies FR-GOV-004.
- Database, migration '0001', routes, workers, screens and application code: prohibited before M1.

Canonical counts

- Active Requirements: ****336****
- Decisions: ****120****
- Journey Slices: ****16****
- State Machines: ****12****
- Events: ****80****
- Negative Controls: ****44****
- Non Regression Rules: ****96****
- Review Questions: ****14****
- Original Requirement Dispositions: ****500****
- Original Decision Dispositions: ****100****
- Amendments: ****20****
- Findings: ****22****

Non-negotiable product rules

- Phase 1 is dine-in customer service and outlet execution only.
- English, Amharic and Arabic are the exact customer launch languages; Arabic is true RTL.
- Bill, payment and tip are separate records; no tip is preselected.
- Cash, external-terminal result recording and verified Telebirr/CBE Birr proof confirmation are live

e pilot paths.

- Direct provider APIs remain simulator-only until contracted; raw card data is prohibited.
- Minimum real receipt printing begins at M4; resilient local print queueing begins at M5a.
- Local authority continuity begins at M5b, not M5a; the cloud is never a writable dine-in fallback.
- The outlet node generates and retains its private key; only a CSR leaves the node.
- Phase 2/3 entities, routes, tables, workers, screens and positive tests are physically absent.

Milestone sequence

Gate	Name	Requirements	Exit criterion
M0	Package M0	2	Codex independent review of this exact ZIP.
M0R	Repository Conformance	30	No database, migration, route, worker or UI exists.
M1	Foundation	77	Real PostgreSQL and production-role isolation tests pass.
M2	Menu, QR and customer session	49	English/Amharic/Arabic customer surfaces and true Arabic RTL pass.
M3	Ordering and service	75	The three M3 language journeys pass without billing or local authority.
M4	POS, billing and settlement	55	Live pilot payment paths and physical/digital receipts pass.
M5a	Outlet execution and resilience	24	Restart, retry, deduplication, reconnect and print recovery pass.
M5b	Same-QR trust and authority	12	Same QR works under supported resolver conditions without split-brain or browser bypass.
M6	Production hardening	12	Destructive restore, production roles, full scans and second-tenant evidence pass.

Phase 1 Product Definition

- Package: 'Hospitality_OS_Phase_1_Clean_Build_Package_v2.0.9'
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Product purpose

Hospitality OS is a configurable multi-tenant operating system for restaurants, caf s, bakeries, bars, food courts, hotel outlets and related hospitality operators. Phase 1 covers the complete dine-in customer-service journey from QR discovery through ordering, kitchen execution, billing, payment, tip and receipt, including controlled outlet continuity.

Personas

- Guest customer
- Waiter and supervisor
- Kitchen, bar and expo staff
- Cashier and manager
- Tenant, legal-entity and outlet administrator
- Platform operations and security staff

Phase 1 capabilities

- QR dine-in
- English/Amharic/Arabic
- waiter
- KDS
- counter POS
- bill
- separate optional tip
- cash
- external terminal recording
- verified Telebirr/CBE Birr proof confirmation
- receipt
- cash shift
- local Wi-Fi continuity
- printing
- operational reports

Explicit exclusions

- pickup
- delivery
- loyalty
- CRM
- purchasing
- inventory
- accounting
- HR/workforce
- operational recipes
- costing
- intelligence
- supplier/Horeca runtime
- Phase 2 data portability product

Customer-language contract

English, Amharic and Arabic are complete customer launch languages. Staff applications launch in English on a translation-ready architecture. Browser language is a suggestion; the customer chooses the session language.

Payment and tip contract

Tips are separate from the bill and from bill allocation. No percentage or amount is selected by default. A payer may add a tip independently, and a tip cannot hide an unpaid bill balance.

Continuity promise and boundary

The same QR can resolve to the cloud or the outlet node under supported system-resolver conditions. Unsupported strict custom encrypted resolvers receive translated captive-portal/signage/staff guidance. No self-signed certificate, browser bypass or writable cloud fallback is permitted.

Implementation Sequence and Gate Closures

- Package: 'Hospitality_OS_Phase_1_Clean_Build_Package_v2.0.9'
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A milestone closes only on behavior executable with approved predecessors. Later capabilities may re validate an earlier mechanism but cannot be used to close an earlier gate.

M0 - Package M0

Purpose. Pin product, architecture, requirements, evidence and review package.

Depends on. None

Requirements introduced. 2

Mandatory journeys. None

Exit criterion. Codex independent review of this exact ZIP.

M0R - Repository Conformance

Purpose. Create an empty documentation-only repository and CI/scanner plans.

Depends on. M0

Requirements introduced. 30

Mandatory journeys. None

Exit criterion. No database, migration, route, worker or UI exists.

M1 - Foundation

Purpose. Create tenancy, identity, security, configuration, data architecture and migration 0001 .

Depends on. M0R

Requirements introduced. 77

Mandatory journeys. None

Exit criterion. Real PostgreSQL and production-role isolation tests pass.

M2 - Menu, QR and customer session

Purpose. Create multilingual menu, QR, tables, guest sessions and safety content.

Depends on. M1

Requirements introduced. 49

Mandatory journeys. None

****Exit criterion.**** English/Amharic/Arabic customer surfaces and true Arabic RTL pass.

M3 - Ordering and service

****Purpose.**** Create customer/waiter orders, service requests, KDS and fulfillment.

****Depends on.**** M2

****Requirements introduced.**** 75

****Mandatory journeys.**** GJ-01A, GJ-02, GJ-03A, GJ-04, GJ-05

****Exit criterion.**** The three M3 language journeys pass without billing or local authority.

M4 - POS, billing and settlement

****Purpose.**** Create checks, separate tips, payments, cash shifts, receipts and minimum real printing
.

****Depends on.**** M3

****Requirements introduced.**** 55

****Mandatory journeys.**** GJ-01B, GJ-02B, GJ-03B, GJ-06, GJ-07

****Exit criterion.**** Live pilot payment paths and physical/digital receipts pass.

M5a - Outlet execution and resilience

****Purpose.**** Create local node services, durable persistence, synchronization and resilient printing
.

****Depends on.**** M4

****Requirements introduced.**** 24

****Mandatory journeys.**** GJ-10

****Exit criterion.**** Restart, retry, deduplication, reconnect and print recovery pass.

M5b - Same-QR trust and authority

****Purpose.**** Create DNS/TLS, bidirectional lease, authority sequence and fencing.

****Depends on.**** M5a

****Requirements introduced.**** 12

****Mandatory journeys.**** GJ-08, GJ-09

****Exit criterion.**** Same QR works under supported resolver conditions without split-brain or browser bypass.

M6 - Production hardening

****Purpose.**** Create production images, backup/restore, deployment, observability and final evidence.

****Depends on.**** M5b

****Requirements introduced.**** 12

****Mandatory journeys.**** GJ-11, GJ-12, GJ-13

****Exit criterion.**** Destructive restore, production roles, full scans and second-tenant evidence pass.

Architecture, Security and Continuity Constraints

- Package: 'Hospitality_OS_Phase_1_Clean_Build_Package_v2.0.9'
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Core architecture

- multi-tenant and outlet-isolated by default
- modular monolith with strict domain boundaries
- versioned APIs and reliable events
- exact money arithmetic
- append-only or reversal-based commerce records
- node-generated private keys and CSR-only certificate issuance
- cloud is never a writable dine-in fallback
- Phase 2/3 surfaces are physically absent from Phase 1 artifacts

Tenant and outlet isolation

- Missing tenant/outlet context must return zero rows and permit zero writes.
- Use populated owner-side fixtures so zero-row tests cannot pass vacuously.
- Test sibling outlets inside the same tenant, not only different tenants.
- Test SELECT, INSERT, UPDATE and DELETE.
- Policies must contain both 'USING' and 'WITH CHECK' where applicable.
- Adding 'outlet_id' later must automatically strengthen existing policies.
- API, jobs, files, caches, reports and sync paths are part of the isolation boundary.
- Production services reject owner, superuser, BYPASSRLS and maintenance database roles.

Money and audit

- Do not use binary floating-point for money, percentages or quantity outcomes.
- Accepted orders, issued bills, payments, tips and cash movements are immutable or reversal-based.
- No tip is selected by default.
- Bill allocation and tip allocation are separate.
- A tip cannot hide an unpaid bill balance.
- Refunds and reversals require purpose-specific step-up, permission, reason and audit.
- Quick PIN cannot authorize sensitive financial actions.
- Split bill participants may choose different tips.

M4/M5a printing boundary

M4 owns the minimum real printer path needed to issue a physical customer receipt. M5a adds durable local queueing, bounded retry, restart recovery, deduplication, printer health, outage continuity and reconciliation.

M5a/M5b authority boundary

M5a provides focused local services but does not claim local authority. M5b adds same-QR DNS/TLS, bi-directional lease, authority sequence and fencing. The cloud remains a control plane/forwarder and is never a writable dine-in fallback.

Certificate custody

The outlet node generates and retains the private key, submits only a certificate signing request, receives only the certificate chain, renews through DNS-01 automation and never exports the private key.

Functional Requirements Register

****336 active requirements****

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Requirements are ordered by introducing gate, domain and ID. Each entry states its executable gate-local behavior and later revalidation ownership.

Gate M0

Build Governance

FR-G0V-004 - Milestone review and adjudication

****Priority:** P0**

****Owner:** Platform & Security**

****Lineage:** retained**

****Introduced:** M0**

****Revalidated:** M0R, M1, M2, M3, M4, M5a, M5b, M6**

****Required behavior:**** Do not begin a later milestone while any P0 remains unresolved, or while any P1 affects product scope, security, money, authority, milestone executability, acceptance completeness or canonical correctness. A P1 limited to publication, projection, wording, identifier or validation or coverage may proceed only through recorded founder adjudication when canonical behavior is correct, runtime behavior cannot change, the defect remains visible and repair is scheduled. An adverse independent verdict pauses progression until the builder response, reviewer rebuttal and founder decision are recorded. Overrides name the affected requirement or rule and are re-examined at the next audit.

****Gate-local behavior:**** Apply the blocker and adjudication rule to Package M0 and record the decision.

on, affected requirement and scheduled repair.

****Later behavior:**** Apply the rule at every later gate and re-examine each override at the next audit.

****Prerequisites:**** None

****Journeys:**** None

****Acceptance tests:**** TST-M0-FR-GOV-004, TST-M0R-FR-GOV-004, TST-M1-FR-GOV-004, TST-M2-FR-GOV-004, TST-M3-FR-GOV-004, TST-M4-FR-GOV-004, TST-M5a-FR-GOV-004, TST-M5b-FR-GOV-004, TST-M6-FR-GOV-004

Testing and Evidence

FR-TST-017 - Anti-false-green and frozen negative controls

****Priority:**** P0

****Owner:**** Quality Engineering

****Lineage:**** retained

****Introduced:**** M0

****Revalidated:**** M0R, M6

****Required behavior:**** Package M0 enumerates every critical planted negative control by milestone, protected property, deliberate break, expected failing test and expected failure signature. Zero tests, skipped tests, empty scans, stale artifacts and unsupported coverage fail. The registry is frozen before M0R and changes require review.

****Gate-local behavior:**** The registry enumerates every control with all five attributes and is frozen before the repository exists.

****Later behavior:**** The controls are planted in CI at M0R and re-proven against the production artifact at M6.

****Prerequisites:**** None

****Journeys:**** GJ-12

****Acceptance tests:**** TST-M0-FR-TST-017, TST-M0R-FR-TST-017, TST-M6-FR-TST-017

Gate M0R

Build Governance

FR-GOV-001A - Empty repository conformance

****Priority:**** P0

****Owner:**** Platform & Security

****Lineage:**** split_successor from 'FR-GOV-001'

****Introduced:**** M0R

****Revalidated:**** None

****Required behavior:**** The repository contains only the approved package and conformance plans; a scan proves no schema, migration, route, worker or UI file exists.

****Gate-local behavior:**** The repository contains only the approved package and conformance plans; a scan proves no schema, migration, route, worker or UI file exists.

****Later behavior:**** None beyond M0R for this clause.

****Prerequisites:**** FR-GOV-006

****Journeys:**** None
****Acceptance tests:**** TST-M0R-FR-GOV-001A

FR-GOV-002 - No dormant future modules

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** retained
****Introduced:**** M0R
****Revalidated:**** M1, M2, M3, M4, M5a, M5b, M6
****Required behavior:**** At Repository Conformance M0R and every later gate, prove the deployable Phase 1 surface contains no purchasing, inventory, accounting, HR, recipe, intelligence, supplier, loyalty, pickup or delivery routes, tables, workers, tests or screens.
****Gate-local behavior:**** Scanner runs over the repository plan with one planted fixture per excluded domain and fails on each.
****Later behavior:**** Every later gate adds deployable surface that must be rescanned.
****Prerequisites:**** None
****Journeys:**** GJ-12
****Acceptance tests:**** TST-M0R-FR-GOV-002, TST-M1-FR-GOV-002, TST-M2-FR-GOV-002, TST-M3-FR-GOV-002, TST-M4-FR-GOV-002, TST-M5a-FR-GOV-002, TST-M5b-FR-GOV-002, TST-M6-FR-GOV-002

FR-GOV-003 - Controlled code reuse

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** retained
****Introduced:**** M0R
****Revalidated:**** M6
****Required behavior:**** Any reused unit from the frozen repository requires isolated review, dependency analysis, provenance recording and new tests written against this package before inclusion.
****Gate-local behavior:**** A provenance register exists and the review/dependency-analysis process is proven on any unit proposed for reuse in the M0R plan.
****Later behavior:**** Reuse decisions recur whenever code is written; the complete provenance audit is only possible against the final artifact.
****Prerequisites:**** FR-GOV-006
****Journeys:**** None
****Acceptance tests:**** TST-M0R-FR-GOV-003, TST-M6-FR-GOV-003

FR-GOV-005 - Traceability authority

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** retained
****Introduced:**** M0R
****Revalidated:**** M1, M2, M3, M4, M5a, M5b, M6
****Required behavior:**** Every implemented route, table, worker, screen and test maps to an active requirement and gate in this package. Deferred requirements cannot appear in production artifacts.

****Gate-local behavior:**** The migration/domain ownership map maps every planned unit to an active requirement and gate; no deferred requirement appears.

****Later behavior:**** Traceability must be re-proven as each gate implements real routes, tables, workers, screens and tests.

****Prerequisites:**** FR-GOV-001A

****Journeys:**** GJ-12

****Acceptance tests:**** TST-M0R-FR-GOV-005, TST-M1-FR-GOV-005, TST-M2-FR-GOV-005, TST-M3-FR-GOV-005, TST-M4-FR-GOV-005, TST-M5a-FR-GOV-005, TST-M5b-FR-GOV-005, TST-M6-FR-GOV-005

FR-GOV-006 - Frozen prototype

****Priority:**** P0

****Owner:**** Platform & Security

****Lineage:**** retained

****Introduced:**** M0R

****Revalidated:**** None

****Required behavior:**** Mark the v1.1 repository as a frozen research and architecture prototype; no Phase 1 release is cut from its branches.

****Gate-local behavior:**** The v1.1 repository is marked frozen and the new repository is established as the only release source.

****Later behavior:**** None beyond M0R for this clause.

****Prerequisites:**** None

****Journeys:**** None

****Acceptance tests:**** TST-M0R-FR-GOV-006

Carts and Orders

FR-ORD-001C - No pickup or delivery order domain

****Priority:**** P0

****Owner:**** Service Execution

****Lineage:**** split_successor from 'FR-ORD-001'

****Introduced:**** M0R

****Revalidated:**** M3, M6

****Required behavior:**** Scanner proves no pickup or delivery policy, fulfillment field, route or screen exists.

****Gate-local behavior:**** Scanner proves no pickup or delivery policy, fulfillment field, route or screen exists.

****Later behavior:**** Absence holds as the order surface grows.

****Prerequisites:**** FR-GOV-002

****Journeys:**** None

****Acceptance tests:**** TST-M0R-FR-ORD-001C, TST-M3-FR-ORD-001C, TST-M6-FR-ORD-001C

FR-ORD-012B - No stock or accounting reversal

****Priority:**** P0

```
**Owner:** Service Execution
**Lineage:** split_successor from 'FR-ORD-012'
**Introduced:** M0R
**Revalidated:** M4, M6
**Required behavior:** Scanner proves no stock or general-ledger posting path exists.

**Gate-local behavior:** Scanner proves no stock or general-ledger posting path exists.

**Later behavior:** Absence holds as the void and refund surfaces grow.

**Prerequisites:** FR-GOV-002
**Journeys:** None
**Acceptance tests:** TST-M0R-FR-ORD-012B, TST-M4-FR-ORD-012B, TST-M6-FR-ORD-012B

### FR-ORD-016B - No delivery, inventory or accounting events

**Priority:** P0
**Owner:** Service Execution
**Lineage:** split_successor from 'FR-ORD-016'
**Introduced:** M0R
**Revalidated:** M3, M6
**Required behavior:** Scanner proves no delivery, inventory or accounting event type exists in the
timeline model.

**Gate-local behavior:** Scanner proves no delivery, inventory or accounting event type exists in th
e timeline model.

**Later behavior:** Absence holds as event catalog grows.

**Prerequisites:** FR-GOV-002
**Journeys:** None
**Acceptance tests:** TST-M0R-FR-ORD-016B, TST-M3-FR-ORD-016B, TST-M6-FR-ORD-016B

### FR-ORD-019B - No campaign or CRM correlation

**Priority:** P0
**Owner:** Service Execution
**Lineage:** split_successor from 'FR-ORD-019'
**Introduced:** M0R
**Revalidated:** M3, M6
**Required behavior:** Scanner proves no campaign or CRM correlation field, route or screen exists,
including morphological variants of "campaign".

**Gate-local behavior:** Scanner proves no campaign or CRM correlation field, route or screen exists
, including morphological variants of "campaign".

**Later behavior:** Absence holds as the correlation model grows.

**Prerequisites:** FR-GOV-002
**Journeys:** None
**Acceptance tests:** TST-M0R-FR-ORD-019B, TST-M3-FR-ORD-019B, TST-M6-FR-ORD-019B

## Configuration and Setup
```

FR-CFG-002B - No deferred policy categories

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** split_successor from 'FR-CFG-002'
****Introduced:**** M0R
****Revalidated:**** M1, M6
****Required behavior:**** Scanner proves no Phase 2/3 policy category is present in the plan and schema.

****Gate-local behavior:**** Scanner proves no Phase 2/3 policy category is present in the plan and schema.

****Later behavior:**** Absence holds as configuration surface grows.

****Prerequisites:**** FR-GOV-002
****Journeys:**** None
****Acceptance tests:**** TST-M0R-FR-CFG-002B, TST-M1-FR-CFG-002B, TST-M6-FR-CFG-002B

FR-CFG-005B - No deferred feature surface

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** split_successor from 'FR-CFG-005'
****Introduced:**** M0R
****Revalidated:**** M1, M6
****Required behavior:**** The Phase 1 artifact contains no deferred Phase 2/3 route, worker, table or screen, regardless of entitlement configuration.

****Gate-local behavior:**** The scanner fails on a planted deferred route, worker, table and screen and confirms their absence from the repository plan.

****Later behavior:**** Re-scan the first executable artifact at M1 and the complete production artifact at M6.

****Prerequisites:**** FR-GOV-002
****Journeys:**** GJ-12
****Acceptance tests:**** TST-M0R-FR-CFG-005B, TST-M1-FR-CFG-005B, TST-M6-FR-CFG-005B

Kitchen/Bar/Expo Fulfillment

FR-FUL-016B - No inventory or recipe-consumption posting

****Priority:**** P0
****Owner:**** Service Execution
****Lineage:**** split_successor from 'FR-FUL-016'
****Introduced:**** M0R
****Revalidated:**** M3, M6
****Required behavior:**** Scanner proves no inventory or recipe-consumption entry is produced by waste recording.

****Gate-local behavior:**** Scanner proves no inventory or recipe-consumption entry is produced by waste recording.

****Later behavior:**** Absence holds as the waste surface grows.

****Prerequisites:**** FR-GOV-002

****Journeys:**** None

****Acceptance tests:**** TST-M0R-FR-FUL-016B, TST-M3-FR-FUL-016B, TST-M6-FR-FUL-016B

Menu and Pricing

FR-MNU-002B - No customer-segment targeting

****Priority:**** P0

****Owner:**** Customer Experience

****Lineage:**** split_successor from 'FR-MNU-002'

****Introduced:**** M0R

****Revalidated:**** M2, M6

****Required behavior:**** Scanner proves no customer-segment field, assignment rule or screen exists.

****Gate-local behavior:**** Scanner proves no customer-segment field, assignment rule or screen exists.

****Later behavior:**** Absence holds as the menu surface grows.

****Prerequisites:**** FR-GOV-002

****Journeys:**** None

****Acceptance tests:**** TST-M0R-FR-MNU-002B, TST-M2-FR-MNU-002B, TST-M6-FR-MNU-002B

Outlet Edge and Synchronization

FR-EDG-002B - No deferred local modules

****Priority:**** P0

****Owner:**** Edge & Operations

****Lineage:**** split_successor from 'FR-EDG-002'

****Introduced:**** M0R

****Revalidated:**** M5a, M6

****Required behavior:**** Scanner proves no ERP or intelligence module is present in the node plan or image manifest.

****Gate-local behavior:**** Scanner proves no ERP or intelligence module is present in the node plan or image manifest.

****Later behavior:**** Absence holds as the node image is built and shipped.

****Prerequisites:**** FR-GOV-002

****Journeys:**** None

****Acceptance tests:**** TST-M0R-FR-EDG-002B, TST-M5a-FR-EDG-002B, TST-M6-FR-EDG-002B

Payments

FR-PAY-010B - No accounting journal posting

****Priority:**** P0

****Owner:**** Commerce & Payments

****Lineage:**** split_successor from 'FR-PAY-010'

****Introduced:**** M0R

****Revalidated:**** M4, M6
****Required behavior:**** Scanner proves no journal entry, ledger posting, accounting entity, field, route, screen or positive accounting test exists, including singular and plural morphological variants.
****Gate-local behavior:**** Scanner proves no journal entry, ledger posting, accounting entity, field, route, screen or positive accounting test exists, including singular and plural morphological variants.
****Later behavior:**** Absence holds as the payment surface grows.
****Prerequisites:**** FR-GOV-002
****Journeys:**** None
****Acceptance tests:**** TST-M0R-FR-PAY-010B, TST-M4-FR-PAY-010B, TST-M6-FR-PAY-010B

Recipes and Costing

FR-RCP-008B - No operational recipe module

****Priority:**** P0
****Owner:**** Platform Engineering
****Lineage:**** split_successor from 'FR-RCP-008'
****Introduced:**** M0R
****Revalidated:**** M2, M6
****Required behavior:**** Scanner proves no recipe module, table or route exists.
****Gate-local behavior:**** Scanner proves no recipe module, table or route exists.
****Later behavior:**** Absence holds as schema grows.
****Prerequisites:**** FR-GOV-002
****Journeys:**** None
****Acceptance tests:**** TST-M0R-FR-RCP-008B, TST-M2-FR-RCP-008B, TST-M6-FR-RCP-008B

Security and Data Protection

FR-SEC-010B - No employee or payroll data

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** split_successor from 'FR-SEC-010'
****Introduced:**** M0R
****Revalidated:**** M1, M6
****Required behavior:**** Scanner proves no employee or payroll field, table or test exists.
****Gate-local behavior:**** Scanner proves no employee or payroll field, table or test exists.
****Later behavior:**** Absence holds as schema grows.
****Prerequisites:**** FR-GOV-002
****Journeys:**** None
****Acceptance tests:**** TST-M0R-FR-SEC-010B, TST-M1-FR-SEC-010B, TST-M6-FR-SEC-010B

FR-SEC-015 - Vulnerability pipeline

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** retained
****Introduced:**** M0R
****Revalidated:**** M1, M6
****Required behavior:**** The vulnerability pipeline fails on unknown, empty, skipped or unsupported coverage and tests the actual production artifacts with planted negative controls.

****Gate-local behavior:**** CI design proves the pipeline fails closed on unknown/empty/skipped/unsupported results, verified with planted controls in the CI plan.

****Later behavior:**** Testing the actual production artifacts requires those artifacts to exist.

****Prerequisites:**** FR-GOV-002
****Journeys:**** GJ-12
****Acceptance tests:**** TST-M0R-FR-SEC-015, TST-M1-FR-SEC-015, TST-M6-FR-SEC-015

Service Requests

FR-SRV-007B - No workforce or attendance record

****Priority:**** P0
****Owner:**** Service Execution
****Lineage:**** split_successor from 'FR-SRV-007'
****Introduced:**** M0R
****Revalidated:**** M3, M6
****Required behavior:**** Scanner proves no roster, shift, break, attendance, timekeeping, payroll or employment-history field or table exists, and presence records carry a retention bound.

****Gate-local behavior:**** Scanner proves no roster, shift, break, attendance, timekeeping, payroll or employment-history field or table exists, and presence records carry a retention bound.

****Later behavior:**** Absence holds as the staff surface grows.

****Prerequisites:**** FR-GOV-002, FR-SEC-010B
****Journeys:**** None
****Acceptance tests:**** TST-M0R-FR-SRV-007B, TST-M3-FR-SRV-007B, TST-M6-FR-SRV-007B

Staff POS and Outlet UX

FR-POS-003C - No pickup or delivery ordering

****Priority:**** P0
****Owner:**** Customer Experience
****Lineage:**** split_successor from 'FR-POS-003'
****Introduced:**** M0R
****Revalidated:**** M3, M4, M6
****Required behavior:**** Scanner proves no pickup or delivery order-creation path, field or screen exists.

****Gate-local behavior:**** Scanner proves no pickup or delivery order-creation path, field or screen exists.

****Later behavior:**** Absence holds as the staff ordering surface grows.

****Prerequisites:**** FR-GOV-002
****Journeys:**** None
****Acceptance tests:**** TST-M0R-FR-POS-003C, TST-M3-FR-POS-003C, TST-M4-FR-POS-003C, TST-M6-FR-POS-003C

FR-POS-010B - No pickup or delivery search

****Priority:**** P0
****Owner:**** Customer Experience
****Lineage:**** split_successor from 'FR-POS-010'
****Introduced:**** M0R
****Revalidated:**** M4, M6
****Required behavior:**** Scanner proves no pickup-code or delivery search field, index or screen exists.

****Gate-local behavior:**** Scanner proves no pickup-code or delivery search field, index or screen exists.

****Later behavior:**** Absence holds as the search surface grows.

****Prerequisites:**** FR-GOV-002
****Journeys:**** None
****Acceptance tests:**** TST-M0R-FR-POS-010B, TST-M4-FR-POS-010B, TST-M6-FR-POS-010B

Tenant and Commercial Control

FR-TEN-002B - No storage-location entity

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** split_successor from 'FR-TEN-002'
****Introduced:**** M0R
****Revalidated:**** M1, M6
****Required behavior:**** Scanner proves absence of any storage-location entity in the repository plan and, from M1, in schema and code.

****Gate-local behavior:**** Scanner proves absence of any storage-location entity in the repository plan and, from M1, in schema and code.

****Later behavior:**** Absence must hold as schema and application surface grow.

****Prerequisites:**** FR-GOV-002
****Journeys:**** None
****Acceptance tests:**** TST-M0R-FR-TEN-002B, TST-M1-FR-TEN-002B, TST-M6-FR-TEN-002B

FR-TEN-009B - No later-domain registry entries

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** split_successor from 'FR-TEN-009'
****Introduced:**** M0R
****Revalidated:**** M1, M6

****Required behavior:**** Scanner proves no excluded-domain enumeration value or registry entry exists.

****Gate-local behavior:**** Scanner proves no excluded-domain enumeration value or registry entry exist S.

****Later behavior:**** Absence holds as configuration surface grows.

****Prerequisites:**** FR-GOV-002

****Journeys:**** None

****Acceptance tests:**** TST-M0R-FR-TEN-009B, TST-M1-FR-TEN-009B, TST-M6-FR-TEN-009B

Testing and Evidence

FR-TST-004B - No supplier or courier contract tests

****Priority:**** P0

****Owner:**** Quality Engineering

****Lineage:**** split_successor from 'FR-TST-004'

****Introduced:**** M0R

****Revalidated:**** M4, M6

****Required behavior:**** Scanner proves no supplier or courier port, simulator or contract test exists .

****Gate-local behavior:**** Scanner proves no supplier or courier port, simulator or contract test exists.

****Later behavior:**** Absence holds as the port set grows.

****Prerequisites:**** FR-GOV-002

****Journeys:**** None

****Acceptance tests:**** TST-M0R-FR-TST-004B, TST-M4-FR-TST-004B, TST-M6-FR-TST-004B

FR-TST-005B - No later-channel or supplier journeys

****Priority:**** P0

****Owner:**** Quality Engineering

****Lineage:**** split_successor from 'FR-TST-005'

****Introduced:**** M0R

****Revalidated:**** M6

****Required behavior:**** Scanner proves no pickup, delivery, supplier or Phase 2 back-office journey or test exists.

****Gate-local behavior:**** Scanner proves no pickup, delivery, supplier or Phase 2 back-office journey or test exists.

****Later behavior:**** Absence holds as the journey set grows.

****Prerequisites:**** FR-GOV-002

****Journeys:**** None

****Acceptance tests:**** TST-M0R-FR-TST-005B, TST-M6-FR-TST-005B

FR-TST-007B - No stock, supplier or delivery races

****Priority:**** P0

****Owner:**** Quality Engineering
****Lineage:**** split_successor from 'FR-TST-007'
****Introduced:**** M0R
****Revalidated:**** M6
****Required behavior:**** Scanner proves no stock, supplier or delivery concurrency test exists.

****Gate-local behavior:**** Scanner proves no stock, supplier or delivery concurrency test exists.

****Later behavior:**** Absence holds as the concurrency suite grows.

****Prerequisites:**** FR-GOV-002
****Journeys:**** None
****Acceptance tests:**** TST-M0R-FR-TST-007B, TST-M6-FR-TST-007B

FR-TST-013 - Golden journeys

****Priority:**** P0
****Owner:**** Quality Engineering
****Lineage:**** retained
****Introduced:**** M0R
****Revalidated:**** M3, M4, M5a, M5b, M6
****Required behavior:**** Map every mandatory journey to test IDs, evidence, owner and defect status.

****Gate-local behavior:**** Every mandatory journey slice is mapped to planned test IDs, owner and evidence slots before coding begins.

****Later behavior:**** Evidence and defect status accrue as each gate executes its journeys.

****Prerequisites:**** FR-GOV-005
****Journeys:**** GJ-12
****Acceptance tests:**** TST-M0R-FR-TST-013, TST-M3-FR-TST-013, TST-M4-FR-TST-013, TST-M5a-FR-TST-013, TST-M5b-FR-TST-013, TST-M6-FR-TST-013

FR-TST-014 - Traceability

****Priority:**** P0
****Owner:**** Quality Engineering
****Lineage:**** retained
****Introduced:**** M0R
****Revalidated:**** M1, M2, M3, M4, M5a, M5b, M6
****Required behavior:**** Map every P0/P1 requirement to implementation component, automated/manual test and acceptance evidence.

****Gate-local behavior:**** Every active requirement maps to a planned component and test before coding begins.

****Later behavior:**** Each implementing gate populates real components, tests and evidence.

****Prerequisites:**** FR-GOV-005
****Journeys:**** GJ-12
****Acceptance tests:**** TST-M0R-FR-TST-014, TST-M1-FR-TST-014, TST-M2-FR-TST-014, TST-M3-FR-TST-014, TST-M4-FR-TST-014, TST-M5a-FR-TST-014, TST-M5b-FR-TST-014, TST-M6-FR-TST-014

FR-TST-019 - Independent milestone audit

****Priority:**** P0
****Owner:**** Quality Engineering
****Lineage:**** retained
****Introduced:**** M0R
****Revalidated:**** M1, M2, M3, M4, M5a, M5b, M6
****Required behavior:**** Each milestone has a Codex audit branch containing immutable commands, evidence, defects and a merge/no-merge decision against the exact commit.

****Gate-local behavior:**** The audit branch structure exists and the M0R review itself produces immutable commands, evidence and a decision against an exact commit.

****Later behavior:**** Every subsequent gate produces its own audit branch and decision.

****Prerequisites:**** FR-GOV-004
****Journeys:**** None
****Acceptance tests:**** TST-M0R-FR-TST-019, TST-M1-FR-TST-019, TST-M2-FR-TST-019, TST-M3-FR-TST-019, TST-M4-FR-TST-019, TST-M5a-FR-TST-019, TST-M5b-FR-TST-019, TST-M6-FR-TST-019

UX and Accessibility

FR-UX-001B - No pickup or delivery UX

****Priority:**** P0
****Owner:**** Customer Experience
****Lineage:**** split_successor from 'FR-UX-001'
****Introduced:**** M0R
****Revalidated:**** M2, M6
****Required behavior:**** Scanner proves no pickup or delivery screen or journey exists.
****Gate-local behavior:**** Scanner proves no pickup or delivery screen or journey exists.
****Later behavior:**** Absence holds as customer UX grows.
****Prerequisites:**** FR-GOV-002
****Journeys:**** None
****Acceptance tests:**** TST-M0R-FR-UX-001B, TST-M2-FR-UX-001B, TST-M6-FR-UX-001B

Gate M1

Build Governance

FR-GOV-001B - Database and migration start

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** split_successor from 'FR-GOV-001'
****Introduced:**** M1
****Revalidated:**** None
****Required behavior:**** Migration 0001 is created and applied at M1 only, with M0R approval recorded as its precondition.
****Gate-local behavior:**** Migration 0001 is created and applied at M1 only, with M0R approval recorded as its precondition.

****Later behavior:**** None beyond M1 for this clause.

****Prerequisites:**** FR-GOV-001A, FR-DAT-001

****Journeys:**** None

****Acceptance tests:**** TST-M1-FR-GOV-001B

Commercial Multi-tenant Product

FR-COM-002 - White labeling

****Priority:**** P0

****Owner:**** Platform & Security

****Lineage:**** retained

****Introduced:**** M1

****Revalidated:**** M4, M5b

****Required behavior:**** Support tenant brand/domain/templates while retaining optional platform attribution by contract.

****Gate-local behavior:**** Brand and template selection resolve per tenant with configurable platform attribution.

****Later behavior:**** Receipt templates render at M4; per-outlet public domains take effect at M5b.

****Prerequisites:**** FR-CFG-006

****Journeys:**** None

****Acceptance tests:**** TST-M1-FR-COM-002, TST-M4-FR-COM-002, TST-M5b-FR-COM-002

FR-COM-007 - Configuration templates

****Priority:**** P0

****Owner:**** Platform & Security

****Lineage:**** retained

****Introduced:**** M1

****Revalidated:**** M2, M4

****Required behavior:**** Offer reusable industry templates without hard-coding one tenant's menu, tax, workflow or language.

****Gate-local behavior:**** Templates exist as data with no tenant-specific constant in code.

****Later behavior:**** Menu and language templates become substantive at M2; tax and workflow templates at M3/M4.

****Prerequisites:**** FR-TEN-003

****Journeys:**** None

****Acceptance tests:**** TST-M1-FR-COM-007, TST-M2-FR-COM-007, TST-M4-FR-COM-007

FR-COM-009 - Version support

****Priority:**** P0

****Owner:**** Platform & Security

****Lineage:**** retained

****Introduced:**** M1

****Revalidated:**** M5a, M5b

****Required behavior:**** Track supported application, schema, edge and connector versions per tenant/deployment.

****Gate-local behavior:**** Application and schema versions are tracked per tenant/deployment.

****Later behavior:**** Edge and connector versions exist only once the node and connectors exist; compatibility posture is part of the M5b lease.

****Prerequisites:**** FR-TEN-003

****Journeys:**** None

****Acceptance tests:**** TST-M1-FR-COM-009, TST-M5a-FR-COM-009, TST-M5b-FR-COM-009

FR-COM-010 - Commercial neutrality

****Priority:**** P0

****Owner:**** Platform & Security

****Lineage:**** retained

****Introduced:**** M1

****Revalidated:**** M2, M4, M6

****Required behavior:**** No product screen or API assumes the first internal tenant, Horeca Addis or a specific payment provider.

****Gate-local behavior:**** No hard-coded tenant identifier or provider name in the M1 API surface.

****Later behavior:**** Customer screens at M2, payment provider surfaces at M4, and the full production artifact at M6.

****Prerequisites:**** FR-COM-007

****Journeys:**** GJ-13

****Acceptance tests:**** TST-M1-FR-COM-010, TST-M2-FR-COM-010, TST-M4-FR-COM-010, TST-M6-FR-COM-010

Configuration and Setup

FR-CFG-001A - Organizational setup

****Priority:**** P0

****Owner:**** Platform & Security

****Lineage:**** split_successor from 'FR-CFG-001'

****Introduced:**** M1

****Revalidated:**** None

****Required behavior:**** A guided setup creates the tenant, brand/legal entity, first outlet and initial administrator.

****Gate-local behavior:**** Complete the organizational setup and verify that the new administrator can access only the created tenant and outlet.

****Later behavior:****

****Prerequisites:**** FR-TEN-002A, FR-AUTH-001

****Journeys:**** None

****Acceptance tests:**** TST-M1-FR-CFG-001A

FR-CFG-002A - Phase 1 policy engine

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** split_successor from 'FR-CFG-002'
****Introduced:**** M1
****Revalidated:**** M3, M4, M5a
****Required behavior:**** Policy records for the named Phase 1 categories exist with validation and effective dating.

****Gate-local behavior:**** Policy records for the named Phase 1 categories exist with validation and effective dating.

****Later behavior:**** Policy enforcement is provable only where the governed behavior exists: ordering /service/cancellation at M3, discount/refund/tip/cash/approval at M4, local continuity at M5a.

****Prerequisites:**** FR-TEN-003
****Journeys:**** None
****Acceptance tests:**** TST-M1-FR-CFG-002A, TST-M3-FR-CFG-002A, TST-M4-FR-CFG-002A, TST-M5a-FR-CFG-002A

FR-CFG-003 - Reason codes

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** retained
****Introduced:**** M1
****Revalidated:**** M3, M4, M5a
****Required behavior:**** Manage localized reason-code sets for order cancellation, void, refund, discount, complimentary item, payment reversal, tip correction, service failure, printer failure and manager override.

****Gate-local behavior:**** Localized reason-code sets are managed as configuration data with the Phase 1 category list.

****Later behavior:**** Each code is exercised only where its action exists: cancellation and service failure at M3, void/refund/discount/complimentary/reversal/tip correction/override at M4, printer failure at M5a.

****Prerequisites:**** FR-TEN-003
****Journeys:**** None
****Acceptance tests:**** TST-M1-FR-CFG-003, TST-M3-FR-CFG-003, TST-M4-FR-CFG-003, TST-M5a-FR-CFG-003

FR-CFG-004 - Numbering

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** retained
****Introduced:**** M1
****Revalidated:**** M4
****Required behavior:**** Generate collision-safe human document numbers by tenant/legal entity/outlet/document type and fiscal period.

****Gate-local behavior:**** Numbering service issues collision-safe numbers under concurrency for the document types defined at M1.

****Later behavior:**** Bill and receipt document types are numbered when they exist.

****Prerequisites:**** FR-DAT-003

****Journeys:**** None

****Acceptance tests:**** TST-M1-FR-CFG-004, TST-M4-FR-CFG-004

FR-CFG-005A - Deny-by-default entitlements

****Priority:**** P0

****Owner:**** Platform & Security

****Lineage:**** split_successor from 'FR-CFG-005'

****Introduced:**** M1

****Revalidated:**** None

****Required behavior:**** Feature entitlements resolve deny-by-default at tenant, legal-entity and outlet scope.

****Gate-local behavior:**** Unknown, absent or disabled entitlement values deny the feature and expose no route or operation.

****Later behavior:****

****Prerequisites:**** FR-TEN-004

****Journeys:**** None

****Acceptance tests:**** TST-M1-FR-CFG-005A

FR-CFG-006 - Branding

****Priority:**** P0

****Owner:**** Platform & Security

****Lineage:**** retained

****Introduced:**** M1

****Revalidated:**** M4, M5b

****Required behavior:**** Configure logos, colors, receipt/footer text, contact data and public-domain mappings without application forks.

****Gate-local behavior:**** Branding attributes are configurable per tenant and consumed from configuration rather than build-time constants.

****Later behavior:**** Receipt/footer text is only rendered on a real receipt at M4; public-domain mapping only takes effect with per-outlet hostname routing at M5b.

****Prerequisites:**** FR-TEN-003

****Journeys:**** None

****Acceptance tests:**** TST-M1-FR-CFG-006, TST-M4-FR-CFG-006, TST-M5b-FR-CFG-006

FR-CFG-007A - Non-production demo fixtures

****Priority:**** P0

****Owner:**** Platform & Security

****Lineage:**** split_successor from 'FR-CFG-007'

****Introduced:**** M1

****Revalidated:**** None

****Required behavior:**** Demo fixtures are explicitly marked and reset works only in non-production environments.

****Gate-local behavior:**** Demo fixtures are explicitly marked and reset works only in non-production environments.

****Later behavior:**** None beyond M1 for this clause.

****Prerequisites:**** FR-DAT-013

****Journeys:**** None

****Acceptance tests:**** TST-M1-FR-CFG-007A

Data Architecture

FR-DAT-001 - SQL-first migrations

****Priority:**** P0

****Owner:**** Platform & Security

****Lineage:**** retained

****Introduced:**** M1

****Revalidated:**** M6

****Required behavior:**** Use a new ordered SQL migration history beginning at 0001. Build from an empty PostgreSQL database and test upgrades only from approved baselines of this package; never import the v1.1 migration history.

****Gate-local behavior:**** Migration 0001 applies to an empty database; ordered history builds cleanly; no v1.1 migration is present.

****Later behavior:**** Upgrade from a prior approved baseline is exercised with a populated schema.

****Prerequisites:**** FR-GOV-001A

****Journeys:**** GJ-12

****Acceptance tests:**** TST-M1-FR-DAT-001, TST-M6-FR-DAT-001

FR-DAT-002 - Constraints

****Priority:**** P0

****Owner:**** Platform & Security

****Lineage:**** retained

****Introduced:**** M1

****Revalidated:**** M3, M4

****Required behavior:**** Enforce tenant keys, foreign keys, uniqueness, money/quantity checks and valid state relationships in the database where practical.

****Gate-local behavior:**** Constraints exist and reject invalid rows for the M1 schema.

****Later behavior:**** Order and fulfillment state relationships at M3; financial state relationships at M4.

****Prerequisites:**** FR-DAT-001

****Journeys:**** None

****Acceptance tests:**** TST-M1-FR-DAT-002, TST-M3-FR-DAT-002, TST-M4-FR-DAT-002

FR-DAT-003 - Identifiers

****Priority:**** P0

****Owner:**** Platform & Security
****Lineage:**** retained
****Introduced:**** M1
****Revalidated:**** None
****Required behavior:**** Use opaque UUID/ULID identifiers and separate human numbers; never rely on mutable external IDs.

****Gate-local behavior:**** All primary identifiers are opaque; human numbers are separate columns.

****Later behavior:**** None beyond M1.

****Prerequisites:**** FR-DAT-001

****Journeys:**** None

****Acceptance tests:**** TST-M1-FR-DAT-003

FR-DAT-004 - Timestamps

****Priority:**** P0

****Owner:**** Platform & Security

****Lineage:**** retained

****Introduced:**** M1

****Revalidated:**** M4, M5a

****Required behavior:**** Store UTC instants plus outlet timezone context and use database/server time for authoritative events.

****Gate-local behavior:**** Timestamps are UTC with outlet timezone context; authoritative events use server time.

****Later behavior:**** Cash-shift and fiscal-period boundaries make timezone correctness financially material at M4; outlet-local event time under outage at M5a.

****Prerequisites:**** FR-TEN-002A

****Journeys:**** None

****Acceptance tests:**** TST-M1-FR-DAT-004, TST-M4-FR-DAT-004, TST-M5a-FR-DAT-004

FR-DAT-005 - Money

****Priority:**** P0

****Owner:**** Platform & Security

****Lineage:**** retained

****Introduced:**** M1

****Revalidated:**** M4

****Required behavior:**** Represent money with exact fixed-point decimal or integer-minor-unit types and explicit currency. JavaScript binary floating point cannot determine financial outcomes.

****Gate-local behavior:**** Money columns use exact types with explicit currency; no float path exists in the M1 code.

****Later behavior:**** Actual financial computation (bill, tax, service charge, tip, allocation) occurs at M4.

****Prerequisites:**** FR-DAT-001

****Journeys:**** GJ-06

****Acceptance tests:**** TST-M1-FR-DAT-005, TST-M4-FR-DAT-005

FR-DAT-006 - Quantity

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** retained
****Introduced:**** M1
****Revalidated:**** M4
****Required behavior:**** Represent quantities and percentages with explicit precision, rounding and validation. Tips and bill totals use deterministic exact arithmetic.

****Gate-local behavior:**** Quantity/percentage precision, rounding mode and validation are defined and enforced.

****Later behavior:**** Tip and bill total arithmetic exists only at M4.

****Prerequisites:**** FR-DAT-005
****Journeys:**** GJ-06
****Acceptance tests:**** TST-M1-FR-DAT-006, TST-M4-FR-DAT-006

FR-DAT-007 - Optimistic concurrency

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** retained
****Introduced:**** M1
****Revalidated:**** M3, M4
****Required behavior:**** Require expected version for mutable aggregates and return explicit conflict outcomes.

****Gate-local behavior:**** Mutable aggregates carry a version and concurrent writes return an explicit conflict.

****Later behavior:**** Order and check aggregates are the operationally contended ones.

****Prerequisites:**** FR-DAT-002
****Journeys:**** None
****Acceptance tests:**** TST-M1-FR-DAT-007, TST-M3-FR-DAT-007, TST-M4-FR-DAT-007

FR-DAT-009 - Soft lifecycle

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** retained
****Introduced:**** M1
****Revalidated:**** M2, M3
****Required behavior:**** Deactivate/archive master data while preserving referenced historical versions.

****Gate-local behavior:**** Master records deactivate/archive without breaking references; historical versions remain readable.

****Later behavior:**** Menu items and prices are the first heavily versioned master data; order snapshots reference them.

****Prerequisites:**** FR-DAT-002
****Journeys:**** None
****Acceptance tests:**** TST-M1-FR-DAT-009, TST-M2-FR-DAT-009, TST-M3-FR-DAT-009

FR-DAT-013 - Seed data

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** retained
****Introduced:**** M1
****Revalidated:**** M2
****Required behavior:**** Seed two differently branded tenants and at least two outlets in one tenant with English, Amharic and Arabic content to prove neutrality and outlet isolation.

****Gate-local behavior:**** Two branded tenants and two outlets are seeded and isolation is provable at the M1 surface.

****Later behavior:**** Three-language content requires the localization and menu surfaces.

****Prerequisites:**** FR-TEN-001
****Journeys:**** None
****Acceptance tests:**** TST-M1-FR-DAT-013, TST-M2-FR-DAT-013

FR-DAT-015 - Schema documentation

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** retained
****Introduced:**** M1
****Revalidated:**** M4, M6
****Required behavior:**** Generate a current schema/domain catalog and relationship diagrams from the implemented model.

****Gate-local behavior:**** Catalog and diagrams generate from the live schema and match it.

****Later behavior:**** The catalog must remain current as the schema grows.

****Prerequisites:**** FR-DAT-001
****Journeys:**** None
****Acceptance tests:**** TST-M1-FR-DAT-015, TST-M4-FR-DAT-015, TST-M6-FR-DAT-015

FR-DAT-016 - Migration rollback

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** retained
****Introduced:**** M1
****Revalidated:**** M6
****Required behavior:**** Migrations are forward-only in production with checksum locking, preflight, backup and tested recovery; do not edit an applied migration.

****Gate-local behavior:**** Checksum locking and preflight reject edited migrations; forward-only policy enforced.

****Later behavior:**** Backup-and-tested-recovery around a production migration requires production backup capability.

****Prerequisites:**** FR-DAT-001

****Journeys:**** GJ-11

****Acceptance tests:**** TST-M1-FR-DAT-016, TST-M6-FR-DAT-016

FR-DAT-017 - Test realism

****Priority:**** P0

****Owner:**** Platform & Security

****Lineage:**** retained

****Introduced:**** M1

****Revalidated:**** M4, M5a

****Required behavior:**** Run integration, migration, RLS, synchronization and financial tests against real PostgreSQL and the actual application roles.

****Gate-local behavior:**** Integration, migration and RLS tests execute against real PostgreSQL using production roles, not a superuser.

****Later behavior:**** Financial tests exist at M4; synchronization tests at M5a.

****Prerequisites:**** FR-DAT-001

****Journeys:**** None

****Acceptance tests:**** TST-M1-FR-DAT-017, TST-M4-FR-DAT-017, TST-M5a-FR-DAT-017

FR-DAT-018 - Retention

****Priority:**** P0

****Owner:**** Platform & Security

****Lineage:**** retained

****Introduced:**** M1

****Revalidated:**** M4, M5a

****Required behavior:**** Apply configurable retention/archive policies without breaking ledger/audit integrity.

****Gate-local behavior:**** Retention/archive policy is configurable and cannot delete append-only ledger or audit rows.

****Later behavior:**** Financial ledgers exist at M4; archived synchronization evidence at M5a.

****Prerequisites:**** FR-DAT-002, FR-SEC-009

****Journeys:**** None

****Acceptance tests:**** TST-M1-FR-DAT-018, TST-M4-FR-DAT-018, TST-M5a-FR-DAT-018

Deployment and Operations

FR-OPS-001 - Environment validation

****Priority:**** P0

****Owner:**** Edge & Operations

****Lineage:**** retained

****Introduced:**** M1

****Revalidated:**** M5a, M6
****Required behavior:**** Validate required environment, role, URLs, certificates, paths, binaries and outlet identity before the process's first healthy/startup claim. Reject owner, superuser, BYPASSRLS and maintenance credentials.

****Gate-local behavior:**** The cloud process refuses to report healthy under owner, superuser, BYPASSRLS or maintenance credentials and validates environment, paths and binaries first.

****Later behavior:**** The outlet node is a second process with its own identity and certificate checks ; the production artifact is the final subject.

****Prerequisites:**** FR-DAT-017, FR-SEC-001
****Journeys:**** GJ-10, GJ-12
****Acceptance tests:**** TST-M1-FR-OPS-001, TST-M5a-FR-OPS-001, TST-M6-FR-OPS-001

FR-OPS-002 - Health endpoints

****Priority:**** P0
****Owner:**** Edge & Operations
****Lineage:**** retained
****Introduced:**** M1
****Revalidated:**** M5a, M6
****Required behavior:**** Health and readiness endpoints truthfully verify database migrations, production-role access, outlet authority, required jobs, local storage, printer status and synchronization dependencies appropriate to each service.

****Gate-local behavior:**** Cloud health truthfully verifies migrations, production-role access and required jobs, and reports unhealthy when any is unavailable.

****Later behavior:**** Outlet authority, local storage, printer status and synchronization dependencies exist on the node.

****Prerequisites:**** FR-OPS-001
****Journeys:**** GJ-10
****Acceptance tests:**** TST-M1-FR-OPS-002, TST-M5a-FR-OPS-002, TST-M6-FR-OPS-002

FR-OPS-003 - Structured logs

****Priority:**** P0
****Owner:**** Edge & Operations
****Lineage:**** retained
****Introduced:**** M1
****Revalidated:**** M5a
****Required behavior:**** Log correlation, tenant/outlet, actor/service, event and error classification with redaction.

****Gate-local behavior:**** Cloud logs carry correlation, tenant/outlet, actor, event and error classification with sensitive values redacted.

****Later behavior:**** The node emits its own log stream that must satisfy the same contract.

****Prerequisites:**** FR-DAT-004, FR-SEC-007
****Journeys:**** GJ-10
****Acceptance tests:**** TST-M1-FR-OPS-003, TST-M5a-FR-OPS-003

FR-OPS-004 - Metrics/tracing

****Priority:**** P0
****Owner:**** Edge & Operations
****Lineage:**** retained
****Introduced:**** M1
****Revalidated:**** M6
****Required behavior:**** Measure request/job/integration latency, error, queue and domain SLA using replaceable observability providers.

****Gate-local behavior:**** Latency, error and queue metrics emit through a replaceable provider interface.

****Later behavior:**** Domain SLA measures and production thresholds are confirmed on the deployed artifact.

****Prerequisites:**** FR-OPS-003
****Journeys:**** None
****Acceptance tests:**** TST-M1-FR-OPS-004, TST-M6-FR-OPS-004

FR-OPS-005 - Background jobs

****Priority:**** P0
****Owner:**** Edge & Operations
****Lineage:**** retained
****Introduced:**** M1
****Revalidated:**** M3, M5a
****Required behavior:**** Only Phase 1 background jobs are included and advertised. Readiness is unhealthy when an advertised job cannot perform real work.

****Gate-local behavior:**** Only Phase 1 jobs are registered and readiness turns unhealthy when an advertised job cannot do real work.

****Later behavior:**** Notification jobs arrive at M3 and synchronization jobs at M5a; each must satisfy the same rule.

****Prerequisites:**** FR-OPS-002, FR-GOV-002
****Journeys:**** GJ-10
****Acceptance tests:**** TST-M1-FR-OPS-005, TST-M3-FR-OPS-005, TST-M5a-FR-OPS-005

FR-OPS-008 - Deployment automation

****Priority:**** P0
****Owner:**** Edge & Operations
****Lineage:**** retained
****Introduced:**** M1
****Revalidated:**** M5a, M6
****Required behavior:**** Build deterministic cloud and outlet production artifacts from clean source; route/module checks run immediately after the canonical build without a second hidden profile build.

****Gate-local behavior:**** The cloud artifact builds deterministically from clean source and route/module checks run against that same build with no second profile build.

****Later behavior:**** The outlet node artifact is built at M5a; both production artifacts are confirmed at M6.

****Prerequisites:**** FR-GOV-002

****Journeys:**** GJ-12

****Acceptance tests:**** TST-M1-FR-OPS-008, TST-M5a-FR-OPS-008, TST-M6-FR-OPS-008

FR-OPS-020 - Production-role readiness

****Priority:**** P0

****Owner:**** Edge & Operations

****Lineage:**** retained

****Introduced:**** M1

****Revalidated:**** M5a, M6

****Required behavior:**** API, outlet node, workers, backup and restore pass readiness using exact least-privileged production identities; owner, superuser or maintenance fallback is forbidden.

****Gate-local behavior:**** API and workers pass readiness under exact least-privileged production identities with no fallback path.

****Later behavior:**** The outlet node is a further identity at M5a; backup and restore identities are exercised at M6.

****Prerequisites:**** FR-OPS-001, FR-DAT-017

****Journeys:**** GJ-11

****Acceptance tests:**** TST-M1-FR-OPS-020, TST-M5a-FR-OPS-020, TST-M6-FR-OPS-020

FR-OPS-021 - Ordinary cross-platform commands

****Priority:**** P0

****Owner:**** Edge & Operations

****Lineage:**** retained

****Introduced:**** M1

****Revalidated:**** M6

****Required behavior:**** The documented ordinary Windows and Linux commands discover required tools or fail clearly; acceptance cannot depend on CI-only PATH or environment injection.

****Gate-local behavior:**** The documented commands run on both platforms from a clean shell and fail clearly when a tool is missing.

****Later behavior:**** The complete command set including node and backup operations is confirmed at M6.

****Prerequisites:**** FR-OPS-008

****Journeys:**** GJ-12

****Acceptance tests:**** TST-M1-FR-OPS-021, TST-M6-FR-OPS-021

Identity and Authentication

FR-AUTH-001 - Staff login

****Priority:**** P0

****Owner:**** Platform & Security

****Lineage:**** retained

****Introduced:**** M1
****Revalidated:**** None
****Required behavior:**** Support verified phone or email login, secure password or OTP flows and provider-adaptor replacement.

****Gate-local behavior:**** Staff authenticate by verified phone or email with password or OTP; the provider adaptor is replaceable behind an interface.

****Later behavior:**** None beyond M1 for this clause.

****Prerequisites:**** FR-DAT-001
****Journeys:**** None
****Acceptance tests:**** TST-M1-FR-AUTH-001

FR-AUTH-004 - Session management

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** retained
****Introduced:**** M1
****Revalidated:**** M5a
****Required behavior:**** List and revoke sessions/devices; rotate tokens; invalidate on security events and role removal.

****Gate-local behavior:**** Sessions and devices are listable and revocable; tokens rotate; role removal invalidates access immediately.

****Later behavior:**** Edge-node and print-agent principals become revocable subjects when they exist.

****Prerequisites:**** FR-AUTH-001, FR-AUTH-008
****Journeys:**** None
****Acceptance tests:**** TST-M1-FR-AUTH-004, TST-M5a-FR-AUTH-004

FR-AUTH-005 - Quick PIN

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** retained
****Introduced:**** M1
****Revalidated:**** M4
****Required behavior:**** Trusted outlet terminals may use staff quick PIN for low-risk re-entry only; refunds, voids, high discounts, configuration and other sensitive actions require step-up authentication.

****Gate-local behavior:**** Quick PIN re-entry works only on registered trusted terminals and only for low-risk actions. | Configuration changes at M1 require step-up; the sensitive-action list is enforced for actions that exist.

****Later behavior:**** None beyond M1 for this clause. | Refunds, voids and high discounts exist only at M4 and must be proven to demand step-up there.

****Prerequisites:**** FR-AUTH-001, FR-SEC-014, FR-AUTH-006
****Journeys:**** GJ-07
****Acceptance tests:**** TST-M1-FR-AUTH-005A, TST-M1-FR-AUTH-005B, TST-M4-FR-AUTH-005B

FR-AUTH-006 - Step-up authentication

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** retained
****Introduced:**** M1
****Revalidated:**** M4, M6
****Required behavior:**** Require recent stronger authentication for refunds, reversals, payout, role changes, exports and other configured sensitive actions.

****Gate-local behavior:**** Step-up challenge with recency window enforced for role changes and configuration at M1.

****Later behavior:**** Refunds, reversals and payout exist at M4; export actions at M6.

****Prerequisites:**** FR-AUTH-001
****Journeys:**** GJ-07
****Acceptance tests:**** TST-M1-FR-AUTH-006, TST-M4-FR-AUTH-006, TST-M6-FR-AUTH-006

FR-AUTH-007 - Password/OTP security

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** retained
****Introduced:**** M1
****Revalidated:**** M6
****Required behavior:**** Apply rate limits, lockout and secure credential/OTP handling for the identity methods active in Phase 1. CI and simulators do not count as distributed production rate limiting.

****Gate-local behavior:**** Authentication and OTP endpoints enforce rate limits and lockout with secure credential handling.

****Later behavior:**** Distributed production enforcement is only demonstrable on the deployed artifact.

****Prerequisites:**** FR-AUTH-001
****Journeys:**** None
****Acceptance tests:**** TST-M1-FR-AUTH-007, TST-M6-FR-AUTH-007

FR-AUTH-008 - Memberships

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** retained
****Introduced:**** M1
****Revalidated:**** M3, M4, M5a
****Required behavior:**** Staff access is derived from explicit tenant/outlet memberships and role assignments; missing tenant or outlet context fails closed.

****Gate-local behavior:**** Access resolves only through membership records; absent tenant/outlet context returns no data and permits no write.

****Later behavior:**** New authorized surfaces must continue to fail closed.

****Prerequisites:**** FR-TEN-001
****Journeys:**** None
****Acceptance tests:**** TST-M1-FR-AUTH-008, TST-M3-FR-AUTH-008, TST-M4-FR-AUTH-008, TST-M5a-FR-AUTH-008

FR-AUTH-009 - Service accounts

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** retained
****Introduced:**** M1
****Revalidated:**** M5a
****Required behavior:**** Use scoped service principals for workers, edge nodes, print agents and integrations with rotation and revocation.

****Gate-local behavior:**** Cloud workers and integrations authenticate as scoped service principals with rotation and revocation.

****Later behavior:**** Edge node and print agent principals exist only at M5a.

****Prerequisites:**** FR-AUTH-004
****Journeys:**** GJ-10
****Acceptance tests:**** TST-M1-FR-AUTH-009, TST-M5a-FR-AUTH-009

FR-AUTH-010 - Recovery

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** retained
****Introduced:**** M1
****Revalidated:**** None
****Required behavior:**** Provide a documented administrator-controlled recovery process for lost credentials or factors, with identity verification, revocation and audit. Self-service advanced recovery may follow later.

****Gate-local behavior:**** Administrator-controlled recovery executes with identity verification, revocation of old factors and an audit record.

****Later behavior:**** None beyond M1 for this clause.

****Prerequisites:**** FR-AUTH-001, FR-SEC-009
****Journeys:**** None
****Acceptance tests:**** TST-M1-FR-AUTH-010

Integration Runtime

FR-INT-010 - Credential isolation

****Priority:**** P0
****Owner:**** Edge & Operations
****Lineage:**** retained
****Introduced:**** M1
****Revalidated:**** M4, M5a

****Required behavior:**** Store cloud, outlet-node, printer and payment-adapter credentials separately by tenant/outlet/environment; never include real secrets in source, images or fixtures.

****Gate-local behavior:**** Cloud credentials are stored per tenant/outlet/environment and no real secret appears in source, images or fixtures.

****Later behavior:**** Payment-adapter credentials exist at M4; outlet-node and printer credentials at M5a.

****Prerequisites:**** FR-SEC-007

****Journeys:**** GJ-10

****Acceptance tests:**** TST-M1-FR-INT-010, TST-M4-FR-INT-010, TST-M5a-FR-INT-010

Internationalization and Localization

FR-I18N-007 - Staff and back-office language

****Priority:**** P0

****Owner:**** Customer Experience

****Lineage:**** retained

****Introduced:**** M1

****Revalidated:**** M3, M4

****Required behavior:**** Waiter, POS, KDS and administration launch in English. Their architecture remains translation-ready, but no non-English staff UI is acceptance-critical in Phase 1.

****Gate-local behavior:**** The administration surface launches in English on a translation-ready framework.

****Later behavior:**** Waiter and KDS surfaces exist at M3; POS at M4. Each must launch in English on the same framework.

****Prerequisites:**** FR-CFG-006

****Journeys:**** None

****Acceptance tests:**** TST-M1-FR-I18N-007, TST-M3-FR-I18N-007, TST-M4-FR-I18N-007

Security and Data Protection

FR-SEC-001 - Deny by default

****Priority:**** P0

****Owner:**** Platform & Security

****Lineage:**** retained

****Introduced:**** M1

****Revalidated:**** M2, M3, M4, M5a

****Required behavior:**** Every API, database policy, job, cache, file and local/cloud synchronization path is deny-by-default. Missing tenant, outlet, session or actor context returns no data and permits no write.

****Gate-local behavior:**** All M1 API routes, policies, jobs, cache and file paths deny without full context.

****Later behavior:**** Each later gate adds new paths; the synchronization path clause is only executable when sync exists.

****Prerequisites:**** FR-TEN-001, FR-AUTH-008
****Journeys:**** None
****Acceptance tests:**** TST-M1-FR-SEC-001, TST-M2-FR-SEC-001, TST-M3-FR-SEC-001, TST-M4-FR-SEC-001, TST-M5a-FR-SEC-001

FR-SEC-002A - Cloud IDOR defense

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** split_successor from 'FR-SEC-002'
****Introduced:**** M1
****Revalidated:**** M3, M4, M6
****Required behavior:**** Populated foreign-tenant and sibling-outlet fixtures fail closed for all four verbs on cloud routes.

****Gate-local behavior:**** Populated foreign-tenant and sibling-outlet fixtures fail closed for all four verbs on cloud routes.

****Later behavior:**** New cloud resources at later gates must be covered; the complete production surface is revalidated at M6.

****Prerequisites:**** FR-TEN-001, FR-SEC-001
****Journeys:**** GJ-13
****Acceptance tests:**** TST-M1-FR-SEC-002A, TST-M3-FR-SEC-002A, TST-M4-FR-SEC-002A, TST-M6-FR-SEC-002A

FR-SEC-003 - Input validation

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** retained
****Introduced:**** M1
****Revalidated:**** M3, M4, M6
****Required behavior:**** Validate request schemas, lengths, enumerations, money, quantity, dates, files and URLs before domain execution.

****Gate-local behavior:**** Enforced across every route, template and data access path existing at M1.

****Later behavior:**** Later gates add routes and rendering surfaces subject to the same rule.

****Prerequisites:**** FR-DAT-001
****Journeys:**** None
****Acceptance tests:**** TST-M1-FR-SEC-003, TST-M3-FR-SEC-003, TST-M4-FR-SEC-003, TST-M6-FR-SEC-003

FR-SEC-004 - Injection defense

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** retained
****Introduced:**** M1
****Revalidated:**** M3, M4, M6
****Required behavior:**** Use parameterized data access, output escaping, CSP and safe template/file handling.

****Gate-local behavior:**** Enforced across every route, template and data access path existing at M1.

****Later behavior:**** Later gates add routes and rendering surfaces subject to the same rule.

****Prerequisites:**** FR-DAT-001

****Journeys:**** None

****Acceptance tests:**** TST-M1-FR-SEC-004, TST-M3-FR-SEC-004, TST-M4-FR-SEC-004, TST-M6-FR-SEC-004

FR-SEC-005 - CSRF/session

****Priority:**** P0

****Owner:**** Platform & Security

****Lineage:**** retained

****Introduced:**** M1

****Revalidated:**** M3, M4, M6

****Required behavior:**** Protect cookie-based actions against CSRF and use secure, httpOnly, same-site cookies where applicable.

****Gate-local behavior:**** Enforced across every route, template and data access path existing at M1.

****Later behavior:**** Later gates add routes and rendering surfaces subject to the same rule.

****Prerequisites:**** FR-DAT-001

****Journeys:**** None

****Acceptance tests:**** TST-M1-FR-SEC-005, TST-M3-FR-SEC-005, TST-M4-FR-SEC-005, TST-M6-FR-SEC-005

FR-SEC-006 - Rate limiting

****Priority:**** P0

****Owner:**** Platform & Security

****Lineage:**** retained

****Introduced:**** M1

****Revalidated:**** M2, M3, M4, M6

****Required behavior:**** Apply contextual rate limits to auth, QR resolution, ordering, payment, search, exports and webhooks.

****Gate-local behavior:**** Contextual limits enforced on the authentication and search surfaces existing at M1.

****Later behavior:**** QR resolution at M2, ordering at M3, payment and webhooks at M4, exports at M6.

****Prerequisites:**** FR-AUTH-007

****Journeys:**** None

****Acceptance tests:**** TST-M1-FR-SEC-006, TST-M2-FR-SEC-006, TST-M3-FR-SEC-006, TST-M4-FR-SEC-006, TST-M6-FR-SEC-006

FR-SEC-007 - Secret management

****Priority:**** P0

****Owner:**** Platform & Security

****Lineage:**** retained

****Introduced:**** M1

****Revalidated:**** M3, M4, M6

****Required behavior:**** Keep keys, passwords and tokens out of source, database plaintext, logs, screenshots and reports.

****Gate-local behavior:**** Enforced across every route, template and data access path existing at M1.

****Later behavior:**** Later gates add routes and rendering surfaces subject to the same rule.

****Prerequisites:**** FR-DAT-001

****Journeys:**** None

****Acceptance tests:**** TST-M1-FR-SEC-007, TST-M3-FR-SEC-007, TST-M4-FR-SEC-007, TST-M6-FR-SEC-007

FR-SEC-008 - Encryption

****Priority:**** P0

****Owner:**** Platform & Security

****Lineage:**** retained

****Introduced:**** M1

****Revalidated:**** M5b, M6

****Required behavior:**** Use TLS in transit and provider/platform encryption at rest; encrypt selected high-risk fields or backups.

****Gate-local behavior:**** Cloud transport uses TLS and data at rest is encrypted; selected high-risk fields are encrypted.

****Later behavior:**** Per-outlet public-trust TLS on the local node is an M5b concern; backup encryption is proven with backup/restore.

****Prerequisites:**** FR-DAT-001

****Journeys:**** GJ-11

****Acceptance tests:**** TST-M1-FR-SEC-008, TST-M5b-FR-SEC-008, TST-M6-FR-SEC-008

FR-SEC-009 - Audit separation

****Priority:**** P0

****Owner:**** Platform & Security

****Lineage:**** retained

****Introduced:**** M1

****Revalidated:**** M4, M5a

****Required behavior:**** Security audit storage and operational audit storage are append-only and cannot be changed by ordinary application roles.

****Gate-local behavior:**** Application roles cannot UPDATE or DELETE audit rows; append-only enforced at database level.

****Later behavior:**** Later gates write new audit categories under the same constraint.

****Prerequisites:**** FR-DAT-002, FR-DAT-017

****Journeys:**** None

****Acceptance tests:**** TST-M1-FR-SEC-009, TST-M4-FR-SEC-009, TST-M5a-FR-SEC-009

FR-SEC-010A - Customer PII classification

****Priority:**** P0

****Owner:**** Platform & Security

****Lineage:**** split_successor from 'FR-SEC-010'

****Introduced:**** M1

****Revalidated:** M3**
****Required behavior:**** Data classification exists and customer contact/complaint fields are restricted by role.

****Gate-local behavior:**** Data classification exists and customer contact/complaint fields are restricted by role.

****Later behavior:**** Complaint and service-failure data is generated at M3; restriction is revalidated once real records exist.

****Prerequisites:**** FR-AUTH-008
****Journeys:**** None
****Acceptance tests:**** TST-M1-FR-SEC-010A, TST-M3-FR-SEC-010A

FR-SEC-011 - Payment boundary

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** retained
****Introduced:**** M1
****Revalidated:**** M4
****Required behavior:**** Prohibit storage/logging of raw PAN, CVV and sensitive authentication data.

****Gate-local behavior:**** No schema field or log sink accepts PAN/CVV; scanner and log assertions prove absence.

****Later behavior:**** The payment surface where such data could appear exists at M4; external terminal recording at M4 and proof images must be re-proven.

****Prerequisites:**** FR-SEC-007
****Journeys:**** GJ-01B
****Acceptance tests:**** TST-M1-FR-SEC-011, TST-M4-FR-SEC-011

FR-SEC-012 - File security

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** retained
****Introduced:**** M1
****Revalidated:**** M4
****Required behavior:**** Validate, scan, hash, authorize, sign and expire access to private files.

****Gate-local behavior:**** Uploads are validated, scanned, hashed and served through signed expiring authorized URLs.

****Later behavior:**** The first real private-file class is the Telebirr/CBE payment proof at M4.

****Prerequisites:**** FR-SEC-003
****Journeys:**** GJ-02B
****Acceptance tests:**** TST-M1-FR-SEC-012, TST-M4-FR-SEC-012

FR-SEC-013 - MFA

****Priority:**** P0

****Owner:**** Platform & Security
****Lineage:**** retained
****Introduced:**** M1
****Revalidated:**** M4
****Required behavior:**** Support step-up MFA for high-risk staff actions. Phase 1 may use an administrator-controlled enrolment/recovery process, but cannot falsely claim every role is forced to enrol.

****Gate-local behavior:**** Step-up MFA is available for high-risk actions with administrator-controlled enrolment; documentation states the true enrolment posture.

****Later behavior:**** High-risk financial actions appear at M4.

****Prerequisites:**** FR-AUTH-006, FR-AUTH-010
****Journeys:**** None
****Acceptance tests:**** TST-M1-FR-SEC-013, TST-M4-FR-SEC-013

FR-SEC-014 - Device security

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** retained
****Introduced:**** M1
****Revalidated:**** M5a
****Required behavior:**** Register, name, revoke and monitor staff devices/terminals/edge nodes.

****Gate-local behavior:**** Staff devices and terminals are registered, named, revocable and monitored.

****Later behavior:**** Edge nodes are registrable subjects only once they exist.

****Prerequisites:**** FR-AUTH-004
****Journeys:**** GJ-10
****Acceptance tests:**** TST-M1-FR-SEC-014, TST-M5a-FR-SEC-014

FR-SEC-016 - Security headers

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** retained
****Introduced:**** M1
****Revalidated:**** M2, M5b
****Required behavior:**** Set CSP, HSTS, frame, referrer, permissions and content-type protections appropriate to each app.

****Gate-local behavior:**** Security headers are set correctly for each application surface existing at M1.

****Later behavior:**** Customer PWA at M2 and the locally served same-QR surface at M5b are additional apps with their own header posture.

****Prerequisites:**** FR-SEC-004
****Journeys:**** GJ-08
****Acceptance tests:**** TST-M1-FR-SEC-016, TST-M2-FR-SEC-016, TST-M5b-FR-SEC-016

FR-SEC-018 - Data deletion

****Priority:** P0**
****Owner:** Platform & Security**
****Lineage:** retained**
****Introduced:** M1**
****Revalidated:** M4, M6**
****Required behavior:** Use retention-aware anonymization/erasure workflows and legal holds.**

****Gate-local behavior:** Anonymization/erasure workflow and legal hold flags exist and respect ledger integrity for M1 data.**

****Later behavior:** Customer session and order data subject to erasure exist from M2 and M3; ledger-protected financial records at M4.**

****Prerequisites:** FR-DAT-018**
****Journeys:** None**
****Acceptance tests:** TST-M1-FR-SEC-018, TST-M4-FR-SEC-018, TST-M6-FR-SEC-018**

FR-SEC-020 - Incident response

****Priority:** P0**
****Owner:** Platform & Security**
****Lineage:** retained**
****Introduced:** M1**
****Revalidated:** M6**
****Required behavior:** Provide detection, containment, credential rotation, tenant notification and evidence-preservation runbook.**

****Gate-local behavior:** Runbook exists and credential rotation, containment and evidence preservation are executable against M1 capability.**

****Later behavior:** Detection depends on production monitoring; tenant notification on the deployed multi-tenant artifact.**

****Prerequisites:** FR-AUTH-004, FR-SEC-009**
****Journeys:** None**
****Acceptance tests:** TST-M1-FR-SEC-020, TST-M6-FR-SEC-020**

Tenant and Commercial Control

FR-TEN-001 - Tenant isolation

****Priority:** P0**
****Owner:** Platform & Security**
****Lineage:** retained**
****Introduced:** M1**
****Revalidated:** M2, M3, M4, M5a, M6**
****Required behavior:** Every tenant-owned record and operation is scoped to a tenant; cross-tenant reads/writes are denied in API, database policy, jobs, cache, files and exports.**

****Gate-local behavior:** Deny cross-tenant SELECT/INSERT/UPDATE/DELETE on the tables existing at M1 through production roles, with populated foreign-tenant fixtures.**

****Later behavior:** Each later gate adds new tenant-owned records and new access paths that the same**

rule must cover.

****Prerequisites:**** FR-DAT-001
****Journeys:**** None
****Acceptance tests:**** TST-M1-FR-TEN-001, TST-M2-FR-TEN-001, TST-M3-FR-TEN-001, TST-M4-FR-TEN-001, TST-M5a-FR-TEN-001, TST-M6-FR-TEN-001

FR-TEN-002A - Phase 1 organizational hierarchy

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** split_successor from 'FR-TEN-002'
****Introduced:**** M1
****Revalidated:**** M2, M5a
****Required behavior:**** Brand, legal entity, outlet, service area, preparation station and device entities exist with configurable depth; no fixed-level assumption in schema or queries.

****Gate-local behavior:**** Brand, legal entity, outlet, service area, preparation station and device entities exist with configurable depth; no fixed-level assumption in schema or queries.

****Later behavior:**** Table entity becomes operationally exercised when QR/table sessions exist; device registration extends to outlet nodes and print agents.

****Prerequisites:**** FR-DAT-001
****Journeys:**** None
****Acceptance tests:**** TST-M1-FR-TEN-002A, TST-M2-FR-TEN-002A, TST-M5a-FR-TEN-002A

FR-TEN-003 - Tenant configuration

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** retained
****Introduced:**** M1
****Revalidated:**** M4, M5a
****Required behavior:**** Store versioned branding, locale, currency, timezone, tax, calendar, numbering, payment, service, feature and connector configuration.

****Gate-local behavior:**** Versioned configuration records exist and are readable/writable per tenant with effective dating for the categories above.

****Later behavior:**** Payment and connector configuration values are only exercised once payment execution (M4) and outlet connectors (M5a) exist.

****Prerequisites:**** FR-DAT-001, FR-TEN-010
****Journeys:**** None
****Acceptance tests:**** TST-M1-FR-TEN-003, TST-M4-FR-TEN-003, TST-M5a-FR-TEN-003

FR-TEN-004 - Module entitlements

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** retained
****Introduced:**** M1
****Revalidated:**** M6

****Required behavior:**** Enable or disable modules and features by tenant and optionally legal entity/outlet without code forks.

****Gate-local behavior:**** Entitlement records resolve per tenant/entity/outlet and gate feature availability at runtime with no forked build.

****Later behavior:**** Entitlement effect on later feature surfaces is confirmed as those surfaces appear.

****Prerequisites:**** FR-TEN-003

****Journeys:**** None

****Acceptance tests:**** TST-M1-FR-TEN-004, TST-M6-FR-TEN-004

FR-TEN-005 - Second tenant

****Priority:**** P0

****Owner:**** Platform & Security

****Lineage:**** retained

****Introduced:**** M1

****Revalidated:**** M2, M4, M6

****Required behavior:**** Seed and test a second tenant with different branding, language defaults, tax settings and module entitlements.

****Gate-local behavior:**** A second tenant is seeded with distinct branding and entitlements and proven isolated at the M1 surface.

****Later behavior:**** Different language defaults require the localization surface; tax settings require billing; full commercial isolation is proven by the M6 second-tenant journey.

****Prerequisites:**** FR-TEN-001, FR-DAT-013

****Journeys:**** GJ-13

****Acceptance tests:**** TST-M1-FR-TEN-005, TST-M2-FR-TEN-005, TST-M4-FR-TEN-005, TST-M6-FR-TEN-005

FR-TEN-009A - Phase 1 system-of-record registry

****Priority:**** P0

****Owner:**** Platform & Security

****Lineage:**** split_successor from 'FR-TEN-009'

****Introduced:**** M1

****Revalidated:**** M4

****Required behavior:**** A system-of-record registry entry exists per tenant/legal entity with the Phase 1 enumeration only.

****Gate-local behavior:**** A system-of-record registry entry exists per tenant/legal entity with the Phase 1 enumeration only.

****Later behavior:**** The registry governs actual fiscal document and payment behavior once those exist.

****Prerequisites:**** FR-TEN-003

****Journeys:**** None

****Acceptance tests:**** TST-M1-FR-TEN-009A, TST-M4-FR-TEN-009A

FR-TEN-010 - Configuration history

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** retained
****Introduced:**** M1
****Revalidated:**** M4
****Required behavior:**** Version and audit policy changes, including effective dates and the actor who approved them.

****Gate-local behavior:**** Each configuration change writes an append-only audit record with actor, approval and effective date.

****Later behavior:**** Policy categories introduced later produce the same audit evidence.

****Prerequisites:**** FR-AUTH-008, FR-SEC-009
****Journeys:**** None
****Acceptance tests:**** TST-M1-FR-TEN-010, TST-M4-FR-TEN-010

Testing and Evidence

FR-TST-001 - Unit tests

****Priority:**** P0
****Owner:**** Quality Engineering
****Lineage:**** retained
****Introduced:**** M1
****Revalidated:**** M2, M3, M4
****Required behavior:**** Test pure domain policies, calculations, state machines, money, sellable-item quantity units (portion, each and sold-by-weight presentation), translations and permissions. Operational inventory or recipe units of measure are outside Phase 1 and have no test obligation.

****Gate-local behavior:**** Unit tests cover M1 domain policies, money types and permissions with meaningful assertions.

****Later behavior:**** State machines and translations mature at M2 and M3; money calculations become substantive at M4.

****Prerequisites:**** FR-DAT-005, FR-AUTH-008
****Journeys:**** None
****Acceptance tests:**** TST-M1-FR-TST-001, TST-M2-FR-TST-001, TST-M3-FR-TST-001, TST-M4-FR-TST-001

FR-TST-002 - Database integration

****Priority:**** P0
****Owner:**** Quality Engineering
****Lineage:**** retained
****Introduced:**** M1
****Revalidated:**** M3, M4, M5a
****Required behavior:**** Test constraints, transactions, locks, tenant/outlet RLS, later-added outlet columns, append-only rules and migrations on real PostgreSQL using raw production-role pools.

****Gate-local behavior:**** Constraints, transactions, locks, RLS and migrations are tested on real PostgreSQL through raw production-role pools.

****Later behavior:**** Order and financial state relationships at M3 and M4; the local database at M5a.

****Prerequisites:**** FR-DAT-017, FR-SEC-002A

****Journeys:**** GJ-10

****Acceptance tests:**** TST-M1-FR-TST-002, TST-M3-FR-TST-002, TST-M4-FR-TST-002, TST-M5a-FR-TST-002

FR-TST-003 - API tests

****Priority:**** P0

****Owner:**** Quality Engineering

****Lineage:**** retained

****Introduced:**** M1

****Revalidated:**** M3, M4

****Required behavior:**** Test validation, auth, idempotency, concurrency, error contracts and pagination.

****Gate-local behavior:**** API tests cover validation, auth, idempotency, concurrency, error contracts and pagination for M1 routes.

****Later behavior:**** Each later gate adds substantial route surface subject to the same suite.

****Prerequisites:**** FR-DAT-001, FR-AUTH-001

****Journeys:**** None

****Acceptance tests:**** TST-M1-FR-TST-003, TST-M3-FR-TST-003, TST-M4-FR-TST-003

FR-TST-004A - Phase 1 adapter contract tests

****Priority:**** P0

****Owner:**** Quality Engineering

****Lineage:**** split_successor from 'FR-TST-004'

****Introduced:**** M1

****Revalidated:**** M4, M5a

****Required behavior:**** Messaging and storage ports are tested against production-equivalent simulators.

****Gate-local behavior:**** Messaging and storage ports are tested against production-equivalent simulators.

****Later behavior:**** Payment and fiscal ports exist at M4; outlet-node, synchronization and print ports at M5a.

****Prerequisites:**** FR-AUTH-001, FR-SEC-012

****Journeys:**** GJ-10

****Acceptance tests:**** TST-M1-FR-TST-004A, TST-M4-FR-TST-004A, TST-M5a-FR-TST-004A

FR-TST-006 - Security tests

****Priority:**** P0

****Owner:**** Quality Engineering

****Lineage:**** retained

****Introduced:**** M1

****Revalidated:**** M3, M4, M5a, M6

****Required behavior:**** Include non-vacuous security tests with owner-confirmed populated fixtures, a bsent/malformed context, sibling-outlet attempts and deliberately weakened policies.

****Gate-local behavior:**** Security tests run against owner-confirmed populated fixtures and fail when a policy is deliberately weakened.

****Later behavior:**** Each gate adds new policy surface that must be tested the same way; the complete production surface is revalidated at M6.

****Prerequisites:**** FR-SEC-002A, FR-TST-017

****Journeys:**** GJ-13

****Acceptance tests:**** TST-M1-FR-TST-006, TST-M3-FR-TST-006, TST-M4-FR-TST-006, TST-M5a-FR-TST-006, TST-M6-FR-TST-006

FR-TST-012 - Migration and cross-platform upgrade

****Priority:**** P0

****Owner:**** Quality Engineering

****Lineage:**** retained

****Introduced:**** M1

****Revalidated:**** M6

****Required behavior:**** Test a fresh migration from this package baseline and a supported upgrade from a prior approved baseline of this package, checksum lock, line endings and ordinary Windows/Linux tool invocation.

****Gate-local behavior:**** A fresh migration applies cleanly with checksum lock and correct line endings under ordinary Windows and Linux invocation.

****Later behavior:**** The supported upgrade path from a prior baseline is exercised against the production schema.

****Prerequisites:**** FR-DAT-001, FR-DAT-016

****Journeys:**** GJ-12

****Acceptance tests:**** TST-M1-FR-TST-012, TST-M6-FR-TST-012

FR-TST-015 - Defect gates

****Priority:**** P0

****Owner:**** Quality Engineering

****Lineage:**** retained

****Introduced:**** M1

****Revalidated:**** M6

****Required behavior:**** Every confirmed defect receives an unchanged regression gate plus an independent stronger negative control when the original assertion could be vacuous or self-invalidating.

****Gate-local behavior:**** Each confirmed defect produces an unchanged regression gate and, where the assertion could be vacuous, an independent stronger negative control.

****Later behavior:**** The complete defect and control set is audited against the final artifact.

****Prerequisites:**** FR-TST-017

****Journeys:**** None

****Acceptance tests:**** TST-M1-FR-TST-015, TST-M6-FR-TST-015

FR-TST-016 - Evidence report

****Priority:**** P0
****Owner:**** Quality Engineering
****Lineage:**** retained
****Introduced:**** M1
****Revalidated:**** M6
****Required behavior:**** Produce commit, versions, migration list, test results, screenshots/artifacts, known nonblocking limitations and deployment commands.

****Gate-local behavior:**** Each gate produces an evidence report containing all seven named elements.

****Later behavior:**** The consolidated pilot evidence report is produced at handover.

****Prerequisites:**** FR-GOV-004
****Journeys:**** GJ-12
****Acceptance tests:**** TST-M1-FR-TST-016, TST-M6-FR-TST-016

FR-TST-020 - Order-independent validation

****Priority:**** P0
****Owner:**** Quality Engineering
****Lineage:**** retained
****Introduced:**** M1
****Revalidated:**** M6
****Required behavior:**** Clean build and test results are identical regardless of prior command order; stale dist, .next, test databases and generated artifacts cannot influence a pass.

****Gate-local behavior:**** Results are identical from a clean checkout regardless of command order and stale artifacts cannot influence a pass.

****Later behavior:**** Order independence is re-proven across the full production build including the node artifact.

****Prerequisites:**** FR-OPS-008
****Journeys:**** GJ-12
****Acceptance tests:**** TST-M1-FR-TST-020, TST-M6-FR-TST-020

UX and Accessibility

FR-UX-003 - Back office desktop first

****Priority:**** P0
****Owner:**** Customer Experience
****Lineage:**** retained
****Introduced:**** M1
****Revalidated:**** M6
****Required behavior:**** Use dense but responsive information architecture, filters, saved views and keyboard support.

****Gate-local behavior:**** The administration surface supports dense responsive layout, filters, saved views and keyboard navigation.

****Later behavior:**** The reporting and back-office surfaces extend this at M6.

****Prerequisites:**** FR-I18N-007

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**Journeys:** None
**Acceptance tests:** TST-M1-FR-UX-003, TST-M6-FR-UX-003

# Gate M2

## Allergens and Dietary Safety

### FR-SAF-001 - Allergen catalog

**Priority:** P0
**Owner:** Platform Engineering
**Lineage:** retained
**Introduced:** M2
**Revalidated:** M3
**Required behavior:** Provide a tenant/jurisdiction-configurable allergen catalog with translated customer text, approved universal supporting icons and English kitchen codes. Icons supplement but never replace written warnings.

**Gate-local behavior:** The catalog is configurable per tenant/jurisdiction with translated text, icons and English kitchen codes, and no icon-only warning path exists.

**Later behavior:** English kitchen codes are consumed by kitchen tickets at M3.

**Prerequisites:** FR-I18N-003
**Journeys:** GJ-05
**Acceptance tests:** TST-M2-FR-SAF-001, TST-M3-FR-SAF-001

### FR-SAF-002 - Item allergen declaration

**Priority:** P0
**Owner:** Platform Engineering
**Lineage:** retained
**Introduced:** M2
**Revalidated:** M3
**Required behavior:** Record contains, may-contain and cross-contact warnings by item/variant/modifier and effective version.

**Gate-local behavior:** All three warning classes record per item/variant/modifier with effective versioning.

**Later behavior:** Declarations are propagated to order and kitchen surfaces at M3.

**Prerequisites:** FR-SAF-001, FR-MNU-005
**Journeys:** None
**Acceptance tests:** TST-M2-FR-SAF-002, TST-M3-FR-SAF-002

### FR-SAF-003 - Allergy customer input

**Priority:** P0
**Owner:** Platform Engineering
**Lineage:** retained
**Introduced:** M2
**Revalidated:** M3
**Required behavior:** Allow a customer or waiter to flag an allergy concern for an order/table with
```


explicit acknowledgement text.

****Gate-local behavior:**** A table-level allergy flag is recorded with explicit acknowledgement text on the session.

****Later behavior:**** The order-level flag and waiter-initiated flagging require the M3 order and waiter surfaces.

****Prerequisites:**** FR-TAB-003, FR-SAF-001

****Journeys:**** GJ-01A, GJ-05

****Acceptance tests:**** TST-M2-FR-SAF-003, TST-M3-FR-SAF-003

FR-SAF-005 - Change detection

****Priority:**** P0

****Owner:**** Platform Engineering

****Lineage:**** retained

****Introduced:**** M2

****Revalidated:**** M3

****Required behavior:**** Re-evaluate allergen/dietary declarations when modifiers or substitutions alter ingredients.

****Gate-local behavior:**** Selecting a modifier that alters ingredients re-evaluates the declaration in the menu/cart surface.

****Later behavior:**** Substitutions during an accepted order are an M3 behavior.

****Prerequisites:**** FR-SAF-002, FR-MNU-006

****Journeys:**** GJ-01A

****Acceptance tests:**** TST-M2-FR-SAF-005, TST-M3-FR-SAF-005

FR-SAF-006 - Dietary claims

****Priority:**** P0

****Owner:**** Platform Engineering

****Lineage:**** retained

****Introduced:**** M2

****Revalidated:**** None

****Required behavior:**** Manage vegetarian, vegan, fasting, halal and other claims with definitions, evidence owner, review date and outlet applicability.

****Gate-local behavior:**** Dietary claims carry definitions, evidence owner, review date and outlet applicability.

****Later behavior:**** None beyond M2 for this clause.

****Prerequisites:**** FR-SAF-001, FR-TEN-002A

****Journeys:**** None

****Acceptance tests:**** TST-M2-FR-SAF-006

FR-SAF-007 - Publication block

****Priority:**** P0

****Owner:**** Platform Engineering

****Lineage:**** retained
****Introduced:**** M2
****Revalidated:**** None
****Required behavior:**** Block publication when required safety translation or allergen review is incomplete.

****Gate-local behavior:**** Publication is blocked when safety translation or allergen review is incomplete, in all three locales.

****Later behavior:**** None beyond M2 for this clause.

****Prerequisites:**** FR-MNU-003, FR-SAF-002

****Journeys:**** None

****Acceptance tests:**** TST-M2-FR-SAF-007

FR-SAF-008 - Audit

****Priority:**** P0
****Owner:**** Platform Engineering
****Lineage:**** retained
****Introduced:**** M2
****Revalidated:**** None
****Required behavior:**** Record who created, reviewed, approved and published safety-critical content.

****Gate-local behavior:**** Every safety-content state transition records actor and timestamp in append-only audit.

****Later behavior:**** None beyond M2 for this clause.

****Prerequisites:**** FR-SEC-009, FR-MNU-003

****Journeys:**** None

****Acceptance tests:**** TST-M2-FR-SAF-008

FR-SAF-009 - Disclaimer

****Priority:**** P0
****Owner:**** Platform Engineering
****Lineage:**** retained
****Introduced:**** M2
****Revalidated:**** M4
****Required behavior:**** Display tenant-approved wording that information supports informed choice but cannot eliminate all cross-contact risk.

****Gate-local behavior:**** The tenant-approved disclaimer renders on the customer menu surface in all three locales.

****Later behavior:**** The disclaimer must also appear on the printed and digital receipt where the tenant requires it.

****Prerequisites:**** FR-SAF-001, FR-CFG-006

****Journeys:**** None

****Acceptance tests:**** TST-M2-FR-SAF-009, TST-M4-FR-SAF-009

Configuration and Setup

FR-CFG-001B - Locale, table and QR setup

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** split_successor from 'FR-CFG-001'
****Introduced:**** M2
****Revalidated:**** None
****Required behavior:**** The guided setup configures the three customer locales, currency, timezone, preparation stations, tables and opaque QR codes.

****Gate-local behavior:**** Complete locale, currency, timezone, station, table and QR setup and prove that the generated QR resolves to the configured outlet and table.

****Later behavior:****

****Prerequisites:**** FR-I18N-001A, FR-TAB-001
****Journeys:**** None
****Acceptance tests:**** TST-M2-FR-CFG-001B

Customer, Privacy and Consent

FR-CST-002 - Guest privacy

****Priority:**** P0
****Owner:**** Customer Experience
****Lineage:**** retained
****Introduced:**** M2
****Revalidated:**** M6
****Required behavior:**** Create guest profiles only when operationally necessary and expire/anonymize optional data according to policy.

****Gate-local behavior:**** A guest profile is created only where operationally required by the session, and optional data carries an expiry/anonymization policy.

****Later behavior:**** Retention execution over accumulated customer data is proven with the retention workflow.

****Prerequisites:**** FR-AUTH-003
****Journeys:**** None
****Acceptance tests:**** TST-M2-FR-CST-002, TST-M6-FR-CST-002

Identity and Authentication

FR-AUTH-003 - Guest session

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** retained
****Introduced:**** M2
****Revalidated:**** M3
****Required behavior:**** Allow a customer to create a privacy-minimized guest session for QR ordering without phone, email or loyalty registration.

****Gate-local behavior:**** Scanning a table QR creates a guest session holding no phone, email or account identifier.

****Later behavior:**** The M3 dine-in ordering journey revalidates that the privacy-minimized session remains anonymous when it submits an order.

****Prerequisites:**** FR-TAB-001, FR-TAB-003

****Journeys:**** GJ-01A

****Acceptance tests:**** TST-M2-FR-AUTH-003, TST-M3-FR-AUTH-003

Internationalization and Localization

FR-I18N-001A - Customer menu and QR languages

****Priority:**** P0

****Owner:**** Customer Experience

****Lineage:**** split_successor from 'FR-I18N-001'

****Introduced:**** M2

****Revalidated:**** None

****Required behavior:**** Exactly three customer locales are offered and menu content plus the QR/session surface renders completely in the selected language.

****Gate-local behavior:**** Exactly three customer locales are offered and menu content plus the QR/session surface renders completely in the selected language.

****Later behavior:**** Service text, bills and receipts do not exist at M2.

****Prerequisites:**** FR-I18N-011, FR-MNU-004

****Journeys:**** None

****Acceptance tests:**** TST-M2-FR-I18N-001A

FR-I18N-002 - Arabic RTL

****Priority:**** P0

****Owner:**** Customer Experience

****Lineage:**** retained

****Introduced:**** M2

****Revalidated:**** M3, M4

****Required behavior:**** Arabic uses true right-to-left layout, mirrored navigation where appropriate, correct number/currency rendering and mixed-script handling without breaking prices, modifiers or status timelines.

****Gate-local behavior:**** Menu, modifier and price surfaces render true RTL with correct ETB and numeral formatting and mixed Arabic/Latin strings.

****Later behavior:**** Status timelines exist at M3; bill and receipt layouts at M4.

****Prerequisites:**** FR-I18N-001A, FR-MNU-006

****Journeys:**** GJ-03A, GJ-03B

****Acceptance tests:**** TST-M2-FR-I18N-002, TST-M3-FR-I18N-002, TST-M4-FR-I18N-002

FR-I18N-003 - Customer translation records

****Priority:**** P0

****Owner:**** Customer Experience
****Lineage:**** retained
****Introduced:**** M2
****Revalidated:**** M3, M4
****Required behavior:**** Store human-approved customer translations separately from canonical records for menu, modifier, allergen, dietary, service, order-status, bill, tip and receipt content.

****Gate-local behavior:**** Translation records exist separately from canonical records and are populated and approved for menu, modifier, allergen and dietary content.

****Later behavior:**** Service and order-status content types exist at M3; bill, tip and receipt content types at M4.

****Prerequisites:**** FR-DAT-009
****Journeys:**** None
****Acceptance tests:**** TST-M2-FR-I18N-003, TST-M3-FR-I18N-003, TST-M4-FR-I18N-003

FR-I18N-004 - Language selection and fallback

****Priority:**** P0
****Owner:**** Customer Experience
****Lineage:**** retained
****Introduced:**** M2
****Revalidated:**** None
****Required behavior:**** Offer English, Amharic and Arabic explicitly. Browser language is a suggestion only; remember the customer's choice for the table session and permit switching without losing the cart.

****Gate-local behavior:**** Explicit three-language selection persists on the table session and switching preserves cart contents.

****Later behavior:**** None beyond M2 for this clause.

****Prerequisites:**** FR-TAB-003, FR-TAB-005
****Journeys:**** None
****Acceptance tests:**** TST-M2-FR-I18N-004

FR-I18N-005 - Locale formatting and order snapshot

****Priority:**** P0
****Owner:**** Customer Experience
****Lineage:**** retained
****Introduced:**** M2
****Revalidated:**** M3, M4
****Required behavior:**** Localize dates, times, numbers, currencies and plural forms while storing canonical values, and snapshot the customer-selected locale on the table session/order for communications, receipts, analytics and support evidence.

****Gate-local behavior:**** Formatting is locale-correct with canonical storage, and the selected locale is snapshot on the table session.

****Later behavior:**** The order-level snapshot exists at M3; receipt consumption of the snapshot at M4.

****Prerequisites:**** FR-DAT-004, FR-TAB-003
****Journeys:**** GJ-02B
****Acceptance tests:**** TST-M2-FR-I18N-005, TST-M3-FR-I18N-005, TST-M4-FR-I18N-005

FR-I18N-006 - Completeness and safety approval

****Priority:**** P0
****Owner:**** Customer Experience
****Lineage:**** retained
****Introduced:**** M2
****Revalidated:**** None
****Required behavior:**** Block publication when required English/Amharic/Arabic customer or safety translations are incomplete. No live runtime machine translation may fill missing safety content.

****Gate-local behavior:**** Publication is blocked when any of the three locales lacks required customer or safety text, and no runtime translation path exists.

****Later behavior:**** None beyond M2 for this clause.

****Prerequisites:**** FR-MNU-003, FR-SAF-007
****Journeys:**** None
****Acceptance tests:**** TST-M2-FR-I18N-006

FR-I18N-010 - Machine assistance boundary

****Priority:**** P1
****Owner:**** Customer Experience
****Lineage:**** retained
****Introduced:**** M2
****Revalidated:**** None
****Required behavior:**** Permit AI/machine-assisted draft translation with provenance and human review, but prohibit live runtime customer translation and automatic publication of safety-critical text.

****Gate-local behavior:**** Draft translations carry provenance and require human approval; no runtime translation path and no automatic safety publication exists.

****Later behavior:**** None beyond M2 for this clause.

****Prerequisites:**** FR-I18N-003, FR-I18N-006
****Journeys:**** None
****Acceptance tests:**** TST-M2-FR-I18N-010

FR-I18N-011 - Exact launch locale set

****Priority:**** P0
****Owner:**** Customer Experience
****Lineage:**** retained
****Introduced:**** M2
****Revalidated:**** M3, M4
****Required behavior:**** The customer launch locale set is exactly English (en), Amharic (am) and Arabic (ar); staff and administration launch in English; Arabic renders true RTL.

****Gate-local behavior:**** The locale registry contains exactly en, am and ar for customers, and Arabic renders true RTL on the customer surface.

****Later behavior:**** The staff-English clause is revalidated wherever a staff surface is introduced.

****Prerequisites:**** None

****Journeys:**** GJ-03A

****Acceptance tests:**** TST-M2-FR-I18N-011, TST-M3-FR-I18N-011, TST-M4-FR-I18N-011

Menu and Pricing

FR-MNU-001 - Menu hierarchy

****Priority:**** P0

****Owner:**** Customer Experience

****Lineage:**** retained

****Introduced:**** M2

****Revalidated:**** None

****Required behavior:**** Manage menus, categories, groups, sellable items and variants independently of recipe and inventory identities.

****Gate-local behavior:**** The menu hierarchy exists and carries no recipe or inventory identity reference.

****Later behavior:**** None beyond M2 for this clause.

****Prerequisites:**** FR-DAT-009

****Journeys:**** None

****Acceptance tests:**** TST-M2-FR-MNU-001

FR-MNU-002A - Menu assignment

****Priority:**** P0

****Owner:**** Customer Experience

****Lineage:**** split_successor from 'FR-MNU-002'

****Introduced:**** M2

****Revalidated:**** None

****Required behavior:**** Menu assignment resolves correctly by outlet, service area, channel, daypart and date range.

****Gate-local behavior:**** Menu assignment resolves correctly by outlet, service area, channel, daypart and date range.

****Later behavior:**** None beyond M2 for this clause.

****Prerequisites:**** FR-MNU-001, FR-TEN-002A

****Journeys:**** None

****Acceptance tests:**** TST-M2-FR-MNU-002A

FR-MNU-003 - Menu publishing

****Priority:**** P0

****Owner:**** Customer Experience

****Lineage:**** retained

****Introduced:**** M2

****Revalidated:**** M3

****Required behavior:**** Use draft, review, scheduled, published, paused and archived states with immutable publication snapshots.

****Gate-local behavior:**** All six publication states and transitions work, and published snapshots are immutable.

****Later behavior:**** Order-time price/content snapshots reference these publication snapshots.

****Prerequisites:**** FR-MNU-001, FR-DAT-009

****Journeys:**** None

****Acceptance tests:**** TST-M2-FR-MNU-003, TST-M3-FR-MNU-003

FR-MNU-004 - Item content

****Priority:**** P0

****Owner:**** Customer Experience

****Lineage:**** retained

****Introduced:**** M2

****Revalidated:**** None

****Required behavior:**** Store translated name, short/long description, customer-visible ingredients, images, preparation time and display order.

****Gate-local behavior:**** All named content fields store and render per locale.

****Later behavior:**** None beyond M2 for this clause.

****Prerequisites:**** FR-MNU-001, FR-I18N-003

****Journeys:**** None

****Acceptance tests:**** TST-M2-FR-MNU-004

FR-MNU-005 - Variants

****Priority:**** P0

****Owner:**** Customer Experience

****Lineage:**** retained

****Introduced:**** M2

****Revalidated:**** None

****Required behavior:**** Support size, portion, temperature, preparation style and other sellable variants with independent prices and availability.

****Gate-local behavior:**** Variants carry independent prices and availability states.

****Later behavior:**** None beyond M2 for this clause.

****Prerequisites:**** FR-MNU-001

****Journeys:**** None

****Acceptance tests:**** TST-M2-FR-MNU-005

FR-MNU-006 - Modifier sets

****Priority:**** P0

****Owner:**** Customer Experience

****Lineage:**** retained

****Introduced:**** M2

****Revalidated:** M3**
****Required behavior:**** Support required/optional groups, min/max selection, defaults, included quantity, price deltas and incompatibilities.

****Gate-local behavior:**** Modifier group rules, price deltas and incompatibility constraints validate correctly.

****Later behavior:**** Modifier selection is exercised against real submitted orders at M3.

****Prerequisites:**** FR-MNU-005
****Journeys:**** GJ-01A
****Acceptance tests:**** TST-M2-FR-MNU-006, TST-M3-FR-MNU-006

FR-MNU-007 - Availability

****Priority:**** P0
****Owner:**** Customer Experience
****Lineage:**** retained
****Introduced:**** M2
****Revalidated:**** M3
****Required behavior:**** Publish available, limited, temporarily unavailable, scheduled later and hidden states without exposing exact stock.

****Gate-local behavior:**** All five availability states publish correctly and no exact stock quantity is exposed anywhere.

****Later behavior:**** Availability revalidation at order submission occurs at M3.

****Prerequisites:**** FR-MNU-001
****Journeys:**** GJ-01A
****Acceptance tests:**** TST-M2-FR-MNU-007, TST-M3-FR-MNU-007

FR-MNU-008 - Sold-out action

****Priority:**** P0
****Owner:**** Customer Experience
****Lineage:**** retained
****Introduced:**** M2
****Revalidated:**** M3
****Required behavior:**** Allow authorized station/outlet staff to pause an item/variant/modifier with reason and optional expected return.

****Gate-local behavior:**** Authorized staff pause an item/variant/modifier with a reason code and optional expected return time.

****Later behavior:**** Station-initiated sold-out during service is exercised with the kitchen surface.

****Prerequisites:**** FR-MNU-007, FR-CFG-003
****Journeys:**** GJ-05
****Acceptance tests:**** TST-M2-FR-MNU-008, TST-M3-FR-MNU-008

FR-MNU-009 - Price versions

****Priority:**** P0

****Owner:**** Customer Experience
****Lineage:**** retained
****Introduced:**** M2
****Revalidated:**** M3
****Required behavior:**** Store effective-dated prices by outlet/channel/variant/currency/tax context and preserve price snapshots on orders.

****Gate-local behavior:**** Effective-dated prices resolve correctly across outlet, channel, variant, currency and tax context.

****Later behavior:**** Price snapshots on orders require orders to exist.

****Prerequisites:**** FR-MNU-005, FR-DAT-005
****Journeys:**** GJ-01A
****Acceptance tests:**** TST-M2-FR-MNU-009, TST-M3-FR-MNU-009

FR-MNU-010 - Daypart scheduling

****Priority:**** P0
****Owner:**** Customer Experience
****Lineage:**** retained
****Introduced:**** M2
****Revalidated:**** None
****Required behavior:**** Activate breakfast, lunch, dinner, late-night and tenant-defined windows using outlet-local time.

****Gate-local behavior:**** Daypart windows activate and deactivate menus using outlet-local time including boundary cases.

****Later behavior:**** None beyond M2 for this clause.

****Prerequisites:**** FR-MNU-002A, FR-DAT-004
****Journeys:**** None
****Acceptance tests:**** TST-M2-FR-MNU-010

FR-MNU-011 - Images

****Priority:**** P0
****Owner:**** Customer Experience
****Lineage:**** retained
****Introduced:**** M2
****Revalidated:**** None
****Required behavior:**** Support optimized responsive images, alt text, focal crop and private source asset management.

****Gate-local behavior:**** Responsive derivatives, alt text and focal crop work; source assets remain private and authorized.

****Later behavior:**** None beyond M2 for this clause.

****Prerequisites:**** FR-MNU-004, FR-SEC-012
****Journeys:**** None
****Acceptance tests:**** TST-M2-FR-MNU-011

FR-MNU-012 - Search and filtering

****Priority:**** P0
****Owner:**** Customer Experience
****Lineage:**** retained
****Introduced:**** M2
****Revalidated:**** M3
****Required behavior:**** Search translated names/descriptions and filter by category, dietary tags, allergens, availability, price and preparation time.

****Gate-local behavior:**** Search returns correct results across all three locales including mixed-script queries, and every named filter works.

****Later behavior:**** The M3 Arabic ordering journey revalidates mixed-script search against the live order-entry surface.

****Prerequisites:**** FR-MNU-004, FR-SAF-002
****Journeys:**** GJ-03A
****Acceptance tests:**** TST-M2-FR-MNU-012, TST-M3-FR-MNU-012

Recipes and Costing

FR-RCP-008A - Customer ingredient and allergen content

****Priority:**** P0
****Owner:**** Platform Engineering
****Lineage:**** split_successor from 'FR-RCP-008'
****Introduced:**** M2
****Revalidated:**** None
****Required behavior:**** Customer-visible ingredient text and allergen declarations are stored as customer content with no operational recipe linkage.

****Gate-local behavior:**** Customer-visible ingredient text and allergen declarations are stored as customer content with no operational recipe linkage.

****Later behavior:**** None beyond M2 for this clause.

****Prerequisites:**** FR-MNU-004, FR-SAF-002
****Journeys:**** None
****Acceptance tests:**** TST-M2-FR-RCP-008A

Tables and Sessions

FR-TAB-001 - QR resolution

****Priority:**** P0
****Owner:**** Customer Experience
****Lineage:**** retained
****Introduced:**** M2
****Revalidated:**** M5b
****Required behavior:**** Resolve a signed QR into tenant, outlet, service area and table without exposing internal IDs.

****Gate-local behavior:**** A signed opaque QR resolves to the correct tenant/outlet/service area/table

with no internal identifier in the URL or payload.

****Later behavior:**** The same QR must also resolve locally during outage under split-horizon DNS.

****Prerequisites:**** FR-TEN-002A, FR-DAT-003

****Journeys:**** GJ-08

****Acceptance tests:**** TST-M2-FR-TAB-001, TST-M5b-FR-TAB-001

FR-TAB-002 - QR rotation

****Priority:**** P0

****Owner:**** Customer Experience

****Lineage:**** retained

****Introduced:**** M2

****Revalidated:**** None

****Required behavior:**** Rotate/revoke table QR tokens and keep printable version history.

****Gate-local behavior:**** Tokens rotate and revoke; revoked tokens fail resolution; printable version history is retained.

****Later behavior:**** None beyond M2 for this clause.

****Prerequisites:**** FR-TAB-001

****Journeys:**** None

****Acceptance tests:**** TST-M2-FR-TAB-002

FR-TAB-003 - Table session creation

****Priority:**** P0

****Owner:**** Customer Experience

****Lineage:**** retained

****Introduced:**** M2

****Revalidated:**** None

****Required behavior:**** Create a table session with occupancy state, opening source, host staff, participants and timestamps.

****Gate-local behavior:**** A table session is created with all named attributes and correct UTC/outlet-time handling.

****Later behavior:**** None beyond M2 for this clause.

****Prerequisites:**** FR-TAB-001, FR-DAT-004

****Journeys:**** None

****Acceptance tests:**** TST-M2-FR-TAB-003

FR-TAB-004 - Participant joining

****Priority:**** P0

****Owner:**** Customer Experience

****Lineage:**** retained

****Introduced:**** M2

****Revalidated:**** M3

****Required behavior:**** Allow multiple devices/guests to join a session using the same active QR under configured visibility rules.

****Gate-local behavior:**** Multiple devices join one session under the configured visibility rule and see only permitted content.

****Later behavior:**** Multi-participant behavior under real ordering and service requests is proven operationally at M3.

****Prerequisites:**** FR-TAB-003

****Journeys:**** GJ-04

****Acceptance tests:**** TST-M2-FR-TAB-004, TST-M3-FR-TAB-004

FR-TAB-005 - Personal and shared carts

****Priority:**** P0

****Owner:**** Customer Experience

****Lineage:**** retained

****Introduced:**** M2

****Revalidated:**** M3

****Required behavior:**** Support personal baskets, shared baskets and transfer of items before submission.

****Gate-local behavior:**** Personal and shared baskets exist and items transfer between them prior to submission.

****Later behavior:**** Submission boundary behavior is exercised when orders exist.

****Prerequisites:**** FR-TAB-004, FR-MNU-005

****Journeys:**** GJ-04

****Acceptance tests:**** TST-M2-FR-TAB-005, TST-M3-FR-TAB-005

FR-TAB-006 - Table ownership

****Priority:**** P0

****Owner:**** Customer Experience

****Lineage:**** retained

****Introduced:**** M2

****Revalidated:**** M3

****Required behavior:**** Assign primary waiter/section and allow acknowledged transfer or supervisor reassignment.

****Gate-local behavior:**** A primary waiter/section is assignable to a table session with the ownership record stored.

****Later behavior:**** Acknowledged transfer and supervisor reassignment are service operations exercised with real waiter workflow.

****Prerequisites:**** FR-AUTH-008, FR-TAB-003

****Journeys:**** GJ-05

****Acceptance tests:**** TST-M2-FR-TAB-006, TST-M3-FR-TAB-006

FR-TAB-010 - Stale QR protection

****Priority:**** P0

****Owner:**** Customer Experience

****Lineage:**** retained
****Introduced:**** M2
****Revalidated:**** None
****Required behavior:**** Prevent a historic or photographed QR session from silently joining a later occupied table without configured verification.

****Gate-local behavior:**** A QR captured from an earlier occupancy cannot silently join a new session; configured verification is required.

****Later behavior:**** None beyond M2 for this clause.

****Prerequisites:**** FR-TAB-001, FR-TAB-004
****Journeys:**** None
****Acceptance tests:**** TST-M2-FR-TAB-010

Testing and Evidence

FR-TST-011 - Accessibility

****Priority:**** P0
****Owner:**** Quality Engineering
****Lineage:**** retained
****Introduced:**** M2
****Revalidated:**** M3, M6
****Required behavior:**** Automate common checks and perform manual keyboard/screen-reader/RTL reviews.

****Gate-local behavior:**** Automated accessibility checks and manual keyboard, screen-reader and RTL reviews run on the customer surface.

****Later behavior:**** Staff surfaces and the final production artifact are reviewed the same way.

****Prerequisites:**** FR-UX-011, FR-I18N-002
****Journeys:**** GJ-03A
****Acceptance tests:**** TST-M2-FR-TST-011, TST-M3-FR-TST-011, TST-M6-FR-TST-011

UX and Accessibility

FR-UX-001A - Customer dine-in mobile experience

****Priority:**** P0
****Owner:**** Customer Experience
****Lineage:**** split_successor from 'FR-UX-001'
****Introduced:**** M2
****Revalidated:**** M3, M4
****Required behavior:**** The QR entry and menu journey is one-handed, fast to first content and free of unnecessary mandatory fields.

****Gate-local behavior:**** The QR entry and menu journey is one-handed, fast to first content and free of unnecessary mandatory fields.

****Later behavior:**** Service-request journeys exist at M3; bill, tip and payment-request journeys at M4.

****Prerequisites:**** FR-TAB-001, FR-MNU-004

****Journeys:**** GJ-01A, GJ-01B
****Acceptance tests:**** TST-M2-FR-UX-001A, TST-M3-FR-UX-001A, TST-M4-FR-UX-001A

FR-UX-005 - Status clarity

****Priority:**** P0
****Owner:**** Customer Experience
****Lineage:**** retained
****Introduced:**** M2
****Revalidated:**** M3, M4, M5a
****Required behavior:**** Display text/icon status and timestamps; never communicate critical state by color alone.

****Gate-local behavior:**** Customer status uses text and icon with timestamps, never colour alone.

****Later behavior:**** Kitchen allergy salience at M3, payment state at M4, and connectivity state at M5a are all critical-state surfaces.

****Prerequisites:**** FR-I18N-001A, FR-TAB-003
****Journeys:**** GJ-05, GJ-10
****Acceptance tests:**** TST-M2-FR-UX-005, TST-M3-FR-UX-005, TST-M4-FR-UX-005, TST-M5a-FR-UX-005

FR-UX-006 - Error recovery

****Priority:**** P0
****Owner:**** Customer Experience
****Lineage:**** retained
****Introduced:**** M2
****Revalidated:**** M3, M4
****Required behavior:**** Preserve entered work, explain the issue and offer safe retry/correction without duplicate commitment.

****Gate-local behavior:**** A failure preserves cart contents, explains the issue and offers safe retry without duplicating anything.

****Later behavior:**** Duplicate commitment becomes materially harmful once orders and payments exist.

****Prerequisites:**** FR-TAB-005
****Journeys:**** GJ-01A, GJ-01B
****Acceptance tests:**** TST-M2-FR-UX-006, TST-M3-FR-UX-006, TST-M4-FR-UX-006

FR-UX-007 - Loading/offline

****Priority:**** P0
****Owner:**** Customer Experience
****Lineage:**** retained
****Introduced:**** M2
****Revalidated:**** M5a
****Required behavior:**** Distinguish loading, locally saved, queued, synchronized, stale, failed and blocked states.

****Gate-local behavior:**** Loading, stale and failed states are distinguishable on the customer surface.

****Later behavior:**** Locally saved, queued and synchronized states are produced by the synchronization subsystem.

****Prerequisites:**** FR-UX-006

****Journeys:**** GJ-10

****Acceptance tests:**** TST-M2-FR-UX-007, TST-M5a-FR-UX-007

FR-UX-011 - Accessibility

****Priority:**** P0

****Owner:**** Customer Experience

****Lineage:**** retained

****Introduced:**** M2

****Revalidated:**** M3, M4, M6

****Required behavior:**** Meet WCAG-aligned keyboard, focus, labels, contrast, zoom and screen-reader essentials.

****Gate-local behavior:**** The customer surface meets WCAG-aligned keyboard, focus, label, contrast, zoom and screen-reader essentials in all three locales.

****Later behavior:**** Staff surfaces at M3 and M4, and the reporting surface at M6, must meet the same standard.

****Prerequisites:**** FR-UX-005

****Journeys:**** GJ-03A

****Acceptance tests:**** TST-M2-FR-UX-011, TST-M3-FR-UX-011, TST-M4-FR-UX-011, TST-M6-FR-UX-011

FR-UX-012 - Performance budgets

****Priority:**** P0

****Owner:**** Customer Experience

****Lineage:**** retained

****Introduced:**** M2

****Revalidated:**** M3, M6

****Required behavior:**** Define and test key-page load, interaction and queue update budgets under realistic devices/network.

****Gate-local behavior:**** Menu and QR entry pages meet defined budgets on realistic devices and networks.

****Later behavior:**** Queue update budgets require the KDS realtime surface; production thresholds are confirmed on the deployment.

****Prerequisites:**** FR-UX-001A

****Journeys:**** GJ-01A

****Acceptance tests:**** TST-M2-FR-UX-012, TST-M3-FR-UX-012, TST-M6-FR-UX-012

FR-UX-013 - Localization fit

****Priority:**** P0

****Owner:**** Customer Experience

****Lineage:**** retained

****Introduced:**** M2

****Revalidated:**** M4

****Required behavior:**** Allow longer French/Amharic text and Arabic RTL without clipping or hard-coded widths.

****Gate-local behavior:**** Longer Amharic strings and Arabic RTL render without clipping or hard-coded widths on the menu surface.

****Later behavior:**** Bill, tip and receipt layouts are the tightest constrained surfaces.

****Prerequisites:**** FR-I18N-002

****Journeys:**** GJ-02B, GJ-03B

****Acceptance tests:**** TST-M2-FR-UX-013, TST-M4-FR-UX-013

FR-UX-014 - Empty states

****Priority:**** P0

****Owner:**** Customer Experience

****Lineage:**** retained

****Introduced:**** M2

****Revalidated:**** M6

****Required behavior:**** Use instructive empty/error states and never display fabricated analytics.

****Gate-local behavior:**** Empty and error states are instructive and no placeholder or fabricated figure is displayed.

****Later behavior:**** The no-fabricated-analytics rule bites hardest on the reporting surface.

****Prerequisites:**** FR-UX-006

****Journeys:**** GJ-12

****Acceptance tests:**** TST-M2-FR-UX-014, TST-M6-FR-UX-014

FR-UX-020 - Consistent design system

****Priority:**** P0

****Owner:**** Customer Experience

****Lineage:**** retained

****Introduced:**** M2

****Revalidated:**** M3, M4

****Required behavior:**** Use a shared component/token system across customer and staff apps while allowing tenant branding boundaries.

****Gate-local behavior:**** A shared token and component system drives the customer surface with tenant branding applied through configuration.

****Later behavior:**** Staff surfaces at M3 and M4 consume the same system.

****Prerequisites:**** FR-CFG-006

****Journeys:**** None

****Acceptance tests:**** TST-M2-FR-UX-020, TST-M3-FR-UX-020, TST-M4-FR-UX-020

Gate M3

Allergens and Dietary Safety

FR-SAF-004 - Prominent routing

****Priority:**** P0
****Owner:**** Platform Engineering
****Lineage:**** retained
****Introduced:**** M3
****Revalidated:**** None
****Required behavior:**** Display allergy flags prominently on confirmation, kitchen tickets, KDS, expo and service handoff.

****Gate-local behavior:**** Allergy flags render prominently on the order confirmation, kitchen ticket, KDS, expo and service handoff surfaces.

****Later behavior:**** None beyond M3 for this clause.

****Prerequisites:**** FR-SAF-003, FR-FUL-001
****Journeys:**** GJ-01A, GJ-05
****Acceptance tests:**** TST-M3-FR-SAF-004

Carts and Orders

FR-ORD-001A - Dine-in order aggregate

****Priority:**** P0
****Owner:**** Service Execution
****Lineage:**** split_successor from 'FR-ORD-001'
****Introduced:**** M3
****Revalidated:**** None
****Required behavior:**** A single order aggregate serves both QR dine-in and waiter-entered dine-in with channel-specific policy fields.

****Gate-local behavior:**** A single order aggregate serves both QR dine-in and waiter-entered dine-in with channel-specific policy fields.

****Later behavior:**** The counter POS channel does not exist until the POS surface is built.

****Prerequisites:**** FR-TAB-005, FR-MNU-009
****Journeys:**** GJ-01A, GJ-05
****Acceptance tests:**** TST-M3-FR-ORD-001A

FR-ORD-002 - Draft cart

****Priority:**** P0
****Owner:**** Service Execution
****Lineage:**** retained
****Introduced:**** M3
****Revalidated:**** M4
****Required behavior:**** Maintain draft carts without reserving stock or creating financial commitment.

****Gate-local behavior:**** Draft carts persist with no reservation record and no financial obligation created.

****Later behavior:**** The no-financial-commitment claim is re-proven once checks and payments exist.

****Prerequisites:**** FR-TAB-005
****Journeys:**** GJ-01A
****Acceptance tests:**** TST-M3-FR-ORD-002, TST-M4-FR-ORD-002

FR-ORD-003 - Order preview

****Priority:**** P0
****Owner:**** Service Execution
****Lineage:**** retained
****Introduced:**** M3
****Revalidated:**** M4
****Required behavior:**** Return a server-calculated preview with line prices, modifiers, tax, fees, discounts, availability, timing and policy warnings.

****Gate-local behavior:**** Preview is calculated server-side and returns every named component with no client-side price authority.

****Later behavior:**** Tax, fee and discount figures become financially binding when they flow into a check.

****Prerequisites:**** FR-MNU-009, FR-MNU-007
****Journeys:**** GJ-01A
****Acceptance tests:**** TST-M3-FR-ORD-003, TST-M4-FR-ORD-003

FR-ORD-004 - Submit idempotency

****Priority:**** P0
****Owner:**** Service Execution
****Lineage:**** retained
****Introduced:**** M3
****Revalidated:**** M5a
****Required behavior:**** Require idempotency key for submission and return the original outcome for safe retries.

****Gate-local behavior:**** Duplicate submission with the same key returns the original outcome and creates no second order.

****Later behavior:**** The same guarantee must hold when submissions are replayed through the outlet or tbox after an outage.

****Prerequisites:**** FR-DAT-007
****Journeys:**** GJ-01A, GJ-10
****Acceptance tests:**** TST-M3-FR-ORD-004, TST-M5a-FR-ORD-004

FR-ORD-005 - Commercial and language snapshot

****Priority:**** P0
****Owner:**** Service Execution
****Lineage:**** retained
****Introduced:**** M3
****Revalidated:**** M4
****Required behavior:**** Persist item, modifier, discount, tax, fee and total snapshots used at acceptance together with the customer-selected locale used for communications and receipt evidence.

****Gate-local behavior:**** Acceptance snapshots persist for every named component with the locale recorded on the order.

****Later behavior:**** The receipt evidence use of these snapshots exists only at M4.

****Prerequisites:**** FR-MNU-009, FR-I18N-005

****Journeys:**** GJ-01A, GJ-02B

****Acceptance tests:**** TST-M3-FR-ORD-005, TST-M4-FR-ORD-005

FR-ORD-006 - Availability revalidation

****Priority:**** P0

****Owner:**** Service Execution

****Lineage:**** retained

****Introduced:**** M3

****Revalidated:**** None

****Required behavior:**** Revalidate item/modifier availability, hours, channel, quantity and capacity at submission.

****Gate-local behavior:**** Submission re-checks availability, hours, channel, quantity and station capacity and rejects stale carts.

****Later behavior:**** None beyond M3 for this clause.

****Prerequisites:**** FR-MNU-007, FR-FUL-013

****Journeys:**** GJ-01A

****Acceptance tests:**** TST-M3-FR-ORD-006

FR-ORD-007A - Order acceptance

****Priority:**** P0

****Owner:**** Service Execution

****Lineage:**** split_successor from 'FR-ORD-007'

****Introduced:**** M3

****Revalidated:**** None

****Required behavior:**** Automatic and staff-confirmed acceptance resolve per channel and outlet policy.

****Gate-local behavior:**** Automatic and staff-confirmed acceptance resolve per channel and outlet policy.

****Later behavior:**** Payment-dependent acceptance requires payment authorization to exist.

****Prerequisites:**** FR-CFG-002A

****Journeys:**** GJ-01A, GJ-05

****Acceptance tests:**** TST-M3-FR-ORD-007A

FR-ORD-008 - Line ownership

****Priority:**** P0

****Owner:**** Service Execution

****Lineage:**** retained

****Introduced:**** M3

****Revalidated:**** M4

****Required behavior:**** Optionally associate lines with table participants without requiring every guest to register.

****Gate-local behavior:**** Lines optionally carry a participant reference and unregistered guests can still order.

****Later behavior:**** Participant attribution becomes financially material when bills are split per payer.

****Prerequisites:**** FR-TAB-004, FR-ORD-001A

****Journeys:**** GJ-04, GJ-06

****Acceptance tests:**** TST-M3-FR-ORD-008, TST-M4-FR-ORD-008

FR-ORD-009 - Add-on orders

****Priority:**** P0

****Owner:**** Service Execution

****Lineage:**** retained

****Introduced:**** M3

****Revalidated:**** M4

****Required behavior:**** Allow additional orders in the same table session while preserving separate timestamps and fulfillment.

****Gate-local behavior:**** Add-on orders create distinct orders and tickets in one session with independent timestamps.

****Later behavior:**** Add-on orders must consolidate correctly onto one check.

****Prerequisites:**** FR-ORD-001A, FR-FUL-002

****Journeys:**** GJ-04

****Acceptance tests:**** TST-M3-FR-ORD-009, TST-M4-FR-ORD-009

FR-ORD-010 - Amendment

****Priority:**** P0

****Owner:**** Service Execution

****Lineage:**** retained

****Introduced:**** M3

****Revalidated:**** None

****Required behavior:**** Use explicit amendment events for allowed pre-preparation changes and retain before/after details.

****Gate-local behavior:**** Pre-preparation amendments emit explicit events retaining before/after detail; post-preparation changes are refused.

****Later behavior:**** None beyond M3 for this clause.

****Prerequisites:**** FR-DAT-008A, FR-FUL-003

****Journeys:**** GJ-01A

****Acceptance tests:**** TST-M3-FR-ORD-010

FR-ORD-011 - Cancellation

****Priority:**** P0

****Owner:**** Service Execution
****Lineage:**** retained
****Introduced:**** M3
****Revalidated:**** M4
****Required behavior:**** Apply item/order cancellation policy by state, channel, reason, payment and preparation progress.

****Gate-local behavior:**** Cancellation is permitted or refused per state, channel, reason and preparation progress.

****Later behavior:**** The payment dimension of the policy requires payments to exist.

****Prerequisites:**** FR-CFG-002A, FR-CFG-003
****Journeys:**** GJ-07
****Acceptance tests:**** TST-M3-FR-ORD-011, TST-M4-FR-ORD-011

FR-ORD-012A - Operational void

****Priority:**** P0
****Owner:**** Service Execution
****Lineage:**** split_successor from 'FR-ORD-012'
****Introduced:**** M3
****Revalidated:**** M4
****Required behavior:**** Authorized void after acceptance records reason and immutable audit for an unpaid order.

****Gate-local behavior:**** Authorized void after acceptance records reason and immutable audit for an unpaid order.

****Later behavior:**** Linked payment and tip correction cannot be proven before payments and tips exist.

****Prerequisites:**** FR-AUTH-006, FR-DAT-008A
****Journeys:**** GJ-07
****Acceptance tests:**** TST-M3-FR-ORD-012A, TST-M4-FR-ORD-012A

FR-ORD-013 - Notes

****Priority:**** P0
****Owner:**** Service Execution
****Lineage:**** retained
****Introduced:**** M3
****Revalidated:**** None
****Required behavior:**** Separate customer notes, allergy declarations, kitchen instructions and private staff notes.

****Gate-local behavior:**** Four note classes are stored separately with correct audience filtering; private staff notes never reach the customer surface.

****Later behavior:**** None beyond M3 for this clause.

****Prerequisites:**** FR-SAF-003, FR-AUTH-008
****Journeys:**** GJ-01A, GJ-05
****Acceptance tests:**** TST-M3-FR-ORD-013

FR-ORD-016A - Dine-in order timeline

****Priority:**** P0
****Owner:**** Service Execution
****Lineage:**** split_successor from 'FR-ORD-016'
****Introduced:**** M3
****Revalidated:**** M4
****Required behavior:**** Timeline renders order, station and service events chronologically with audience filtering.

****Gate-local behavior:**** Timeline renders order, station and service events chronologically with audience filtering.

****Later behavior:**** Check and payment events do not exist until M4.

****Prerequisites:**** FR-DAT-010, FR-ORD-013
****Journeys:**** GJ-01A, GJ-01B
****Acceptance tests:**** TST-M3-FR-ORD-016A, TST-M4-FR-ORD-016A

FR-ORD-017 - Duplicate detection

****Priority:**** P0
****Owner:**** Service Execution
****Lineage:**** retained
****Introduced:**** M3
****Revalidated:**** M5a
****Required behavior:**** Detect suspicious duplicate orders while allowing intentional repeat/add-on orders.

****Gate-local behavior:**** Suspicious duplicates are flagged while deliberate repeat and add-on orders pass unimpeded.

****Later behavior:**** Detection must not produce false positives when orders replay through the outlet outbox.

****Prerequisites:**** FR-ORD-004, FR-ORD-009
****Journeys:**** GJ-10
****Acceptance tests:**** TST-M3-FR-ORD-017, TST-M5a-FR-ORD-017

FR-ORD-019A - Operational source attribution

****Priority:**** P0
****Owner:**** Service Execution
****Lineage:**** split_successor from 'FR-ORD-019'
****Introduced:**** M3
****Revalidated:**** M4, M5a
****Required behavior:**** A stable correlation chain links all six named artifacts and survives projection rebuild.

****Gate-local behavior:**** A stable correlation chain links all six named artifacts and survives projection rebuild.

****Later behavior:**** Correlation extends to check and payment records once they exist, and across the

sync boundary.

****Prerequisites:**** FR-DAT-003, FR-FUL-002
****Journeys:**** GJ-01A
****Acceptance tests:**** TST-M3-FR-ORD-019A, TST-M4-FR-ORD-019A, TST-M5a-FR-ORD-019A

Data Architecture

FR-DAT-008A - Accepted-order ledger

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** split_successor from 'FR-DAT-008'
****Introduced:**** M3
****Revalidated:**** None
****Required behavior:**** Accepted orders are append-only or reversal-based; no destructive edit path exists.

****Gate-local behavior:**** Accepted orders are append-only or reversal-based; no destructive edit path exists.

****Later behavior:**** None beyond M3 for this clause.

****Prerequisites:**** FR-DAT-002, FR-SEC-009
****Journeys:**** GJ-01A
****Acceptance tests:**** TST-M3-FR-DAT-008A

FR-DAT-010 - Projection rebuild

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** retained
****Introduced:**** M3
****Revalidated:**** M4, M5a
****Required behavior:**** Rebuild key projections from authoritative events/ledgers and compare deterministically.

****Gate-local behavior:**** Order projections rebuild from authoritative events and compare byte-deterministically.

****Later behavior:**** Financial projections at M4; projections rebuilt after outage replay at M5a.

****Prerequisites:**** FR-DAT-008A
****Journeys:**** GJ-10
****Acceptance tests:**** TST-M3-FR-DAT-010, TST-M4-FR-DAT-010, TST-M5a-FR-DAT-010

Integration Runtime

FR-INT-005 - Idempotency

****Priority:**** P0
****Owner:**** Edge & Operations
****Lineage:**** retained
****Introduced:**** M3

****Revalidated:**** M4, M5a
****Required behavior:**** Every externally repeated Phase 1 command and event has an idempotency key and returns the original outcome without duplicate commercial effect.

****Gate-local behavior:**** Repeated M3 commands return the original outcome with no duplicate order or ticket.

****Later behavior:**** Payment and tip commands carry commercial effect from M4; replayed sync events from M5a.

****Prerequisites:**** FR-ORD-004
****Journeys:**** GJ-01A, GJ-10
****Acceptance tests:**** TST-M3-FR-INT-005, TST-M4-FR-INT-005, TST-M5a-FR-INT-005

FR-INT-007 - Dead letter

****Priority:**** P0
****Owner:**** Edge & Operations
****Lineage:**** retained
****Introduced:**** M3
****Revalidated:**** M4, M5a
****Required behavior:**** Move repeatedly failing Phase 1 integration or synchronization work to an operator-visible dead-letter queue with reason and safe replay controls.

****Gate-local behavior:**** Repeatedly failing notification jobs move to an operator-visible dead-letter queue with reason and safe replay.

****Later behavior:**** Payment-adapter failures at M4 and synchronization work at M5a use the same queue.

****Prerequisites:**** FR-NOT-001
****Journeys:**** GJ-10
****Acceptance tests:**** TST-M3-FR-INT-007, TST-M4-FR-INT-007, TST-M5a-FR-INT-007

FR-INT-014 - Correlation

****Priority:**** P0
****Owner:**** Edge & Operations
****Lineage:**** retained
****Introduced:**** M3
****Revalidated:**** M4, M5a
****Required behavior:**** Propagate correlation IDs across customer command, outlet persistence, kitchen ticket, bill, payment, tip, sync and audit records.

****Gate-local behavior:**** Correlation IDs propagate from customer command through persistence, kitchen ticket and audit records.

****Later behavior:**** Bill, payment and tip records exist at M4; outlet persistence and sync records at M5a.

****Prerequisites:**** FR-ORD-019A
****Journeys:**** GJ-01A, GJ-10
****Acceptance tests:**** TST-M3-FR-INT-014, TST-M4-FR-INT-014, TST-M5a-FR-INT-014

Internationalization and Localization

FR-I18N-001B - Customer service languages

****Priority:**** P0
****Owner:**** Customer Experience
****Lineage:**** split_successor from 'FR-I18N-001'
****Introduced:**** M3
****Revalidated:**** None
****Required behavior:**** Order status and service text render in the session language across the ordering and service journeys.

****Gate-local behavior:**** Order status and service text render in the session language across the ordering and service journeys.

****Later behavior:**** Bills and receipts remain outside this clause.

****Prerequisites:**** FR-I18N-001A, FR-ORD-016A
****Journeys:**** GJ-01A, GJ-02, GJ-03A
****Acceptance tests:**** TST-M3-FR-I18N-001B

FR-I18N-008 - Customer communications

****Priority:**** P0
****Owner:**** Customer Experience
****Lineage:**** retained
****Introduced:**** M3
****Revalidated:**** M4
****Required behavior:**** Customer status, service and receipt communications use the language snapshot on the table session or order, with approved fallback rules.

****Gate-local behavior:**** Status and service communications render in the snapshot language with approved fallback.

****Later behavior:**** Receipt communications exist at M4.

****Prerequisites:**** FR-I18N-005, FR-NOT-001
****Journeys:**** GJ-02, GJ-02B
****Acceptance tests:**** TST-M3-FR-I18N-008, TST-M4-FR-I18N-008

Kitchen/Bar/Expo Fulfillment

FR-FUL-001 - Station routing

****Priority:**** P0
****Owner:**** Service Execution
****Lineage:**** retained
****Introduced:**** M3
****Revalidated:**** M5a
****Required behavior:**** Route accepted line units to kitchen, bar, coffee, bakery, dessert, expo or configured stations using versioned rules.

****Gate-local behavior:**** Accepted line units route to the correct station by versioned rule, including multi-station orders.

****Later behavior:**** Routing must continue to function on the outlet node during an outage.

****Prerequisites:**** FR-TEN-002A, FR-ORD-001A

****Journeys:**** GJ-01A, GJ-05

****Acceptance tests:**** TST-M3-FR-FUL-001, TST-M5a-FR-FUL-001

FR-FUL-002 - Ticket identity

****Priority:**** P0

****Owner:**** Service Execution

****Lineage:**** retained

****Introduced:**** M3

****Revalidated:**** None

****Required behavior:**** Create fulfillment tickets separate from the commercial order and support multiple stations per order.

****Gate-local behavior:**** Tickets exist as records distinct from the order and one order fans out to multiple station tickets.

****Later behavior:**** None beyond M3 for this clause.

****Prerequisites:**** FR-FUL-001

****Journeys:**** GJ-01A

****Acceptance tests:**** TST-M3-FR-FUL-002

FR-FUL-003 - KDS queue

****Priority:**** P0

****Owner:**** Service Execution

****Lineage:**** retained

****Introduced:**** M3

****Revalidated:**** M5a

****Required behavior:**** Show new, acknowledged, preparing, held, ready, completed and exception tickets with elapsed/SLA time.

****Gate-local behavior:**** All seven ticket states render on KDS with correct elapsed and SLA time.

****Later behavior:**** The same KDS must operate from the local node during an outage.

****Prerequisites:**** FR-FUL-002

****Journeys:**** GJ-01A, GJ-05, GJ-10

****Acceptance tests:**** TST-M3-FR-FUL-003, TST-M5a-FR-FUL-003

FR-FUL-004 - Line-level progress

****Priority:**** P0

****Owner:**** Service Execution

****Lineage:**** retained

****Introduced:**** M3

****Revalidated:**** None

****Required behavior:**** Track quantities and statuses at line-unit level so partial preparation and partial readiness are visible.

****Gate-local behavior:**** Line-unit quantities and statuses support partial preparation and partial readiness.

****Later behavior:**** None beyond M3 for this clause.

****Prerequisites:**** FR-FUL-003

****Journeys:**** GJ-01A

****Acceptance tests:**** TST-M3-FR-FUL-004

FR-FUL-005 - Accept/recall

****Priority:**** P0

****Owner:**** Service Execution

****Lineage:**** retained

****Introduced:**** M3

****Revalidated:**** None

****Required behavior:**** Allow station acceptance and controlled recall of recently completed tickets with audit.

****Gate-local behavior:**** Stations accept tickets and recall recently completed ones within policy, with immutable audit.

****Later behavior:**** None beyond M3 for this clause.

****Prerequisites:**** FR-FUL-003, FR-DAT-008A

****Journeys:**** GJ-05

****Acceptance tests:**** TST-M3-FR-FUL-005

FR-FUL-006 - Course firing

****Priority:**** P0

****Owner:**** Service Execution

****Lineage:**** retained

****Introduced:**** M3

****Revalidated:**** None

****Required behavior:**** Support held courses and authorized fire commands from waiter/expo or configured timing.

****Gate-local behavior:**** Courses hold and fire on authorized command or configured timing.

****Later behavior:**** None beyond M3 for this clause.

****Prerequisites:**** FR-FUL-003

****Journeys:**** GJ-05

****Acceptance tests:**** TST-M3-FR-FUL-006

FR-FUL-007 - Priority

****Priority:**** P0

****Owner:**** Service Execution

****Lineage:**** retained

****Introduced:**** M3

****Revalidated:**** None

****Required behavior:**** Apply ordinary, rush and accessibility/service priority using authorized reas

on and visible attribution.

****Gate-local behavior:**** Three priority levels apply with authorized reason and visible attribution on KDS.

****Later behavior:**** None beyond M3 for this clause.

****Prerequisites:**** FR-FUL-003, FR-CFG-003

****Journeys:**** GJ-04

****Acceptance tests:**** TST-M3-FR-FUL-007

FR-FUL-008 - Allergy emphasis

****Priority:**** P0

****Owner:**** Service Execution

****Lineage:**** retained

****Introduced:**** M3

****Revalidated:**** M5a

****Required behavior:**** Render allergy concerns in high-salience non-color-only form and require acknowledgement where configured.

****Gate-local behavior:**** Allergy concerns render with non-color-only salience and require station acknowledgement where configured.

****Later behavior:**** The same salience must survive the printed station ticket.

****Prerequisites:**** FR-SAF-004

****Journeys:**** GJ-05

****Acceptance tests:**** TST-M3-FR-FUL-008, TST-M5a-FR-FUL-008

FR-FUL-009 - Expo coordination

****Priority:**** P0

****Owner:**** Service Execution

****Lineage:**** retained

****Introduced:**** M3

****Revalidated:**** None

****Required behavior:**** Provide an expo view combining station readiness and blocking incomplete sets before service.

****Gate-local behavior:**** Expo view aggregates station readiness and blocks service of incomplete sets.

****Later behavior:**** None beyond M3 for this clause.

****Prerequisites:**** FR-FUL-003, FR-FUL-004

****Journeys:**** GJ-01A

****Acceptance tests:**** TST-M3-FR-FUL-009

FR-FUL-010 - Ready notification

****Priority:**** P0

****Owner:**** Service Execution

****Lineage:**** retained

****Introduced:**** M3
****Revalidated:**** M5a
****Required behavior:**** Notify assigned waiter/runner when all or selected items are ready and escalate uncollected items.

****Gate-local behavior:**** Ready notifications reach the assigned waiter and uncollected items escalate per policy.

****Later behavior:**** Notification delivery must continue from the local node during an outage.

****Prerequisites:**** FR-NOT-001, FR-FUL-009
****Journeys:**** GJ-01A, GJ-10
****Acceptance tests:**** TST-M3-FR-FUL-010, TST-M5a-FR-FUL-010

FR-FUL-011 - Serve confirmation

****Priority:**** P0
****Owner:**** Service Execution
****Lineage:**** retained
****Introduced:**** M3
****Revalidated:**** None
****Required behavior:**** Record who collected and served items and allow missing/wrong item exception.

****Gate-local behavior:**** Collection and service are recorded with actor identity and missing/wrong-item exceptions are capturable.

****Later behavior:**** None beyond M3 for this clause.

****Prerequisites:**** FR-FUL-009, FR-AUTH-008
****Journeys:**** GJ-01A, GJ-05
****Acceptance tests:**** TST-M3-FR-FUL-011

FR-FUL-012 - Preparation timer

****Priority:**** P0
****Owner:**** Service Execution
****Lineage:**** retained
****Introduced:**** M3
****Revalidated:**** M6
****Required behavior:**** Calculate prep and wait times per station/item/order for live operations and later analytics.

****Gate-local behavior:**** Prep and wait times compute correctly per station, item and order for live operational use.

****Later behavior:**** Analytical consumption of these measures belongs to operational reporting.

****Prerequisites:**** FR-FUL-003, FR-DAT-004
****Journeys:**** None
****Acceptance tests:**** TST-M3-FR-FUL-012, TST-M6-FR-FUL-012

FR-FUL-013 - Capacity controls

****Priority:**** P0

****Owner:**** Service Execution
****Lineage:**** retained
****Introduced:**** M3
****Revalidated:**** None
****Required behavior:**** Allow configurable throttling or promise-time adjustment when station workload exceeds thresholds.

****Gate-local behavior:**** Throttling or promise-time adjustment triggers when configured workload thresholds are exceeded.

****Later behavior:**** None beyond M3 for this clause.

****Prerequisites:**** FR-FUL-012, FR-CFG-002A

****Journeys:**** GJ-01A

****Acceptance tests:**** TST-M3-FR-FUL-013

FR-FUL-014 - Printer fallback

****Priority:**** P0
****Owner:**** Service Execution
****Lineage:**** retained
****Introduced:**** M3
****Revalidated:**** M5a
****Required behavior:**** Generate deduplicated station tickets when KDS is unavailable or policy requires paper.

****Gate-local behavior:**** Station ticket documents render with deduplication logic and correct content when KDS is unavailable or policy requires paper.

****Later behavior:**** Physical printing, print-agent delivery and print-job deduplication across rest of the local print path.

****Prerequisites:**** FR-FUL-002

****Journeys:**** GJ-10

****Acceptance tests:**** TST-M3-FR-FUL-014, TST-M5a-FR-FUL-014

FR-FUL-015 - Station transfer

****Priority:**** P0
****Owner:**** Service Execution
****Lineage:**** retained
****Introduced:**** M3
****Revalidated:**** M5a
****Required behavior:**** Allow authorized rerouting if a station/device is unavailable without duplicating work.

****Gate-local behavior:**** Authorized reroute moves a ticket to another station without duplicating line units.

****Later behavior:**** Rerouting must behave correctly when the outlet node is authoritative during an outage.

****Prerequisites:**** FR-FUL-001

****Journeys:**** GJ-10

****Acceptance tests:**** TST-M3-FR-FUL-015, TST-M5a-FR-FUL-015

FR-FUL-016A - Waste and rework operations

****Priority:**** P0

****Owner:**** Service Execution

****Lineage:**** split_successor from 'FR-FUL-016'

****Introduced:**** M3

****Revalidated:**** None

****Required behavior:**** Rework, remake and service-waste events record reason, actor and the linked order/ticket.

****Gate-local behavior:**** Rework, remake and service-waste events record reason, actor and the linked order/ticket.

****Later behavior:**** None beyond M3 for this clause.

****Prerequisites:**** FR-CFG-003, FR-FUL-002

****Journeys:**** GJ-05

****Acceptance tests:**** TST-M3-FR-FUL-016A

Notifications

FR-NOT-001 - Event-driven notifications

****Priority:**** P0

****Owner:**** Platform Engineering

****Lineage:**** retained

****Introduced:**** M3

****Revalidated:**** M4, M5a, M5b

****Required behavior:**** Generate event-driven local/in-app notifications for order, kitchen, service request, bill, payment, tip, outage and synchronization events.

****Gate-local behavior:**** Order, kitchen and service-request events generate in-app notifications to the correct recipients.

****Later behavior:**** Bill, payment and tip events exist at M4; outage and synchronization events at M5a and M5b.

****Prerequisites:**** FR-ORD-016A, FR-FUL-003

****Journeys:**** GJ-01A, GJ-04

****Acceptance tests:**** TST-M3-FR-NOT-001, TST-M4-FR-NOT-001, TST-M5a-FR-NOT-001, TST-M5b-FR-NOT-001

FR-NOT-003 - Templates

****Priority:**** P0

****Owner:**** Platform Engineering

****Lineage:**** retained

****Introduced:**** M3

****Revalidated:**** M4

****Required behavior:**** Store human-approved English, Amharic and Arabic customer notification templates plus English staff templates for Phase 1.

****Gate-local behavior:**** Approved templates exist in three customer languages plus English staff templates

plates and render for M3 events.

****Later behavior:**** Bill, payment and tip templates are added when those events exist.

****Prerequisites:**** FR-I18N-003, FR-NOT-001

****Journeys:**** GJ-02

****Acceptance tests:**** TST-M3-FR-NOT-003, TST-M4-FR-NOT-003

FR-NOT-005 - Critical alerts

****Priority:**** P0

****Owner:**** Platform Engineering

****Lineage:**** retained

****Introduced:**** M3

****Revalidated:**** M4, M5a

****Required behavior:**** Route critical operational alerts for payment mismatch, printer failure, local-node failure, sync conflict and security events to accountable staff.

****Gate-local behavior:**** Security and operational alerts existing at M3 route to accountable staff with correct escalation.

****Later behavior:**** Payment mismatch requires payments; printer failure, local-node failure and sync conflict require the outlet node.

****Prerequisites:**** FR-NOT-001, FR-AUTH-008

****Journeys:**** GJ-10

****Acceptance tests:**** TST-M3-FR-NOT-005, TST-M4-FR-NOT-005, TST-M5a-FR-NOT-005

FR-NOT-007 - Deduplication

****Priority:**** P0

****Owner:**** Platform Engineering

****Lineage:**** retained

****Introduced:**** M3

****Revalidated:**** M5a

****Required behavior:**** Deduplicate repeated notifications by event and recipient while allowing deliberate repeated service requests.

****Gate-local behavior:**** Repeated notifications for one event and recipient collapse while deliberate repeat requests still notify.

****Later behavior:**** Deduplication must hold when notifications replay after reconnection.

****Prerequisites:**** FR-NOT-001

****Journeys:**** GJ-10

****Acceptance tests:**** TST-M3-FR-NOT-007, TST-M5a-FR-NOT-007

FR-NOT-009 - Deep links

****Priority:**** P0

****Owner:**** Platform Engineering

****Lineage:**** retained

****Introduced:**** M3

****Revalidated:**** M5a

****Required behavior:**** Notification links open the correct local or cloud screen for the authorized outlet/session.

****Gate-local behavior:**** Notification links open the correct cloud screen and enforce outlet/session authorization.

****Later behavior:**** The local screen target does not exist until the outlet node serves screens.

****Prerequisites:**** FR-NOT-001, FR-AUTH-008

****Journeys:**** GJ-10

****Acceptance tests:**** TST-M3-FR-NOT-009, TST-M5a-FR-NOT-009

FR-NOT-010 - No sensitive leakage

****Priority:**** P0

****Owner:**** Platform Engineering

****Lineage:**** retained

****Introduced:**** M3

****Revalidated:**** M4, M5a

****Required behavior:**** Do not include sensitive customer, payment or authentication data in notification payloads or logs.

****Gate-local behavior:**** Notification payloads and logs contain no sensitive customer or authentication data for M3 event classes.

****Later behavior:**** Payment data does not exist in any payload until payments exist.

****Prerequisites:**** FR-SEC-011, FR-NOT-001

****Journeys:**** None

****Acceptance tests:**** TST-M3-FR-NOT-010, TST-M4-FR-NOT-010, TST-M5a-FR-NOT-010

FR-NOT-011 - Escalation

****Priority:**** P0

****Owner:**** Platform Engineering

****Lineage:**** retained

****Introduced:**** M3

****Revalidated:**** M5a

****Required behavior:**** Escalate unacknowledged service and operational alerts according to outlet policy.

****Gate-local behavior:**** Unacknowledged service and operational alerts escalate per outlet policy with recorded escalation events.

****Later behavior:**** Operational alert classes originating at the outlet node escalate through the same policy.

****Prerequisites:**** FR-NOT-005, FR-SRV-004

****Journeys:**** GJ-04

****Acceptance tests:**** TST-M3-FR-NOT-011, TST-M5a-FR-NOT-011

FR-NOT-012 - Notification center

****Priority:**** P0

****Owner:**** Platform Engineering
****Lineage:**** retained
****Introduced:**** M3
****Revalidated:**** M4
****Required behavior:**** Provide an English staff notification center and localized customer status timeline for Phase 1.

****Gate-local behavior:**** Staff notification center renders in English and the customer status timeline renders in the session language.

****Later behavior:**** The customer timeline gains check and payment entries at M4.

****Prerequisites:**** FR-NOT-001, FR-ORD-016A
****Journeys:**** GJ-02
****Acceptance tests:**** TST-M3-FR-NOT-012, TST-M4-FR-NOT-012

Service Requests

FR-SRV-001 - Request types

****Priority:**** P0
****Owner:**** Service Execution
****Lineage:**** retained
****Introduced:**** M3
****Revalidated:**** M4
****Required behavior:**** Configure translated customer requests such as call waiter, water, cutlery, assistance, missing item, packaging and bill.

****Gate-local behavior:**** The translated request catalog exists and every named request type can be raised from a table session.

****Later behavior:**** The bill request only produces a check once billing exists.

****Prerequisites:**** FR-I18N-003, FR-TAB-003
****Journeys:**** GJ-04, GJ-01B
****Acceptance tests:**** TST-M3-FR-SRV-001, TST-M4-FR-SRV-001

FR-SRV-002 - Routing

****Priority:**** P0
****Owner:**** Service Execution
****Lineage:**** retained
****Introduced:**** M3
****Revalidated:**** None
****Required behavior:**** Route requests by table assignment, service area, role and availability.

****Gate-local behavior:**** Requests route to the correct staff by table assignment, service area, role and current presence.

****Later behavior:**** None beyond M3 for this clause.

****Prerequisites:**** FR-TAB-006, FR-SRV-007A
****Journeys:**** GJ-04
****Acceptance tests:**** TST-M3-FR-SRV-002

FR-SRV-003 - Acknowledgement

****Priority:**** P0
****Owner:**** Service Execution
****Lineage:**** retained
****Introduced:**** M3
****Revalidated:**** None
****Required behavior:**** Allow an employee to accept a request and show customer-visible received/being-handled status.

****Gate-local behavior:**** Acceptance transitions the request and the customer sees received/being-handled status in the session language.

****Later behavior:**** None beyond M3 for this clause.

****Prerequisites:**** FR-SRV-002
****Journeys:**** GJ-04
****Acceptance tests:**** TST-M3-FR-SRV-003

FR-SRV-004 - SLA timer

****Priority:**** P0
****Owner:**** Service Execution
****Lineage:**** retained
****Introduced:**** M3
****Revalidated:**** None
****Required behavior:**** Track response deadline, reminders and escalation to supervisor/alternate staff.

****Gate-local behavior:**** Deadlines, reminders and escalation to supervisor or alternate staff fire correctly.

****Later behavior:**** None beyond M3 for this clause.

****Prerequisites:**** FR-SRV-003
****Journeys:**** GJ-04
****Acceptance tests:**** TST-M3-FR-SRV-004

FR-SRV-005 - Completion

****Priority:**** P0
****Owner:**** Service Execution
****Lineage:**** retained
****Introduced:**** M3
****Revalidated:**** None
****Required behavior:**** Require completion status and optional result/reason for exceptions.

****Gate-local behavior:**** A request cannot close without completion status; exceptions capture result and reason.

****Later behavior:**** None beyond M3 for this clause.

****Prerequisites:**** FR-SRV-003, FR-CFG-003

****Journeys:**** GJ-04
****Acceptance tests:**** TST-M3-FR-SRV-005

FR-SRV-006 - Deduplication

****Priority:**** P0
****Owner:**** Service Execution
****Lineage:**** retained
****Introduced:**** M3
****Revalidated:**** None
****Required behavior:**** Avoid accidental repeated taps creating uncontrolled duplicate alerts while preserving deliberate repeats.

****Gate-local behavior:**** Rapid repeated taps collapse into one alert while a deliberate later repeat still raises a new request.

****Later behavior:**** None beyond M3 for this clause.

****Prerequisites:**** FR-SRV-001
****Journeys:**** GJ-04
****Acceptance tests:**** TST-M3-FR-SRV-006

FR-SRV-007A - Ephemeral waiter availability

****Priority:**** P0
****Owner:**** Service Execution
****Lineage:**** split_successor from 'FR-SRV-007'
****Introduced:**** M3
****Revalidated:**** None
****Required behavior:**** Three presence states drive request routing and are discarded per operational-session retention.

****Gate-local behavior:**** Three presence states drive request routing and are discarded per operational-session retention.

****Later behavior:**** None beyond M3 for this clause.

****Prerequisites:**** FR-AUTH-008
****Journeys:**** GJ-04
****Acceptance tests:**** TST-M3-FR-SRV-007A

FR-SRV-008 - Internal tasks

****Priority:**** P0
****Owner:**** Service Execution
****Lineage:**** retained
****Introduced:**** M3
****Revalidated:**** None
****Required behavior:**** Allow staff-generated service tasks linked to table/order/customer issue.

****Gate-local behavior:**** Staff create tasks linked to a table, order or customer issue with correct correlation.

****Later behavior:**** None beyond M3 for this clause.

****Prerequisites:**** FR-SRV-001, FR-ORD-019A
****Journeys:**** GJ-04
****Acceptance tests:**** TST-M3-FR-SRV-008

FR-SRV-009 - Customer messaging

****Priority:**** P0
****Owner:**** Service Execution
****Lineage:**** retained
****Introduced:**** M3
****Revalidated:**** None
****Required behavior:**** Provide limited translated status messages without exposing staff identity unless configured.

****Gate-local behavior:**** Customer status messages render translated and omit staff identity unless configuration permits it.

****Later behavior:**** None beyond M3 for this clause.

****Prerequisites:**** FR-I18N-008, FR-SRV-003
****Journeys:**** GJ-02
****Acceptance tests:**** TST-M3-FR-SRV-009

Staff POS and Outlet UX

FR-POS-001 - Terminal registration

****Priority:**** P0
****Owner:**** Customer Experience
****Lineage:**** retained
****Introduced:**** M3
****Revalidated:**** M4
****Required behavior:**** Register POS/waiter/KDS devices to a tenant/outlet/device profile and revoke compromised terminals.

****Gate-local behavior:**** Waiter and KDS devices register to a tenant/outlet/device profile and revocation immediately blocks them.

****Later behavior:**** POS terminals are a further device class introduced with the POS surface.

****Prerequisites:**** FR-SEC-014, FR-TEN-002A
****Journeys:**** GJ-05
****Acceptance tests:**** TST-M3-FR-POS-001, TST-M4-FR-POS-001

FR-POS-002 - Role home

****Priority:**** P0
****Owner:**** Customer Experience
****Lineage:**** retained
****Introduced:**** M3
****Revalidated:**** M4
****Required behavior:**** Present role-specific queues and next actions rather than generic CRUD menus.

****Gate-local behavior:**** Waiter and kitchen roles land on task queues with next actions, not generic menus.

****Later behavior:**** Cashier and manager role homes exist with the POS surface.

****Prerequisites:**** FR-AUTH-008, FR-FUL-003

****Journeys:**** GJ-05

****Acceptance tests:**** TST-M3-FR-POS-002, TST-M4-FR-POS-002

FR-POS-003A - Waiter dine-in ordering

****Priority:**** P0

****Owner:**** Customer Experience

****Lineage:**** split_successor from 'FR-POS-003'

****Introduced:**** M3

****Revalidated:**** M4

****Required behavior:**** Waiter-entered dine-in orders apply identical menu, modifier, price, safety and authorization rules as QR ordering.

****Gate-local behavior:**** Waiter-entered dine-in orders apply identical menu, modifier, price, safety and authorization rules as QR ordering.

****Later behavior:**** The tax rule comparison is only fully provable once tax reaches a bill; counter orders need POS.

****Prerequisites:**** FR-ORD-001A, FR-MNU-006

****Journeys:**** GJ-05

****Acceptance tests:**** TST-M3-FR-POS-003A, TST-M4-FR-POS-003A

FR-POS-004 - Table view

****Priority:**** P0

****Owner:**** Customer Experience

****Lineage:**** retained

****Introduced:**** M3

****Revalidated:**** M4

****Required behavior:**** Show floor/list occupancy, assigned waiter, open requests, order progress, unpaid balance and attention flags.

****Gate-local behavior:**** Occupancy, assigned waiter, open requests, order progress and attention flags render correctly.

****Later behavior:**** Unpaid balance requires checks to exist.

****Prerequisites:**** FR-TAB-006, FR-SRV-003

****Journeys:**** GJ-05

****Acceptance tests:**** TST-M3-FR-POS-004, TST-M4-FR-POS-004

FR-POS-005 - Fast item entry

****Priority:**** P0

****Owner:**** Customer Experience

****Lineage:**** retained

****Introduced:**** M3

****Revalidated:** M4**
****Required behavior:**** Support touch search, categories, favorites, barcode where appropriate and keyboard shortcuts.

****Gate-local behavior:**** Fast item entry works on the waiter surface with search, categories, favorites and shortcuts.

****Later behavior:**** The same entry affordances are required on the POS terminal.

****Prerequisites:**** FR-MNU-012, FR-POS-003A
****Journeys:**** GJ-05
****Acceptance tests:**** TST-M3-FR-POS-005, TST-M4-FR-POS-005

FR-POS-006 - Manager override

****Priority:** P0**
****Owner:**** Customer Experience
****Lineage:**** retained
****Introduced:**** M3
****Revalidated:**** M4
****Required behavior:**** Request and record supervisor approval without sharing credentials or bypassing audit.

****Gate-local behavior:**** Supervisor approval for M3 overrides (void, priority, reroute) is recorded without credential sharing.

****Later behavior:**** Financial overrides such as discount, refund and tip correction exist at M4.

****Prerequisites:**** FR-AUTH-006, FR-ORD-012A
****Journeys:**** GJ-07
****Acceptance tests:**** TST-M3-FR-POS-006, TST-M4-FR-POS-006

FR-POS-007 - Shift handover

****Priority:** P0**
****Owner:**** Customer Experience
****Lineage:**** retained
****Introduced:**** M3
****Revalidated:**** M4
****Required behavior:**** Transfer responsibility for open tables/tasks/cash only through explicit handover workflow.

****Gate-local behavior:**** Open tables and tasks transfer only through an explicit acknowledged handover.

****Later behavior:**** Cash responsibility handover requires cash shifts.

****Prerequisites:**** FR-TAB-006, FR-SRV-008
****Journeys:**** GJ-05
****Acceptance tests:**** TST-M3-FR-POS-007, TST-M4-FR-POS-007

FR-POS-009 - Accessibility mode

****Priority:** P0**

****Owner:**** Customer Experience
****Lineage:**** retained
****Introduced:**** M3
****Revalidated:**** M4
****Required behavior:**** Support larger targets/text and reduced-motion/high-contrast preferences where configured.

****Gate-local behavior:**** Accessibility preferences apply on the waiter and KDS surfaces.

****Later behavior:**** The POS terminal surface must honour the same preferences.

****Prerequisites:**** FR-POS-002

****Journeys:**** None

****Acceptance tests:**** TST-M3-FR-POS-009, TST-M4-FR-POS-009

FR-POS-010A - Operational search

****Priority:**** P0

****Owner:**** Customer Experience

****Lineage:**** split_successor from 'FR-POS-010'

****Introduced:**** M3

****Revalidated:**** M4

****Required behavior:**** Order and table search works with tenant/outlet and role filtering and no cross-tenant leakage.

****Gate-local behavior:**** Order and table search works with tenant/outlet and role filtering and no cross-tenant leakage.

****Later behavior:**** Check, receipt and payment/reference identifiers do not exist until M4.

****Prerequisites:**** FR-AUTH-008, FR-ORD-019A

****Journeys:**** GJ-07

****Acceptance tests:**** TST-M3-FR-POS-010A, TST-M4-FR-POS-010A

Tables and Sessions

FR-TAB-007A - Merge tables for service

****Priority:**** P0

****Owner:**** Customer Experience

****Lineage:**** split_successor from 'FR-TAB-007'

****Introduced:**** M3

****Revalidated:**** None

****Required behavior:**** Two physical tables merge into one service session with orders and requests correctly consolidated and audited.

****Gate-local behavior:**** Two physical tables merge into one service session with orders and requests correctly consolidated and audited.

****Later behavior:**** The check-splitting half of this row belongs to billing.

****Prerequisites:**** FR-TAB-003, FR-TAB-006

****Journeys:**** GJ-04

****Acceptance tests:**** TST-M3-FR-TAB-007A

FR-TAB-008 - Move session

****Priority:**** P0
****Owner:**** Customer Experience
****Lineage:**** retained
****Introduced:**** M3
****Revalidated:**** M4
****Required behavior:**** Transfer an open session between tables without losing orders, requests or checks.

****Gate-local behavior:**** An open session moves between tables preserving session identity, orders and service requests.

****Later behavior:**** Preservation of checks can only be proven once checks exist.

****Prerequisites:**** FR-TAB-003, FR-ORD-001A
****Journeys:**** GJ-04
****Acceptance tests:**** TST-M3-FR-TAB-008, TST-M4-FR-TAB-008

FR-TAB-009 - Session closure

****Priority:**** P0
****Owner:**** Customer Experience
****Lineage:**** retained
****Introduced:**** M3
****Revalidated:**** M4
****Required behavior:**** Close only after service and financial conditions are met or an authorized exception is recorded.

****Gate-local behavior:**** A session cannot close with outstanding service obligations unless an authorized exception is recorded.

****Later behavior:**** The financial condition is unprovable before checks and settlement exist.

****Prerequisites:**** FR-TAB-003, FR-FUL-001
****Journeys:**** GJ-01B
****Acceptance tests:**** TST-M3-FR-TAB-009, TST-M4-FR-TAB-009

Testing and Evidence

FR-TST-005A - Phase 1 end-to-end journeys

****Priority:**** P0
****Owner:**** Quality Engineering
****Lineage:**** split_successor from 'FR-TST-005'
****Introduced:**** M3
****Revalidated:**** M4, M5a, M5b, M6
****Required behavior:**** Customer, waiter and KDS journeys run in browser/device automation against real persistence.

****Gate-local behavior:**** Customer, waiter and KDS journeys run in browser/device automation against real persistence.

****Later behavior:**** Cashier journeys at M4, outlet-continuity journeys at M5a and M5b, manager and administration journeys at M6.

****Prerequisites:**** FR-TST-013

****Journeys:**** GJ-01A, GJ-01B, GJ-08, GJ-10, GJ-12

****Acceptance tests:**** TST-M3-FR-TST-005A, TST-M4-FR-TST-005A, TST-M5a-FR-TST-005A, TST-M5b-FR-TST-005A, TST-M6-FR-TST-005A

FR-TST-007A - Phase 1 concurrency tests

****Priority:**** P0

****Owner:**** Quality Engineering

****Lineage:**** split_successor from 'FR-TST-007'

****Introduced:**** M3

****Revalidated:**** M4, M5a, M5b

****Required behavior:**** Duplicate order submit races are exercised and produce no duplicate commercial effect.

****Gate-local behavior:**** Duplicate order submit races are exercised and produce no duplicate commercial effect.

****Later behavior:**** Payment, proof, tip, split and cash races at M4; synchronization replay at M5a; authority-lease races at M5b.

****Prerequisites:**** FR-ORD-004, FR-ORD-017

****Journeys:**** GJ-06, GJ-09, GJ-10

****Acceptance tests:**** TST-M3-FR-TST-007A, TST-M4-FR-TST-007A, TST-M5a-FR-TST-007A, TST-M5b-FR-TST-007A

UX and Accessibility

FR-UX-002 - Staff touch first

****Priority:**** P0

****Owner:**** Customer Experience

****Lineage:**** retained

****Introduced:**** M3

****Revalidated:**** M4

****Required behavior:**** Design waiter/POS/KDS controls for quick touch use, large targets and readable distance.

****Gate-local behavior:**** Waiter and KDS controls use large touch targets readable at working distance.

****Later behavior:**** The POS terminal surface must meet the same standard.

****Prerequisites:**** FR-POS-002

****Journeys:**** GJ-05

****Acceptance tests:**** TST-M3-FR-UX-002, TST-M4-FR-UX-002

FR-UX-004 - Next action

****Priority:**** P0

****Owner:**** Customer Experience

****Lineage:**** retained
****Introduced:**** M3
****Revalidated:**** M4, M6
****Required behavior:**** Every operational screen prioritizes the next required action, active exception and elapsed time.

****Gate-local behavior:**** Waiter and KDS screens lead with next action, active exception and elapsed time.

****Later behavior:**** POS and cashier screens at M4; operator screens at M6.

****Prerequisites:**** FR-POS-002, FR-FUL-003
****Journeys:**** GJ-05
****Acceptance tests:**** TST-M3-FR-UX-004, TST-M4-FR-UX-004, TST-M6-FR-UX-004

FR-UX-008 - Confirmation friction

****Priority:**** P0
****Owner:**** Customer Experience
****Lineage:**** retained
****Introduced:**** M3
****Revalidated:**** M4
****Required behavior:**** Use low friction for ordinary service and deliberate friction for allergy, cancellation, refund, override and payment.

****Gate-local behavior:**** Allergy declaration and cancellation carry deliberate confirmation friction while ordinary service does not.

****Later behavior:**** Refund, override and payment confirmations exist at M4.

****Prerequisites:**** FR-SAF-003, FR-ORD-011
****Journeys:**** GJ-07
****Acceptance tests:**** TST-M3-FR-UX-008, TST-M4-FR-UX-008

FR-UX-015 - Destructive actions

****Priority:**** P0
****Owner:**** Customer Experience
****Lineage:**** retained
****Introduced:**** M3
****Revalidated:**** M4
****Required behavior:**** Require explicit confirmation and reason where action cannot be trivially undone.

****Gate-local behavior:**** Irreversible M3 actions require explicit confirmation and a reason code.

****Later behavior:**** Financial irreversible actions exist at M4.

****Prerequisites:**** FR-ORD-012A, FR-CFG-003
****Journeys:**** GJ-07
****Acceptance tests:**** TST-M3-FR-UX-015, TST-M4-FR-UX-015

Gate M4

Carts and Orders

FR-ORD-001B - Counter order aggregate

****Priority:**** P0
****Owner:**** Service Execution
****Lineage:**** split_successor from 'FR-ORD-001'
****Introduced:**** M4
****Revalidated:**** None
****Required behavior:**** The counter POS channel uses the same aggregate and policy model as the dine-in channels with no divergent order path.

****Gate-local behavior:**** The counter POS channel uses the same aggregate and policy model as the dine-in channels with no divergent order path.

****Later behavior:**** None beyond M4 for this clause.

****Prerequisites:**** FR-ORD-001A, FR-MNU-006
****Journeys:**** GJ-06
****Acceptance tests:**** TST-M4-FR-ORD-001B

FR-ORD-007B - Financial acceptance revalidation

****Priority:**** P0
****Owner:**** Service Execution
****Lineage:**** split_successor from 'FR-ORD-007'
****Introduced:**** M4
****Revalidated:**** None
****Required behavior:**** An order configured for payment-dependent acceptance is accepted only after a verified payment outcome.

****Gate-local behavior:**** An order configured for payment-dependent acceptance is accepted only after a verified payment outcome.

****Later behavior:**** None beyond M4 for this clause.

****Prerequisites:**** FR-ORD-007A, FR-PAY-015
****Journeys:**** GJ-01B
****Acceptance tests:**** TST-M4-FR-ORD-007B

Cash Management

FR-CSH-001 - Cash shift open

****Priority:**** P0
****Owner:**** Commerce & Payments
****Lineage:**** retained
****Introduced:**** M4
****Revalidated:**** None
****Required behavior:**** Open a cashier/drawer session with counted float, assigned terminal and approval policy.

****Gate-local behavior:**** A cash shift opens with counted float, assigned terminal and configured approval policy.

****Later behavior:**** None beyond M4 for this clause.

****Prerequisites:**** FR-POS-001, FR-CFG-002A

****Journeys:**** GJ-07

****Acceptance tests:**** TST-M4-FR-CSH-001

FR-CSH-002 - Cash movements

****Priority:**** P0

****Owner:**** Commerce & Payments

****Lineage:**** retained

****Introduced:**** M4

****Revalidated:**** None

****Required behavior:**** Record sales receipts, refunds, payouts, drops, float adjustments and transfers as distinct movements.

****Gate-local behavior:**** All six movement types record as distinct append-only entries.

****Later behavior:**** None beyond M4 for this clause.

****Prerequisites:**** FR-CSH-001, FR-DAT-008B

****Journeys:**** GJ-07

****Acceptance tests:**** TST-M4-FR-CSH-002

FR-CSH-003 - Cash count

****Priority:**** P0

****Owner:**** Commerce & Payments

****Lineage:**** retained

****Introduced:**** M4

****Revalidated:**** None

****Required behavior:**** Capture denomination count, expected total, actual total and variance.

****Gate-local behavior:**** Denomination counting produces expected, actual and variance figures with exact arithmetic.

****Later behavior:**** None beyond M4 for this clause.

****Prerequisites:**** FR-CSH-002, FR-DAT-005

****Journeys:**** GJ-07

****Acceptance tests:**** TST-M4-FR-CSH-003

FR-CSH-004 - Close approval

****Priority:**** P0

****Owner:**** Commerce & Payments

****Lineage:**** retained

****Introduced:**** M4

****Revalidated:**** None

****Required behavior:**** Require cashier submission and optional manager verification; lock finalized shifts.

****Gate-local behavior:**** Close requires cashier submission, optional manager verification, and a final

alized shift is locked against further movement.

****Later behavior:**** None beyond M4 for this clause.

****Prerequisites:**** FR-CSH-003, FR-AUTH-006

****Journeys:**** GJ-07

****Acceptance tests:**** TST-M4-FR-CSH-004

FR-CSH-007 - Safe transfer

****Priority:**** P0

****Owner:**** Commerce & Payments

****Lineage:**** retained

****Introduced:**** M4

****Revalidated:**** None

****Required behavior:**** Record sealed bag/reference and custody when cash moves to safe or bank.

****Gate-local behavior:**** Safe and bank transfers record sealed bag reference and custody chain.

****Later behavior:**** None beyond M4 for this clause.

****Prerequisites:**** FR-CSH-002

****Journeys:**** GJ-07

****Acceptance tests:**** TST-M4-FR-CSH-007

FR-CSH-008 - Exception report

****Priority:**** P0

****Owner:**** Commerce & Payments

****Lineage:**** retained

****Introduced:**** M4

****Revalidated:**** M6

****Required behavior:**** Report missing close, excessive variance, unusual refunds/payouts and late settlement.

****Gate-local behavior:**** Exception conditions are detected and reported to accountable staff.

****Later behavior:**** Presentation within the operational reporting surface with source and freshness labelling.

****Prerequisites:**** FR-CSH-004, FR-NOT-005

****Journeys:**** GJ-07

****Acceptance tests:**** TST-M4-FR-CSH-008, TST-M6-FR-CSH-008

Checks, Bills and Receipts

FR-BIL-001 - Check creation

****Priority:**** P0

****Owner:**** Platform Engineering

****Lineage:**** retained

****Introduced:**** M4

****Revalidated:**** None

****Required behavior:**** Create one or more checks from accepted/served order lines without changing o

order ownership or history.

****Gate-local behavior:**** Checks are created from accepted/served lines with order ownership and history unchanged.

****Later behavior:**** None beyond M4 for this clause.

****Prerequisites:**** FR-ORD-005, FR-FUL-011

****Journeys:**** GJ-01B, GJ-06

****Acceptance tests:**** TST-M4-FR-BIL-001

FR-BIL-002 - Check line allocation

****Priority:**** P0

****Owner:**** Platform Engineering

****Lineage:**** retained

****Introduced:**** M4

****Revalidated:**** None

****Required behavior:**** Allocate whole or partial line quantities to a check and prevent double billing.

****Gate-local behavior:**** Whole and partial line quantities allocate to checks and a line unit can never be billed twice.

****Later behavior:**** None beyond M4 for this clause.

****Prerequisites:**** FR-BIL-001

****Journeys:**** GJ-06

****Acceptance tests:**** TST-M4-FR-BIL-002

FR-BIL-003 - Split modes

****Priority:**** P0

****Owner:**** Platform Engineering

****Lineage:**** retained

****Introduced:**** M4

****Revalidated:**** None

****Required behavior:**** Split by item, participant, equal share, custom amount or separate orders with deterministic rounding.

****Gate-local behavior:**** All five split modes produce deterministic exact results with configured rounding and no lost or duplicated minor units.

****Later behavior:**** None beyond M4 for this clause.

****Prerequisites:**** FR-BIL-002, FR-DAT-006

****Journeys:**** GJ-06

****Acceptance tests:**** TST-M4-FR-BIL-003

FR-BIL-004 - Merge checks

****Priority:**** P0

****Owner:**** Platform Engineering

****Lineage:**** retained

****Introduced:**** M4
****Revalidated:**** None
****Required behavior:**** Merge eligible open checks with audit and preserve original source relationships.

****Gate-local behavior:**** Eligible open checks merge with full audit and original source relationships preserved.

****Later behavior:**** None beyond M4 for this clause.

****Prerequisites:**** FR-BIL-001, FR-TAB-007B

****Journeys:**** GJ-06

****Acceptance tests:**** TST-M4-FR-BIL-004

FR-BIL-005 - Tax/service calculation

****Priority:**** P0

****Owner:**** Platform Engineering

****Lineage:**** retained

****Introduced:**** M4

****Revalidated:**** None

****Required behavior:**** Calculate item subtotal, discounts, tax, service charge and other bill components exactly. Optional tip is not part of the bill balance and is calculated and recorded separately.

****Gate-local behavior:**** Every bill component computes exactly with fixed-point arithmetic, and tip is excluded from the bill balance entirely.

****Later behavior:**** None beyond M4 for this clause.

****Prerequisites:**** FR-DAT-005, FR-MNU-009

****Journeys:**** GJ-01B, GJ-06

****Acceptance tests:**** TST-M4-FR-BIL-005

FR-BIL-006 - Rounding

****Priority:**** P0

****Owner:**** Platform Engineering

****Lineage:**** retained

****Introduced:**** M4

****Revalidated:**** None

****Required behavior:**** Apply configured currency precision and rounding at defined line/document stages and persist the calculation version.

****Gate-local behavior:**** Rounding applies at the configured stages and the calculation version is persisted on the document.

****Later behavior:**** None beyond M4 for this clause.

****Prerequisites:**** FR-BIL-005, FR-DAT-006

****Journeys:**** GJ-06

****Acceptance tests:**** TST-M4-FR-BIL-006

FR-BIL-007 - Bill preview

****Priority:**** P0
****Owner:**** Platform Engineering
****Lineage:**** retained
****Introduced:**** M4
****Revalidated:**** None
****Required behavior:**** Present a translated bill summary showing items, discounts, tax, service charge and bill total. Display the separate optional Tip box only after or beside the bill summary, never mixed into it.

****Gate-local behavior:**** The bill summary renders in the session language with the tip box visually separate in all three locales including RTL.

****Later behavior:**** None beyond M4 for this clause.

****Prerequisites:**** FR-BIL-005, FR-I18N-001C
****Journeys:**** GJ-01B, GJ-02B, GJ-03B
****Acceptance tests:**** TST-M4-FR-BIL-007

FR-BIL-008 - Finalize

****Priority:**** P0
****Owner:**** Platform Engineering
****Lineage:**** retained
****Introduced:**** M4
****Revalidated:**** None
****Required behavior:**** Finalize the bill only when its bill balance is settled or an authorized disposition exists. Tip completion is recorded separately and cannot hide an unpaid bill balance.

****Gate-local behavior:**** Finalization is refused while the bill balance is outstanding, regardless of tip state, unless an authorized disposition is recorded.

****Later behavior:**** None beyond M4 for this clause.

****Prerequisites:**** FR-BIL-005, FR-PAY-017
****Journeys:**** GJ-01B, GJ-07
****Acceptance tests:**** TST-M4-FR-BIL-008

FR-BIL-009 - Credit/void

****Priority:**** P0
****Owner:**** Platform Engineering
****Lineage:**** retained
****Introduced:**** M4
****Revalidated:**** None
****Required behavior:**** Correct issued bills through authorized void/credit/reissue rather than deletion.

****Gate-local behavior:**** Issued bills are correctable only by authorized void, credit or reissue; no delete path exists.

****Later behavior:**** None beyond M4 for this clause.

****Prerequisites:**** FR-DAT-008B, FR-AUTH-006

****Journeys:**** GJ-07
****Acceptance tests:**** TST-M4-FR-BIL-009

FR-BIL-010 - Digital receipt

****Priority:**** P0
****Owner:**** Platform Engineering
****Lineage:**** retained
****Introduced:**** M4
****Revalidated:**** M5a
****Required behavior:**** Generate digital receipts in the customer's selected language showing bill total, optional tip and total paid as separate lines.

****Gate-local behavior:**** Digital receipts render in the snapshotted locale with bill total, tip and total paid on separate lines.

****Later behavior:**** The physical printed receipt is produced through the local print path.

****Prerequisites:**** FR-I18N-005, FR-BIL-005
****Journeys:**** GJ-01B, GJ-02B, GJ-03B
****Acceptance tests:**** TST-M4-FR-BIL-010, TST-M5a-FR-BIL-010

FR-BIL-011 - Reprint

****Priority:**** P0
****Owner:**** Platform Engineering
****Lineage:**** retained
****Introduced:**** M4
****Revalidated:**** M5a
****Required behavior:**** Mark and audit receipt/bill reprints with operator and reason where required.

****Gate-local behavior:**** Reprints are marked as reprints and record operator and reason.

****Later behavior:**** Physical reprint through the local print path must carry the same marking.

****Prerequisites:**** FR-BIL-010, FR-CFG-003
****Journeys:**** GJ-07
****Acceptance tests:**** TST-M4-FR-BIL-011, TST-M5a-FR-BIL-011

FR-BIL-012 - Fiscal adapter

****Priority:**** P0
****Owner:**** Platform Engineering
****Lineage:**** retained
****Introduced:**** M4
****Revalidated:**** None
****Required behavior:**** Expose fiscal-document port and reconciliation status without embedding one provider's schema.

****Gate-local behavior:**** The fiscal port exposes documents and reconciliation status against a simulator with no provider-specific schema in the domain model.

****Later behavior:**** None beyond M4 for this clause.

****Prerequisites:**** FR-TEN-009A, FR-BIL-010

****Journeys:**** GJ-01B

****Acceptance tests:**** TST-M4-FR-BIL-012

FR-BIL-013 - Separate tip box

****Priority:**** P0

****Owner:**** Platform Engineering

****Lineage:**** retained

****Introduced:**** M4

****Revalidated:**** None

****Required behavior:**** The payment experience presents an optional Tip box visibly separate from the bill summary, with no tip selected by default.

****Gate-local behavior:**** The tip box is visually separate with no preselected amount or percentage in any locale.

****Later behavior:**** None beyond M4 for this clause.

****Prerequisites:**** FR-BIL-007

****Journeys:**** GJ-01B, GJ-02B, GJ-03B

****Acceptance tests:**** TST-M4-FR-BIL-013

FR-BIL-014 - Bill-tip separation

****Priority:**** P0

****Owner:**** Platform Engineering

****Lineage:**** retained

****Introduced:**** M4

****Revalidated:**** None

****Required behavior:**** Bill balance, service charge, tax, tip and total tendered are separate values and records. A tip never changes order lines or bill allocation.

****Gate-local behavior:**** All five values are separately stored and a recorded tip provably leaves order lines and bill allocation unchanged.

****Later behavior:**** None beyond M4 for this clause.

****Prerequisites:**** FR-BIL-005, FR-DAT-005

****Journeys:**** GJ-06, GJ-07

****Acceptance tests:**** TST-M4-FR-BIL-014

FR-BIL-015 - Per-payer tip

****Priority:**** P0

****Owner:**** Platform Engineering

****Lineage:**** retained

****Introduced:**** M4

****Revalidated:**** None

****Required behavior:**** In split or partial settlement, each payer may choose a separate optional tip attached to that payer's payment, without reallocating bill lines.

****Gate-local behavior:**** Each payer in a split settlement records an independent tip attached to their own payment with no bill line reallocation.

****Later behavior:**** None beyond M4 for this clause.

****Prerequisites:**** FR-BIL-003

****Journeys:**** GJ-06

****Acceptance tests:**** TST-M4-FR-BIL-015

FR-BIL-016 - Tip receipt and correction

****Priority:**** P0

****Owner:**** Platform Engineering

****Lineage:**** retained

****Introduced:**** M4

****Revalidated:**** None

****Required behavior:**** Receipts display bill and tip separately. Tip refunds, reversals and corrections are separate auditable records linked to the original tip.

****Gate-local behavior:**** Receipts separate bill and tip, and tip corrections create linked auditable records without touching the bill.

****Later behavior:**** None beyond M4 for this clause.

****Prerequisites:**** FR-BIL-010, FR-PAY-009

****Journeys:**** GJ-07

****Acceptance tests:**** TST-M4-FR-BIL-016

FR-BIL-017 - Minimum physical receipt printing

****Priority:**** P0

****Owner:**** Billing & Payments

****Lineage:**** new_audit_requirement

****Introduced:**** M4

****Revalidated:**** None

****Required behavior:**** At M4, a completed settlement produces a real physical customer receipt through a supported minimum production printer path. The receipt uses the customer language and shows bill total, optional tip, total paid and actual payment method separately.

****Gate-local behavior:**** Configure one supported cashier-connected receipt printer and physically print the M4 settlement receipt once with correct language, exact values, unique receipt number and actual payment method.

****Later behavior:**** M5a adds durable local queueing, retry, restart recovery, deduplication, printer health, outage continuity and reconciliation.

****Prerequisites:**** FR-BIL-010, FR-BIL-016, FR-CFG-001D

****Journeys:**** GJ-01B, GJ-02B, GJ-03B, GJ-06, GJ-07

****Acceptance tests:**** TST-M4-FR-BIL-017

Configuration and Setup

FR-CFG-001C - Billing and payment setup

****Priority:**** P0

****Owner:**** Platform & Security

****Lineage:**** split_successor from 'FR-CFG-001'
****Introduced:**** M4
****Revalidated:**** None
****Required behavior:**** The guided setup configures taxes, service charges, separate optional-tip settings and the permitted payment methods, and those settings drive a real bill.

****Gate-local behavior:**** Create a bill using the configured tax, service-charge, tip and payment settings and verify exact values and the actual permitted payment method.

****Later behavior:****

****Prerequisites:**** FR-PAY-015, FR-BIL-001
****Journeys:**** GJ-01B
****Acceptance tests:**** TST-M4-FR-CFG-001C

FR-CFG-001D - Minimum receipt-printer setup

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** split_successor from 'FR-CFG-001'
****Introduced:**** M4
****Revalidated:**** M5a
****Required behavior:**** At M4 the guided setup registers and tests the minimum production receipt printer used by the cashier for a real physical customer receipt.

****Gate-local behavior:**** Register one supported receipt printer, print a test page and print a settlement receipt without relying on the M5a local print queue.

****Later behavior:**** M5a adds durable local queueing, retry, restart recovery, deduplication, health and outage continuity.

****Prerequisites:**** FR-BIL-010
****Journeys:**** GJ-02B, GJ-03B, GJ-06
****Acceptance tests:**** TST-M4-FR-CFG-001D, TST-M5a-FR-CFG-001D

Data Architecture

FR-DAT-008B - Financial and cash ledgers

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** split_successor from 'FR-DAT-008'
****Introduced:**** M4
****Revalidated:**** None
****Required behavior:**** Bills, receipts, payments, tips and cash movements are append-only or reversal-based with no destructive correction.

****Gate-local behavior:**** Bills, receipts, payments, tips and cash movements are append-only or reversal-based with no destructive correction.

****Later behavior:**** None beyond M4 for this clause.

****Prerequisites:**** FR-DAT-008A, FR-DAT-005
****Journeys:**** GJ-07

****Acceptance tests:**** TST-M4-FR-DAT-008B

Integration Runtime

FR-INT-011 - Health

****Priority:**** P0

****Owner:**** Edge & Operations

****Lineage:**** retained

****Introduced:**** M4

****Revalidated:**** M5a

****Required behavior:**** Expose truthful health for the narrow Phase 1 integration surface, including outlet-node connectivity, sync lag, printer status and active payment adapters.

****Gate-local behavior:**** Health truthfully reports active payment adapters, including that a simulated adapter is reported as simulated.

****Later behavior:**** Outlet-node connectivity, sync lag and printer status require the node and print agent.

****Prerequisites:**** FR-PAY-015

****Journeys:**** GJ-10

****Acceptance tests:**** TST-M4-FR-INT-011, TST-M5a-FR-INT-011

FR-INT-013 - Schema version

****Priority:**** P0

****Owner:**** Edge & Operations

****Lineage:**** retained

****Introduced:**** M4

****Revalidated:**** M5a, M5b

****Required behavior:**** Version local/cloud protocols and Phase 1 adapter messages; incompatible peers stop safely instead of silently accepting unknown shapes.

****Gate-local behavior:**** Adapter messages carry a version and an unknown shape is refused rather than silently accepted.

****Later behavior:**** The local/cloud protocol version exists at M5a; agreement on compatibility is part of the reachability proof at M5b.

****Prerequisites:**** FR-COM-009

****Journeys:**** GJ-10, GJ-09

****Acceptance tests:**** TST-M4-FR-INT-013, TST-M5a-FR-INT-013, TST-M5b-FR-INT-013

Internationalization and Localization

FR-I18N-001C - Bill and receipt languages

****Priority:**** P0

****Owner:**** Customer Experience

****Lineage:**** split_successor from 'FR-I18N-001'

****Introduced:**** M4

****Revalidated:**** None

****Required behavior:**** Bill and receipt render completely in the session language for all three loca

les, including Ethiopic and RTL scripts.

****Gate-local behavior:**** Bill and receipt render completely in the session language for all three locales, including Ethiopic and RTL scripts.

****Later behavior:**** None beyond M4 for this clause.

****Prerequisites:**** FR-I18N-001A, FR-BIL-010, FR-BIL-017

****Journeys:**** GJ-01B, GJ-02B, GJ-03B

****Acceptance tests:**** TST-M4-FR-I18N-001C

Payments

FR-PAY-001 - Payment intent

****Priority:**** P0

****Owner:**** Commerce & Payments

****Lineage:**** retained

****Introduced:**** M4

****Revalidated:**** None

****Required behavior:**** Create a payment intent for a specific check balance and payer, with separate optional tip allocation, expiry, idempotency and permitted tender methods.

****Gate-local behavior:**** Payment intents bind to a check and payer with separate tip allocation, expiry, idempotency key and permitted methods.

****Later behavior:**** None beyond M4 for this clause.

****Prerequisites:**** FR-BIL-001, FR-ORD-004

****Journeys:**** GJ-01B

****Acceptance tests:**** TST-M4-FR-PAY-001

FR-PAY-002 - Cash payment

****Priority:**** P0

****Owner:**** Commerce & Payments

****Lineage:**** retained

****Introduced:**** M4

****Revalidated:**** M5a

****Required behavior:**** Record cash settlement locally with exact bill allocation, separate optional tip, amount tendered and change. Cash service continues during internet outage.

****Gate-local behavior:**** Cash settlement records exact bill allocation, separate tip, tendered amount and change.

****Later behavior:**** Continuation of cash service while the internet is down requires the outlet node to be authoritative.

****Prerequisites:**** FR-PAY-001, FR-BIL-005

****Journeys:**** GJ-01B

****Acceptance tests:**** TST-M4-FR-PAY-002, TST-M5a-FR-PAY-002

FR-PAY-003 - External card terminal

****Priority:**** P0
****Owner:**** Commerce & Payments
****Lineage:**** retained
****Introduced:**** M4
****Revalidated:**** M5a
****Required behavior:**** Record an external card-terminal result without storing card data. During outage, the method remains available only when the terminal itself can complete payment; otherwise staff must choose another method.

****Gate-local behavior:**** External terminal results are recorded with no card data stored anywhere.

****Later behavior:**** Availability of the method during outage depends on outlet-node operation and staff guidance.

****Prerequisites:**** FR-PAY-001, FR-SEC-011
****Journeys:**** GJ-01B
****Acceptance tests:**** TST-M4-FR-PAY-003, TST-M5a-FR-PAY-003

FR-PAY-006 - Mixed tender

****Priority:**** P0
****Owner:**** Commerce & Payments
****Lineage:**** retained
****Introduced:**** M4
****Revalidated:**** None
****Required behavior:**** Support permitted mixed tender while preserving separate allocations to bill balance and optional tip for each payment.

****Gate-local behavior:**** Mixed tender settles one check across methods with each payment carrying its own bill and tip allocation.

****Later behavior:**** None beyond M4 for this clause.

****Prerequisites:**** FR-PAY-017, FR-BIL-014
****Journeys:**** GJ-06
****Acceptance tests:**** TST-M4-FR-PAY-006

FR-PAY-007 - Partial payment

****Priority:**** P0
****Owner:**** Commerce & Payments
****Lineage:**** retained
****Introduced:**** M4
****Revalidated:**** None
****Required behavior:**** Support partial settlement and split bills without modifying the original order history; each payer may add a separate tip.

****Gate-local behavior:**** Partial settlement and split bills leave order history untouched and each payer tip is independent.

****Later behavior:**** None beyond M4 for this clause.

****Prerequisites:**** FR-BIL-003, FR-BIL-015
****Journeys:**** GJ-06

****Acceptance tests:** TST-M4-FR-PAY-007**

FR-PAY-009 - Refund

****Priority:** P0**

****Owner:** Commerce & Payments**

****Lineage:** retained**

****Introduced:** M4**

****Revalidated:** None**

****Required behavior:** Refund bill payments and tip payments through separate linked reversal records, permissions, reason codes and approval thresholds.**

****Gate-local behavior:** Bill and tip refunds create separate linked reversal records under permissions, reason codes and approval thresholds.**

****Later behavior:** None beyond M4 for this clause.**

****Prerequisites:** FR-AUTH-006, FR-PAY-017**

****Journeys:** GJ-07**

****Acceptance tests:** TST-M4-FR-PAY-009**

FR-PAY-010A - Payment reversal

****Priority:** P0**

****Owner:** Commerce & Payments**

****Lineage:** split_successor from 'FR-PAY-010'**

****Introduced:** M4**

****Revalidated:** None**

****Required behavior:** Payment and tip events are emitted in a documented shape that a future accounting consumer could subscribe to, with no consumer in Phase 1.**

****Gate-local behavior:** Payment and tip events are emitted in a documented shape that a future accounting consumer could subscribe to, with no consumer in Phase 1.**

****Later behavior:** None beyond M4 for this clause.**

****Prerequisites:** FR-PAY-017**

****Journeys:** GJ-01B**

****Acceptance tests:** TST-M4-FR-PAY-010A**

FR-PAY-012 - Payment failure

****Priority:** P0**

****Owner:** Commerce & Payments**

****Lineage:** retained**

****Introduced:** M4**

****Revalidated:** None**

****Required behavior:** Present retry/alternate method without duplicating the order or payment.**

****Gate-local behavior:** A failed payment offers retry or an alternate method with no duplicate order or payment created.**

****Later behavior:** None beyond M4 for this clause.**

****Prerequisites:**** FR-PAY-001, FR-ORD-004

****Journeys:**** GJ-01B

****Acceptance tests:**** TST-M4-FR-PAY-012

FR-PAY-013 - Reconciliation

****Priority:**** P0

****Owner:**** Commerce & Payments

****Lineage:**** retained

****Introduced:**** M4

****Revalidated:**** M5a

****Required behavior:**** Reconcile bill allocations, tip allocations, tender totals, cash shifts and provider references without merging tips into sales revenue.

****Gate-local behavior:**** Reconciliation balances all named totals and tips never appear inside sales revenue.

****Later behavior:**** Post-outage reconciliation of locally captured payments occurs after synchronization.

****Prerequisites:**** FR-PAY-017, FR-CSH-003

****Journeys:**** GJ-07

****Acceptance tests:**** TST-M4-FR-PAY-013, TST-M5a-FR-PAY-013

FR-PAY-014 - Payment collection and proof confirmation

****Priority:**** P0

****Owner:**** Commerce & Payments

****Lineage:**** retained

****Introduced:**** M4

****Revalidated:**** M5a

****Required behavior:**** A customer may request cash, external-terminal collection or proof-based manual mobile-money confirmation. For Telebirr and CBE Birr, staff verifies receipt in the provider application, records the actual provider and masked/reference identifier, and leaves unverified proof pending rather than paid.

****Gate-local behavior:**** Proof-based confirmation records provider and masked reference, and an unverified proof remains pending and can never reach captured.

****Later behavior:**** Local verification and recording of proof during outage occurs on the outlet node.

****Prerequisites:**** FR-PAY-001, FR-SEC-012

****Journeys:**** GJ-02B

****Acceptance tests:**** TST-M4-FR-PAY-014, TST-M5a-FR-PAY-014

FR-PAY-015 - Offline payment and adapter boundary

****Priority:**** P0

****Owner:**** Commerce & Payments

****Lineage:**** retained

****Introduced:**** M4

****Revalidated:**** M5a

****Required behavior:**** During internet outage, cash, locally recorded external-terminal payments and

locally verified Telebirr/CBE Birr proof confirmation may continue only when staff can verify the provider result. Direct online-provider APIs remain simulated until contracted and credentialed, are unavailable or queued during outage, and cannot support a live pilot-readiness claim. Receipts always state the actual method.

****Gate-local behavior:**** The live/simulated/prohibited matrix is enforced: simulated results are labelled and cannot be presented as live, and receipts state the actual method and provider.

****Later behavior:**** Continuation and queuing behavior during a real outage is exercised on the outlet node.

****Prerequisites:**** FR-PAY-014, FR-BIL-010

****Journeys:**** GJ-01B, GJ-02B

****Acceptance tests:**** TST-M4-FR-PAY-015, TST-M5a-FR-PAY-015

FR-PAY-016 - PCI boundary

****Priority:**** P0

****Owner:**** Commerce & Payments

****Lineage:**** retained

****Introduced:**** M4

****Revalidated:**** M6

****Required behavior:**** Keep raw PAN/CVV/card cryptograms outside platform storage, logs and analytics.

****Gate-local behavior:**** No PAN, CVV or cryptogram appears in storage, logs or analytics on any payment path.

****Later behavior:**** The prohibition is re-proven against the full production artifact and its logs.

****Prerequisites:**** FR-SEC-011, FR-PAY-003

****Journeys:**** GJ-01B

****Acceptance tests:**** TST-M4-FR-PAY-016, TST-M6-FR-PAY-016

FR-PAY-017 - Dual payment allocation

****Priority:**** P0

****Owner:**** Commerce & Payments

****Lineage:**** retained

****Introduced:**** M4

****Revalidated:**** None

****Required behavior:**** A payment transaction records separate allocations to bill balance and optional tip, with exact arithmetic, independent reversal and no hidden recomputation.

****Gate-local behavior:**** Every payment carries two explicit allocations that reverse independently with exact arithmetic and no recomputation on read.

****Later behavior:**** None beyond M4 for this clause.

****Prerequisites:**** FR-DAT-005, FR-BIL-014

****Journeys:**** GJ-06, GJ-07

****Acceptance tests:**** TST-M4-FR-PAY-017

Reporting and Intelligence

FR-RPT-001 - Role dashboards

****Priority:**** P0
****Owner:**** Quality Engineering
****Lineage:**** retained
****Introduced:**** M4
****Revalidated:**** M5a
****Required behavior:**** Provide role-specific Phase 1 dashboards for outlet manager, waiter, cashier, kitchen/expo and technical operator without exposing deferred modules.

****Gate-local behavior:**** Provide role-specific dashboards for outlet manager, waiter, cashier and kitchen/expo using only active Phase 1 data and without deferred modules.

****Later behavior:**** At M5a add the technical-operator local-node, synchronization and printer dashboard.

****Prerequisites:**** FR-RPT-015, FR-AUTH-008
****Journeys:**** GJ-01B, GJ-10
****Acceptance tests:**** TST-M4-FR-RPT-001, TST-M5a-FR-RPT-001

FR-RPT-002 - Freshness

****Priority:**** P0
****Owner:**** Quality Engineering
****Lineage:**** retained
****Introduced:**** M4
****Revalidated:**** M5a
****Required behavior:**** Label operational metrics with source and freshness; local and cloud views must clearly show synchronization status.

****Gate-local behavior:**** Label every operational metric with its source and freshness in the cloud reporting view.

****Later behavior:**** At M5a show local versus cloud source, synchronization status and staleness explicitly.

****Prerequisites:**** FR-RPT-015, FR-DAT-004
****Journeys:**** GJ-10
****Acceptance tests:**** TST-M4-FR-RPT-002, TST-M5a-FR-RPT-002

FR-RPT-003 - Sales

****Priority:**** P0
****Owner:**** Quality Engineering
****Lineage:**** retained
****Introduced:**** M4
****Revalidated:**** None
****Required behavior:**** Report orders, item sales, discounts, service charges, taxes, bill payments and tips as separately classified values.

****Gate-local behavior:**** Report orders, item sales, discounts, service charges, taxes, bill payments and tips as separately classified exact values.

****Later behavior:****

****Prerequisites:**** FR-RPT-015, FR-DAT-005

****Journeys:**** GJ-02B, GJ-03B

****Acceptance tests:**** TST-M4-FR-RPT-003

FR-RPT-004 - Service

****Priority:**** P0

****Owner:**** Quality Engineering

****Lineage:**** retained

****Introduced:**** M4

****Revalidated:**** None

****Required behavior:**** Report service requests, acknowledgement/completion times, table-session duration and unresolved exceptions.

****Gate-local behavior:**** Report service requests, acknowledgement and completion times, table-session duration and unresolved exceptions.

****Later behavior:****

****Prerequisites:**** FR-RPT-015, FR-SRV-001

****Journeys:**** None

****Acceptance tests:**** TST-M4-FR-RPT-004

FR-RPT-005 - Kitchen

****Priority:**** P0

****Owner:**** Quality Engineering

****Lineage:**** retained

****Introduced:**** M4

****Revalidated:**** None

****Required behavior:**** Report kitchen/bar/expo queue, preparation time, ready-to-serve time, rework and exceptions.

****Gate-local behavior:**** Report kitchen, bar and expo queue state, preparation time, ready-to-serve time, rework and exceptions.

****Later behavior:****

****Prerequisites:**** FR-RPT-015, FR-FUL-001

****Journeys:**** None

****Acceptance tests:**** TST-M4-FR-RPT-005

FR-RPT-013 - Exports

****Priority:**** P0

****Owner:**** Quality Engineering

****Lineage:**** retained

****Introduced:**** M4

****Revalidated:**** M6

****Required behavior:**** Export authorized Phase 1 operational data in documented formats with tenant/outlet scoping.

****Gate-local behavior:**** Export authorized Phase 1 operational data in documented formats with tenant and outlet scoping.

****Later behavior:**** At M6 repeat the export through production roles and the final built artifact.

****Prerequisites:**** FR-RPT-015, FR-AUTH-008

****Journeys:**** GJ-12

****Acceptance tests:**** TST-M4-FR-RPT-013, TST-M6-FR-RPT-013

FR-RPT-014 - Snapshots

****Priority:**** P0

****Owner:**** Quality Engineering

****Lineage:**** retained

****Introduced:**** M4

****Revalidated:**** M6

****Required behavior:**** Snapshot operational metrics used for shift close and later audit; a recomputation cannot silently rewrite a signed-off shift result.

****Gate-local behavior:**** Shift close writes an immutable metric snapshot that later recomputation cannot alter.

****Later behavior:**** The audit consumption of these snapshots occurs in the reporting surface.

****Prerequisites:**** FR-CSH-004, FR-DAT-008B

****Journeys:**** GJ-07

****Acceptance tests:**** TST-M4-FR-RPT-014, TST-M6-FR-RPT-014

FR-RPT-015 - Metric catalog

****Priority:**** P0

****Owner:**** Quality Engineering

****Lineage:**** retained

****Introduced:**** M4

****Revalidated:**** M6

****Required behavior:**** Define each Phase 1 metric, formula, timezone, currency, inclusion rule and data source in a versioned catalog.

****Gate-local behavior:**** Publish a versioned metric catalog defining each Phase 1 formula, timezone, currency, inclusion rule and data source.

****Later behavior:**** At M6 verify the catalog against the complete production reporting surface.

****Prerequisites:**** FR-DAT-004, FR-DAT-005

****Journeys:**** GJ-12

****Acceptance tests:**** TST-M4-FR-RPT-015, TST-M6-FR-RPT-015

Staff POS and Outlet UX

FR-POS-003B - Counter POS ordering

****Priority:**** P0

****Owner:**** Customer Experience

****Lineage:**** split_successor from 'FR-POS-003'

****Introduced:**** M4
****Revalidated:**** None
****Required behavior:**** Counter orders are created at the POS terminal under the same rules as QR and waiter ordering.

****Gate-local behavior:**** Counter orders are created at the POS terminal under the same rules as QR and waiter ordering.

****Later behavior:**** None beyond M4 for this clause.

****Prerequisites:**** FR-POS-003A, FR-ORD-001B
****Journeys:**** GJ-06
****Acceptance tests:**** TST-M4-FR-POS-003B

Tables and Sessions

FR-TAB-007B - Split session and check

****Priority:**** P0
****Owner:**** Customer Experience
****Lineage:**** split_successor from 'FR-TAB-007'
****Introduced:**** M4
****Revalidated:**** None
****Required behavior:**** A merged session splits into separate checks with correct allocation and a complete audit trail.

****Gate-local behavior:**** A merged session splits into separate checks with correct allocation and a complete audit trail.

****Later behavior:**** None beyond M4 for this clause.

****Prerequisites:**** FR-TAB-007A, FR-BIL-003
****Journeys:**** GJ-06
****Acceptance tests:**** TST-M4-FR-TAB-007B

UX and Accessibility

FR-UX-018 - Print preview

****Priority:**** P0
****Owner:**** Customer Experience
****Lineage:**** retained
****Introduced:**** M4
****Revalidated:**** M5a
****Required behavior:**** Preview receipt, kitchen, label and operational documents before configuration publication.

****Gate-local behavior:**** Receipt and document templates preview accurately before publication.

****Later behavior:**** Physical print output must match the preview on the actual printer.

****Prerequisites:**** FR-BIL-010
****Journeys:**** GJ-10
****Acceptance tests:**** TST-M4-FR-UX-018, TST-M5a-FR-UX-018

Gate M5a

Configuration and Setup

FR-CFG-001E - Continuity-node setup

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** split_successor from 'FR-CFG-001'
****Introduced:**** M5a
****Revalidated:**** M5b
****Required behavior:**** The guided setup registers the outlet continuity node and applies its approved outlet, synchronization, authority and local-service configuration.

****Gate-local behavior:**** Register the node, bind it to one tenant and outlet, apply configuration and prove it starts only with the correct outlet identity.

****Later behavior:**** M5b activates the same-QR DNS/TLS and authority-lease configuration.

****Prerequisites:**** FR-EDG-002A
****Journeys:**** GJ-10
****Acceptance tests:**** TST-M5a-FR-CFG-001E, TST-M5b-FR-CFG-001E

Data Architecture

FR-DAT-008C - Synchronization evidence ledger

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** split_successor from 'FR-DAT-008'
****Introduced:**** M5a
****Revalidated:**** None
****Required behavior:**** Synchronization evidence records are append-only across replay and restart.

****Gate-local behavior:**** Synchronization evidence records are append-only across replay and restart.

****Later behavior:**** None beyond M5a for this clause.

****Prerequisites:**** FR-DAT-008A, FR-EDG-001
****Journeys:**** GJ-10
****Acceptance tests:**** TST-M5a-FR-DAT-008C

Deployment and Operations

FR-OPS-010 - Update rollback

****Priority:**** P0
****Owner:**** Edge & Operations
****Lineage:**** retained
****Introduced:**** M5a
****Revalidated:**** M6
****Required behavior:**** Support signed application/node updates, database compatibility checks, staged rollout and rollback without corrupting local queues.

****Gate-local behavior:**** Signed node updates apply with database compatibility checks and roll back without corrupting the outbox or inbox.

****Later behavior:**** Staged rollout across outlets is a production deployment concern.

****Prerequisites:**** FR-EDG-018, FR-INT-003

****Journeys:**** GJ-10, GJ-12

****Acceptance tests:**** TST-M5a-FR-OPS-010, TST-M6-FR-OPS-010

FR-OPS-018 - Printer/device inventory

****Priority:**** P0

****Owner:**** Edge & Operations

****Lineage:**** retained

****Introduced:**** M5a

****Revalidated:**** M6

****Required behavior:**** Maintain an outlet inventory of continuity node, routers, access points, POS terminals, KDS devices and printers with assigned location and support owner.

****Gate-local behavior:**** The outlet asset register lists all six device classes with location and support owner.

****Later behavior:**** The register is confirmed against the deployed pilot outlet.

****Prerequisites:**** FR-POS-001, FR-EDG-001

****Journeys:**** GJ-12

****Acceptance tests:**** TST-M5a-FR-OPS-018, TST-M6-FR-OPS-018

Integration Runtime

FR-INT-003 - Outbox

****Priority:**** P0

****Owner:**** Edge & Operations

****Lineage:**** retained

****Introduced:**** M5a

****Revalidated:**** None

****Required behavior:**** Use a transactional outbox for local-to-cloud operational events, payment-provider events and notification jobs required by Phase 1 only.

****Gate-local behavior:**** Operational events, payment-provider events and notification jobs enqueue transactionally with the originating write and survive process restart.

****Later behavior:**** None beyond M5a for this clause.

****Prerequisites:**** FR-EDG-001, FR-DAT-008C

****Journeys:**** GJ-10

****Acceptance tests:**** TST-M5a-FR-INT-003

FR-INT-004 - Inbox

****Priority:**** P0

****Owner:**** Edge & Operations

****Lineage:**** retained
****Introduced:**** M5a
****Revalidated:**** None
****Required behavior:**** Use an idempotent inbox for cloud-to-outlet configuration and command delivery required by Phase 1 only.

****Gate-local behavior:**** Cloud-to-outlet configuration and commands apply exactly once under repeated delivery.

****Later behavior:**** None beyond M5a for this clause.

****Prerequisites:**** FR-INT-003, FR-INT-005
****Journeys:**** GJ-10
****Acceptance tests:**** TST-M5a-FR-INT-004

Outlet Edge and Synchronization

FR-EDG-001 - Deployment profiles

****Priority:**** P0
****Owner:**** Edge & Operations
****Lineage:**** retained
****Introduced:**** M5a
****Revalidated:**** M6
****Required behavior:**** Every production Phase 1 outlet runs the focused Outlet Continuity Node. Cloud-only mode is permitted only for development, demonstration or explicitly non-production evaluation.

****Gate-local behavior:**** A production outlet profile requires the continuity node and cloud-only mode is refused outside non-production environments.

****Later behavior:**** The profile constraint is re-proven against the production deployment artifact.

****Prerequisites:**** FR-GOV-001B, FR-OPS-001
****Journeys:**** GJ-10, GJ-12
****Acceptance tests:**** TST-M5a-FR-EDG-001, TST-M6-FR-EDG-001

FR-EDG-002A - Phase 1 local services

****Priority:**** P0
****Owner:**** Edge & Operations
****Lineage:**** split_successor from 'FR-EDG-002'
****Introduced:**** M5a
****Revalidated:**** None
****Required behavior:**** The node image contains exactly the five named services and starts them under least privilege.

****Gate-local behavior:**** The node image contains exactly the five named services and starts them under least privilege.

****Later behavior:**** None beyond M5a for this clause.

****Prerequisites:**** FR-EDG-001
****Journeys:**** GJ-10

****Acceptance tests:**** TST-M5a-FR-EDG-002A

FR-EDG-004A - Outlet-network application serving

****Priority:**** P0

****Owner:**** Edge & Operations

****Lineage:**** split_successor from 'FR-EDG-004'

****Introduced:**** M5a

****Revalidated:**** None

****Required behavior:**** All four screen families are served from the outlet node over the LAN at its configured endpoint.

****Gate-local behavior:**** All four screen families are served from the outlet node over the LAN at its configured endpoint.

****Later behavior:**** Reaching those screens through the same public table QR requires split-horizon DNS and public-trust TLS.

****Prerequisites:**** FR-EDG-002A

****Journeys:**** GJ-10

****Acceptance tests:**** TST-M5a-FR-EDG-004A

FR-EDG-005 - Outbox/inbox sync

****Priority:**** P0

****Owner:**** Edge & Operations

****Lineage:**** retained

****Introduced:**** M5a

****Revalidated:**** None

****Required behavior:**** Synchronize through durable outbox/inbox events, ordered cursors, dependency rules and idempotency. Direct database replication is not the business synchronization mechanism.

****Gate-local behavior:**** Synchronization uses durable events with ordered cursors and dependency rules; no database replication path exists for business data.

****Later behavior:**** None beyond M5a for this clause.

****Prerequisites:**** FR-INT-003, FR-INT-004

****Journeys:**** GJ-10

****Acceptance tests:**** TST-M5a-FR-EDG-005

FR-EDG-008 - Conflict policy

****Priority:**** P0

****Owner:**** Edge & Operations

****Lineage:**** retained

****Introduced:**** M5a

****Revalidated:**** None

****Required behavior:**** Resolve conflicts through explicit domain policy and operator evidence; never use silent last-write-wins for orders, bills, payments, tips, cash or permissions.

****Gate-local behavior:**** Conflicts on all six named domains surface to an operator with evidence; no silent overwrite path exists.

****Later behavior:**** None beyond M5a for this clause.

****Prerequisites:**** FR-EDG-005, FR-DAT-007

****Journeys:**** GJ-10

****Acceptance tests:**** TST-M5a-FR-EDG-008

FR-EDG-009 - Offline UI

****Priority:**** P0

****Owner:**** Edge & Operations

****Lineage:**** retained

****Introduced:**** M5a

****Revalidated:**** M5b

****Required behavior:**** Show customer and staff whether the outlet is cloud-connected, local-continuity or reconciling, without blocking ordinary local service.

****Gate-local behavior:**** All three connectivity states display to staff and to customers on the locally served surface without blocking service.

****Later behavior:**** The customer-facing display during a genuine same-QR outage session is proven with the M5b routing.

****Prerequisites:**** FR-EDG-001, FR-EDG-002A

****Journeys:**** GJ-10, GJ-08

****Acceptance tests:**** TST-M5a-FR-EDG-009, TST-M5b-FR-EDG-009

FR-EDG-010 - External-action blocking

****Priority:**** P0

****Owner:**** Edge & Operations

****Lineage:**** retained

****Introduced:**** M5a

****Revalidated:**** None

****Required behavior:**** Block or queue actions requiring external authority and explain the restriction; cash, locally supported card-terminal recording and ordinary service continue.

****Gate-local behavior:**** Actions needing external authority are blocked or queued with a translated explanation while cash, terminal recording and ordinary service continue.

****Later behavior:**** None beyond M5a for this clause.

****Prerequisites:**** FR-EDG-001, FR-CFG-002A

****Journeys:**** GJ-10

****Acceptance tests:**** TST-M5a-FR-EDG-010

FR-EDG-012 - Protocol compatibility

****Priority:**** P0

****Owner:**** Edge & Operations

****Lineage:**** retained

****Introduced:**** M5a

****Revalidated:**** M5b

****Required behavior:**** Check cloud, node, API and event-protocol compatibility before synchronization; incompatible versions pause safely and remain locally operable where safe.

****Gate-local behavior:**** An incompatible peer pauses synchronization safely while local service continues.

****Later behavior:**** Compatibility agreement becomes a field of the bidirectional reachability proof.

****Prerequisites:**** FR-INT-013

****Journeys:**** GJ-10, GJ-09

****Acceptance tests:**** TST-M5a-FR-EDG-012, TST-M5b-FR-EDG-012

FR-EDG-015A - Staff and transaction outage continuity

****Priority:**** P0

****Owner:**** Edge & Operations

****Lineage:**** split_successor from 'FR-EDG-015'

****Introduced:**** M5a

****Revalidated:**** None

****Required behavior:**** With the internet cut, staff complete waiter entry, KDS preparation, bills, separate tips, cash settlement and local printing through the node.

****Gate-local behavior:**** With the internet cut, staff complete waiter entry, KDS preparation, bills, separate tips, cash settlement and local printing through the node.

****Later behavior:**** Customer-side local Wi-Fi QR ordering depends on same-QR routing.

****Prerequisites:**** FR-EDG-001, FR-EDG-002A, FR-EDG-029

****Journeys:**** GJ-10

****Acceptance tests:**** TST-M5a-FR-EDG-015A

FR-EDG-016 - Reconnection

****Priority:**** P0

****Owner:**** Edge & Operations

****Lineage:**** retained

****Introduced:**** M5a

****Revalidated:**** None

****Required behavior:**** On reconnection, upload local operations in dependency order, resume cloud reads, reconcile conflicts and prove no duplicate orders, payments or tips.

****Gate-local behavior:**** Reconnection uploads in dependency order, resumes cloud reads, reconciles conflicts and produces no duplicate order, payment or tip.

****Later behavior:**** None beyond M5a for this clause.

****Prerequisites:**** FR-EDG-005, FR-EDG-008

****Journeys:**** GJ-10

****Acceptance tests:**** TST-M5a-FR-EDG-016

FR-EDG-017 - Edge health

****Priority:**** P0

****Owner:**** Edge & Operations

****Lineage:**** retained

****Introduced:**** M5a

****Revalidated:**** M5b
****Required behavior:**** Expose local node, database, worker, print, storage, certificate and synchronization health to the outlet and cloud operator.

****Gate-local behavior:**** Node, database, worker, print, storage and synchronization health are exposed truthfully to outlet and cloud operators.

****Later behavior:**** Certificate health depends on the per-outlet certificate lifecycle.

****Prerequisites:**** FR-INT-011, FR-EDG-002A
****Journeys:**** GJ-10
****Acceptance tests:**** TST-M5a-FR-EDG-017, TST-M5b-FR-EDG-017

FR-EDG-018 - Edge security

****Priority:**** P0
****Owner:**** Edge & Operations
****Lineage:**** retained
****Introduced:**** M5a
****Revalidated:**** M6
****Required behavior:**** Protect the outlet node with device identity, least-privileged services, encrypted secrets, signed updates, host hardening and audited administrative access.

****Gate-local behavior:**** The node runs with device identity, least-privileged services, encrypted secrets, signed updates, hardened host and audited admin access.

****Later behavior:**** Host hardening and update signing are re-proven against the shipped production node image.

****Prerequisites:**** FR-SEC-014, FR-INT-010
****Journeys:**** GJ-10, GJ-12
****Acceptance tests:**** TST-M5a-FR-EDG-018, TST-M6-FR-EDG-018

FR-EDG-025 - Local readiness dataset

****Priority:**** P0
****Owner:**** Edge & Operations
****Lineage:**** retained
****Introduced:**** M5a
****Revalidated:**** None
****Required behavior:**** Before service, the outlet node stores the active menu, three approved translations, allergens, prices, taxes, service/tip settings, tables, staff access, stations, printers and open sessions.

****Gate-local behavior:**** The node holds a complete readiness dataset covering all eleven named element types before service begins.

****Later behavior:**** None beyond M5a for this clause.

****Prerequisites:**** FR-EDG-002A, FR-MNU-003
****Journeys:**** GJ-10
****Acceptance tests:**** TST-M5a-FR-EDG-025

FR-EDG-027 - Reconnection reconciliation

****Priority:**** P0
****Owner:**** Edge & Operations
****Lineage:**** retained
****Introduced:**** M5a
****Revalidated:**** None
****Required behavior:**** After reconnection, local events synchronize in dependency order, preserve original IDs and timestamps, reject duplicates and expose conflicts instead of silently overwriting.

****Gate-local behavior:**** Replayed events preserve original IDs and timestamps, duplicates are rejected and conflicts surface rather than overwrite.

****Later behavior:**** None beyond M5a for this clause.

****Prerequisites:**** FR-EDG-016, FR-EDG-008
****Journeys:**** GJ-10
****Acceptance tests:**** TST-M5a-FR-EDG-027

FR-EDG-029 - Resilient local receipt printing

****Priority:**** P0
****Owner:**** Outlet Edge & Operations
****Lineage:**** new_audit_requirement
****Introduced:**** M5a
****Revalidated:**** M6
****Required behavior:**** At M5a, customer receipt printing uses a durable local queue with idempotent job identity, bounded retry, restart recovery, deduplication, printer-health visibility, internet-outage continuity and cloud reconciliation without duplicate physical output.

****Gate-local behavior:**** Lose internet, restart the print service and reconnect while a receipt is queued; exactly one physical receipt prints and the job reconciles with complete status evidence.

****Later behavior:**** Repeat in the production-equivalent staging environment.

****Prerequisites:**** FR-BIL-017, FR-EDG-002A, FR-INT-005
****Journeys:**** GJ-10
****Acceptance tests:**** TST-M5a-FR-EDG-029, TST-M6-FR-EDG-029

Security and Data Protection

FR-SEC-002B - Local-node IDOR defense

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** split_successor from 'FR-SEC-002'
****Introduced:**** M5a
****Revalidated:**** None
****Required behavior:**** The local node denies cross-tenant and sibling-outlet access for all four verbs against populated fixtures.

****Gate-local behavior:**** The local node denies cross-tenant and sibling-outlet access for all four verbs against populated fixtures.

****Later behavior:**** None beyond M5a for this clause.

****Prerequisites:**** FR-EDG-001, FR-SEC-002A
****Journeys:**** GJ-10
****Acceptance tests:**** TST-M5a-FR-SEC-002B

Staff POS and Outlet UX

FR-POS-008 - Offline indicator

****Priority:**** P0
****Owner:**** Customer Experience
****Lineage:**** retained
****Introduced:**** M5a
****Revalidated:**** None
****Required behavior:**** Show locally saved, queued, synchronized, conflict and blocked states in plain language.

****Gate-local behavior:**** All five synchronization states render in plain language on the staff surface during and after an outage.

****Later behavior:**** None beyond M5a for this clause.

****Prerequisites:**** FR-EDG-001, FR-INT-003
****Journeys:**** GJ-10
****Acceptance tests:**** TST-M5a-FR-POS-008

Testing and Evidence

FR-TST-008 - Offline tests

****Priority:**** P0
****Owner:**** Quality Engineering
****Lineage:**** retained
****Introduced:**** M5a
****Revalidated:**** M5b
****Required behavior:**** Run customer, waiter, POS, KDS, billing, tip, payment, printing and reconnection journeys through a real cloud/outlet pair with internet loss and local Wi-Fi continuity.

****Gate-local behavior:**** Staff-side outage journeys including billing, tip, payment, printing and reconnection run against a real cloud/outlet pair.

****Later behavior:**** Customer local Wi-Fi continuity requires same-QR routing.

****Prerequisites:**** FR-EDG-015A, FR-EDG-016
****Journeys:**** GJ-10, GJ-08
****Acceptance tests:**** TST-M5a-FR-TST-008, TST-M5b-FR-TST-008

Gate M5b

Deployment and Operations

FR-OPS-017 - Per-outlet public hostname and certificate

****Priority:**** P0

****Owner:**** Edge & Operations
****Lineage:**** retained
****Introduced:**** M5b
****Revalidated:**** M6
****Required behavior:**** Use a managed per-outlet public hostname, split-horizon DNS and a public-CA DNS-01 certificate for cloud and local routing. Provision and rotate the per-outlet key securely. Raw local IP URLs, shared cross-outlet wildcard private keys, self-signed certificates and ignored warnings are prohibited.

****Gate-local behavior:**** The managed per-outlet hostname, split DNS and public-CA certificate operate with secure per-outlet key provisioning and rotation, and none of the four prohibited patterns exist.

****Later behavior:**** The operational rotation procedure runs against the production deployment.

****Prerequisites:**** FR-EDG-022A

****Journeys:**** GJ-08

****Acceptance tests:**** TST-M5b-FR-OPS-017, TST-M6-FR-OPS-017

Outlet Continuity and Edge

FR-EDG-028 - Same-QR client resolution acceptance

****Priority:**** P0

****Owner:**** Edge & Continuity

****Lineage:**** retained

****Introduced:**** M5b

****Revalidated:**** None

****Required behavior:**** Same-QR routing is accepted against real client behavior: a device joining outlet Wi-Fi while holding a cached public DNS answer reaches a correct and trusted endpoint within the documented TTL window; encrypted DNS (Android Private DNS, DNS-over-HTTPS) that bypasses outlet split DNS is detected and handled with a documented supported-network configuration and translated staff guidance; IPv4 and IPv6 answers are consistent so a dual-stack device cannot reach a stale or untrusted endpoint; and resolver-cache flush and TTL behavior during the cloud-to-LAN transition is specified and tested. Every unsupported client configuration fails safe to a clear instruction, never to a certificate warning or a bypass prompt.

****Gate-local behavior:**** Cached-answer, encrypted-DNS and dual-stack devices each reach a trusted endpoint or fail safe to translated guidance, never to a certificate warning or bypass prompt.

****Later behavior:**** None beyond M5b for this clause.

****Prerequisites:**** FR-EDG-022A, FR-EDG-021

****Journeys:**** GJ-08

****Acceptance tests:**** TST-M5b-FR-EDG-028

Outlet Edge and Synchronization

FR-EDG-003 - Local domain authority

****Priority:**** P0

****Owner:**** Edge & Operations

****Lineage:**** retained

****Introduced:**** M5b

****Revalidated:**** M6

****Required behavior:**** The outlet node is the authoritative writer for active dine-in table sessions , orders, fulfillment, service requests, checks, payments, tips and cash shifts for its outlet. Cloud access forwards commands to that authority while connected.

****Gate-local behavior:**** When the bidirectional lease permits local authority, the outlet node is the sole authoritative writer for active dine-in operations and connected cloud access forwards commands to that outlet authority.

****Later behavior:**** Repeat the authority and forwarding proof in staging with production-equivalent fencing and monitoring.

****Prerequisites:**** FR-EDG-021, FR-EDG-018

****Journeys:**** GJ-09

****Acceptance tests:**** TST-M5b-FR-EDG-003, TST-M6-FR-EDG-003

FR-EDG-004B - Same-QR outage access

****Priority:**** P0

****Owner:**** Edge & Operations

****Lineage:**** split_successor from 'FR-EDG-004'

****Introduced:**** M5b

****Revalidated:**** None

****Required behavior:**** The same public table QR resolves to the node with a trusted certificate and no browser warning during outage.

****Gate-local behavior:**** The same public table QR resolves to the node with a trusted certificate and no browser warning during outage.

****Later behavior:**** None beyond M5b for this clause.

****Prerequisites:**** FR-EDG-004A, FR-EDG-022A

****Journeys:**** GJ-08

****Acceptance tests:**** TST-M5b-FR-EDG-004B

FR-EDG-015B - Customer same-QR outage continuity

****Priority:**** P0

****Owner:**** Edge & Operations

****Lineage:**** split_successor from 'FR-EDG-015'

****Introduced:**** M5b

****Revalidated:**** None

****Required behavior:**** A customer on outlet Wi-Fi orders through the same QR while the internet is down.

****Gate-local behavior:**** A customer on outlet Wi-Fi orders through the same QR while the internet is down.

****Later behavior:**** None beyond M5b for this clause.

****Prerequisites:**** FR-EDG-015A, FR-EDG-022A, FR-EDG-025

****Journeys:**** GJ-08

****Acceptance tests:**** TST-M5b-FR-EDG-015B

FR-EDG-021 - Customer local Wi-Fi continuity

****Priority:**** P0
****Owner:**** Edge & Operations
****Lineage:**** retained
****Introduced:**** M5b
****Revalidated:**** None
****Required behavior:**** During internet outage, a customer connected to the restaurant Wi-Fi can use the same QR journey to browse the cached menu, order, call staff, request the bill, tip and settle using available local methods.

****Gate-local behavior:**** A customer on outlet Wi-Fi completes the full same-QR journey during outage : cached menu, order, staff call, bill request, tip and settlement by an available local method.

****Later behavior:**** None beyond M5b for this clause.

****Prerequisites:**** FR-EDG-022A, FR-EDG-025
****Journeys:**** GJ-08
****Acceptance tests:**** TST-M5b-FR-EDG-021

FR-EDG-022A - Same-QR routing and node-key custody

****Priority:**** P0
****Owner:**** Edge & Operations
****Lineage:**** split_successor from 'FR-EDG-022'
****Introduced:**** M5b
****Revalidated:**** M6
****Required behavior:**** The same opaque table QR uses a per-outlet public hostname. Public DNS routes to cloud and outlet split-horizon DNS routes to the LAN node. The node generates and retains its private key, submits only a CSR, and receives the public-CA certificate chain from central DNS-01 automation; the private key is never exported.

****Gate-local behavior:**** The same QR reaches a trusted cloud endpoint normally and a trusted LAN endpoint during supported outlet routing; certificate issuance completes from a node-generated CSR with no private-key export.

****Later behavior:**** Production issuance and rotation evidence is rechecked during staging.

****Prerequisites:**** FR-EDG-004A, FR-TAB-001
****Journeys:**** GJ-08
****Acceptance tests:**** TST-M5b-FR-EDG-022A, TST-M6-FR-EDG-022A

FR-EDG-022B - Certificate lifecycle

****Priority:**** P0
****Owner:**** Edge & Operations
****Lineage:**** split_successor from 'FR-EDG-022'
****Introduced:**** M5b
****Revalidated:**** M6
****Required behavior:**** The per-outlet certificate lifecycle begins renewal 30 days before expiry, uses bounded retry, alerts at 14 and 7 days, records LAN-served fingerprint and expiry before completion, supports compromise revocation and re-issuance, and fails safely to the cloud-served journey with translated guidance.

****Gate-local behavior:**** Exercise renewal, retry, alert, installation verification, revocation and failed-renewal fallback without a browser warning.

****Later behavior:**** Repeat against the production-equivalent outlet environment.

****Prerequisites:**** FR-EDG-022A

****Journeys:**** GJ-08, GJ-12

****Acceptance tests:**** TST-M5b-FR-EDG-022B, TST-M6-FR-EDG-022B

FR-EDG-022C - TLS and URL prohibitions

****Priority:**** P0

****Owner:**** Edge & Operations

****Lineage:**** split_successor from 'FR-EDG-022'

****Introduced:**** M5b

****Revalidated:**** M6

****Required behavior:**** Shared cross-outlet wildcard private keys, raw local-IP customer URLs, self-signed certificate warnings and manual browser bypass are prohibited.

****Gate-local behavior:**** Scanner and client tests prove that no prohibited key, URL, certificate or bypass path exists.

****Later behavior:**** Re-scan the final production artifacts.

****Prerequisites:**** FR-EDG-022A

****Journeys:**** GJ-08, GJ-12

****Acceptance tests:**** TST-M5b-FR-EDG-022C, TST-M6-FR-EDG-022C

FR-EDG-023 - Bidirectional cloud reachability lease

****Priority:**** P0

****Owner:**** Edge & Operations

****Lineage:**** retained

****Introduced:**** M5b

****Revalidated:**** None

****Required behavior:**** Use a signed bidirectional challenge/acknowledgement every 5 seconds by default. A valid proof confirms cloud-to-node reachability, node-to-cloud reachability, the same monotonic authority sequence, protocol compatibility and cursor posture. Mark degraded after 10 seconds and expire the cloud forwarding lease after 20 seconds. Resume only after compatibility checks and three consecutive valid proofs.

****Gate-local behavior:**** Signed bidirectional proofs run at 5s, degrade at 10s and expire the forwarding lease at 20s; a one-way failure in either direction also expires it; recovery requires three consecutive valid proofs.

****Later behavior:**** None beyond M5b for this clause.

****Prerequisites:**** FR-EDG-003, FR-EDG-012

****Journeys:**** GJ-09

****Acceptance tests:**** TST-M5b-FR-EDG-023

FR-EDG-024 - Monotonic authority replacement and fencing

****Priority:**** P0

****Owner:**** Edge & Operations
****Lineage:**** retained
****Introduced:**** M5b
****Revalidated:**** None
****Required behavior:**** Emergency replacement uses the next durable monotonic signed authority sequence. Before a standby node or standby outlet node becomes writable, record step-up, independent approval, old-node power-off or router/firewall/switch-port/VLAN isolation, and an automated LAN-unreachability probe. Every writer persists the highest accepted sequence and rejects rollback. Direct LAN writes to the old node must fail; stale events are quarantined.

****Gate-local behavior:**** A replacement becomes writable only after recorded fence evidence and independent approval; the old node rejects direct LAN writes and its stale events quarantine; rollback is refused.

****Later behavior:**** None beyond M5b for this clause.

****Prerequisites:**** FR-EDG-023, FR-AUTH-006

****Journeys:**** GJ-09

****Acceptance tests:**** TST-M5b-FR-EDG-024

FR-EDG-026 - Session continuity

****Priority:**** P0

****Owner:**** Edge & Operations

****Lineage:**** retained

****Introduced:**** M5b

****Revalidated:**** None

****Required behavior:**** An active table session and its participant tokens remain valid when a customer moves from cloud access to local Wi-Fi, without duplicate orders or loss of cart ownership.

****Gate-local behavior:**** A session started on cellular continues on outlet Wi-Fi with participant tokens intact, no duplicate order and cart ownership preserved.

****Later behavior:**** None beyond M5b for this clause.

****Prerequisites:**** FR-EDG-021, FR-TAB-004

****Journeys:**** GJ-08

****Acceptance tests:**** TST-M5b-FR-EDG-026

Gate M6

Configuration and Setup

FR-CFG-007B - No production reset capability

****Priority:**** P0

****Owner:**** Platform & Security

****Lineage:**** split_successor from 'FR-CFG-007'

****Introduced:**** M6

****Revalidated:**** None

****Required behavior:**** Scan of the built production image and database proves no demo-reset route, job or script is present.

****Gate-local behavior:**** Scan of the built production image and database proves no demo-reset route,

job or script is present.

****Later behavior:**** None; terminal claim.

****Prerequisites:**** FR-OPS-001

****Journeys:**** GJ-12

****Acceptance tests:**** TST-M6-FR-CFG-007B

Deployment and Operations

FR-OPS-006 - Backup schedule

****Priority:**** P0

****Owner:**** Edge & Operations

****Lineage:**** retained

****Introduced:**** M6

****Revalidated:**** None

****Required behavior:**** Back up cloud and outlet databases, configuration and required files on a documented schedule using tools present inside the production artifact.

****Gate-local behavior:**** Scheduled backups of cloud and outlet databases run using only tools present inside the built production artifact.

****Later behavior:**** None beyond M6.

****Prerequisites:**** FR-OPS-019, FR-EDG-001

****Journeys:**** GJ-11

****Acceptance tests:**** TST-M6-FR-OPS-006

FR-OPS-007 - Restore drill

****Priority:**** P0

****Owner:**** Edge & Operations

****Lineage:**** retained

****Introduced:**** M6

****Revalidated:**** None

****Required behavior:**** Restore into clean cloud and outlet environments, start with least-privileged production roles, run the Phase 1 golden journeys and measure recovery time.

****Gate-local behavior:**** A destructive restore into clean cloud and outlet environments starts under production roles, passes the golden journeys and records recovery time.

****Later behavior:**** None beyond M6.

****Prerequisites:**** FR-OPS-006, FR-OPS-020

****Journeys:**** GJ-11

****Acceptance tests:**** TST-M6-FR-OPS-007

FR-OPS-011 - Runbooks

****Priority:**** P0

****Owner:**** Edge & Operations

****Lineage:**** retained

****Introduced:**** M6

****Revalidated:**** None
****Required behavior:**** Provide operator runbooks for installation, menu publish, table QR issue, printer setup, outage, reconnection, backup, restore, update, incident and pilot cutover.

****Gate-local behavior:**** All eleven runbooks exist and each has been executed at least once against the real system.

****Later behavior:**** None beyond M6.

****Prerequisites:**** FR-OPS-007, FR-OPS-015
****Journeys:**** GJ-12
****Acceptance tests:**** TST-M6-FR-OPS-011

FR-OPS-012 - Monitoring ownership

****Priority:**** P0
****Owner:**** Edge & Operations
****Lineage:**** retained
****Introduced:**** M6
****Revalidated:**** None
****Required behavior:**** Assign alert severity, owner, acknowledgement and escalation; avoid unowned dashboards.

****Gate-local behavior:**** Every alert and dashboard has a named owner, severity, acknowledgement path and escalation.

****Later behavior:**** None beyond M6.

****Prerequisites:**** FR-NOT-011, FR-OPS-004
****Journeys:**** None
****Acceptance tests:**** TST-M6-FR-OPS-012

FR-OPS-015 - Go-live cutover

****Priority:**** P0
****Owner:**** Edge & Operations
****Lineage:**** retained
****Introduced:**** M6
****Revalidated:**** None
****Required behavior:**** Use a controlled go-live checklist, pilot tenant/outlet, rollback plan, data owner and named operator. No direct production cutover from an unaudited branch.

****Gate-local behavior:**** Go-live runs from an audited branch with checklist, pilot outlet, rollback plan, data owner and named operator recorded.

****Later behavior:**** None beyond M6.

****Prerequisites:**** FR-GOV-004, FR-TST-019
****Journeys:**** GJ-12
****Acceptance tests:**** TST-M6-FR-OPS-015

FR-OPS-016 - Disaster recovery

****Priority:**** P0

****Owner:**** Edge & Operations
****Lineage:**** retained
****Introduced:**** M6
****Revalidated:**** None
****Required behavior:**** Define cloud and outlet recovery objectives, replacement-node procedure and reconciliation after prolonged outage.

****Gate-local behavior:**** Recovery objectives, the replacement-node procedure and prolonged-outage reconciliation are defined and exercised.

****Later behavior:**** None beyond M6.

****Prerequisites:**** FR-EDG-024, FR-OPS-007

****Journeys:**** GJ-11

****Acceptance tests:**** TST-M6-FR-OPS-016

FR-OPS-019 - Built artifact completeness

****Priority:**** P0
****Owner:**** Edge & Operations
****Lineage:**** retained
****Introduced:**** M6
****Revalidated:**** None
****Required behavior:**** Every advertised runtime job, script, database client and configuration file exists and executes inside the built production image without host development mounts.

****Gate-local behavior:**** Every advertised job, script, database client and config file executes inside the image with no host mount.

****Later behavior:**** None beyond M6.

****Prerequisites:**** FR-OPS-005, FR-OPS-008

****Journeys:**** GJ-12

****Acceptance tests:**** TST-M6-FR-OPS-019

Security and Data Protection

FR-SEC-019 - Backups

****Priority:**** P0
****Owner:**** Platform & Security
****Lineage:**** retained
****Introduced:**** M6
****Revalidated:**** None
****Required behavior:**** Encrypt backups, verify them, retain off-site copies and restore through the exact production roles and built artifacts.

****Gate-local behavior:**** Encrypted backup is taken, verified, stored off-site and destructively restored using only built artifacts and production roles.

****Later behavior:**** None; terminal claim.

****Prerequisites:**** FR-OPS-001, FR-DAT-017

****Journeys:**** GJ-11

****Acceptance tests:** TST-M6-FR-SEC-019**

Testing and Evidence

FR-TST-009 - Backup restore

****Priority:** P0**

****Owner:** Quality Engineering**

****Lineage:** retained**

****Introduced:** M6**

****Revalidated:** None**

****Required behavior:** Back up and destructively restore clean cloud and outlet environments, then run production-role table/order/KDS/bill/tip/payment/sync journeys.**

****Gate-local behavior:** A destructive restore is followed by successful production-role journeys across all named domains.**

****Later behavior:** None beyond M6.**

****Prerequisites:** FR-OPS-007**

****Journeys:** GJ-11**

****Acceptance tests:** TST-M6-FR-TST-009**

FR-TST-010 - Performance

****Priority:** P0**

****Owner:** Quality Engineering**

****Lineage:** retained**

****Introduced:** M6**

****Revalidated:** None**

****Required behavior:** Load test peak ordering/KDS/realtime, menu search, reporting and integration bursts with recorded thresholds.**

****Gate-local behavior:** Load tests record thresholds across ordering, KDS realtime, menu search, reporting and integration bursts.**

****Later behavior:** None beyond M6.**

****Prerequisites:** FR-UX-012, FR-RPT-015**

****Journeys:** GJ-12**

****Acceptance tests:** TST-M6-FR-TST-010**

FR-TST-018 - Built production-path tests

****Priority:** P0**

****Owner:** Quality Engineering**

****Lineage:** retained**

****Introduced:** M6**

****Revalidated:** None**

****Required behavior:** Security, readiness, backup, restore, local continuity and route-surface tests execute against built production artifacts and real production roles, not source-only substitutes.**

****Gate-local behavior:** All six named test families execute against built production artifacts under real production roles.**

****Later behavior:**** None beyond M6.

****Prerequisites:**** FR-OPS-019, FR-OPS-020

****Journeys:**** GJ-11, GJ-12

****Acceptance tests:**** TST-M6-FR-TST-018

Decision Catalog

****120 canonical decisions****

- Package: 'Hospitality_OS_Phase_1_Clean_Build_Package_v2.0.9'
- Canonical source root: 'cd6b8cfc2175004cc2057deb7d9de1ad0d6264a23c82f00e15e96faa526c0c96'
- Status: ****Package M0 candidate - M0R and implementation are not authorized****

ID	Topic	Target	Source	Decision
D-001	Product identity	Core	v1.1	The product is a multi-tenant, multilingual Hospitality Operating System for caf�s, restaurants, bakeries, bars, food courts, hotel outlets and related hospitality operators.
D-002	Commercial posture	Core	v1.1	The product is designed for internal use and commercial sale to independent outlets, groups, franchises and enterprise operators.
D-003	Working name	Core	v1.1	HospitalityOS is a working codename only; product name, trade marks and customer-facing branding are configurable and not hard-coded.
D-004	Architecture horizon	Core	v1.1	The architecture must support 10-20 years of additive evolution without requiring a customer-platform rebuild.
D-005	Implementation style	Core	v1.1	Start as a modular monolith with strict domain boundaries, versioned contracts and reliable events; do not begin with premature microservices.
D-006	API doctrine	Core	v1.1	All business capabilities are exposed through versioned APIs; frontends do not perform direct business writes to the database.
D-007	Tenant hierarchy	Core	v1.1	The hierarchy is platform ? tenant ? brand ? legal entity ? outlet ? service area ? station/table/device, with scopes configurable per tenant.
D-008	Tenant independence	Core	v1.1	Every tenant can operate without Horeca Addis, without another supplier connector and without platform-owned payment services.
D-009	Second-tenant proof	Core	v1.1	A differently branded second tenant must be included in automated tests and demonstration data to prove isolation and neutrality.
D-010	Customer languages	Phase 1	v1.1	English, Amharic and Arabic are the first-class customer launch languages. Arabic uses true right-to-left layout. Each outlet may order or disable launch languages only after approved translations remain complete.
D-011	Localization model	Core	v1.1	Canonical records are language-neutral and use customer translation records with fallback, completeness and approval controls; no duplicated menu per language. Customer interfaces are fully localized, while staff and back-office interfaces are English-first but translation-ready.
D-012	Guest-first ordering	Core	v1.1	A customer may browse and order as a guest; loyalty registration must never block ordinary service.
D-013	Customer identity	Phase 2	v1.1	Phase 1 is guest-first. Phone number remains the preferred future customer identity for loyalty/account features, which activate after the Phase 1 service core is approved.
D-014	Table QR security	Core	v1.1	Table QR codes use opaque signed references and revocable versions; raw sequential table IDs are not exposed.

| D-015 | Table session boundary | Core | v1.1 | A table session is separate from carts, orders, checks and payments and may contain multiple participants, devices and orders. |

| D-016 | Multiple guests | Core | v1.1 | Multiple customers at one table can maintain personal or shared carts, place separate orders and settle together or separately. |

| D-017 | Initial customer choices | Core | v1.1 | After language and guest/loyalty choice, the customer can view the menu, call a waiter or view an active table session. |

| D-018 | Service request ownership | Core | v1.1 | Every customer service request has a type, target role, assignment, acknowledgement timer, escalation and closure evidence. |

| D-019 | Order channels | Phase 1/Core | v1.1 | Phase 1 implements dine-in QR, waiter-entered dine-in and counter/POS through one order engine. Pickup and delivery remain additive channel contracts and are not built until the dine-in release is approved. |

| D-020 | Future channels | Core | v1.1 | Kiosk, marketplace, room service, catering, drive-through and voice/AI ordering are additive channel adapters, not separate order systems. |

| D-021 | Order immutability | Core | v1.1 | Submitted orders and accepted order lines are not silently overwritten; changes use additions, authorized amendments, cancellations, voids or reversals. |

| D-022 | Separate commercial records | Core | v1.1 | Order, fulfillment ticket, check/bill, payment, stock movement and accounting entry are separate linked records. |

| D-023 | Kitchen routing | Core | v1.1 | Each order line is routed to one or more production stations by a versioned preparation profile; one order may produce multiple tickets. |

| D-024 | Customer status model | Core | v1.1 | Customer-facing statuses are simplified and translated in the order/session language, which is snapshotted for communications and support; internal preparation and service statuses remain operationally detailed and launch in English. |

| D-025 | Course control | Core | v1.1 | Dine-in items support fire-now, hold and course sequencing; the customer cannot bypass outlet policies for controlled courses. |

| D-026 | Availability validation | Core | v1.1 | Availability is checked when displaying the menu and revalidated when an order is submitted; stale availability never guarantees fulfillment. |

| D-027 | Menu identity | Core | v1.1 | A menu item is distinct from a recipe, sellable variant, price, modifier set and outlet availability record. |

| D-028 | Recipe confidentiality | Core | v1.1 | Customer-visible ingredient descriptions are separate from confidential operational recipes and quantities. |

| D-029 | Allergen governance | Core | v1.1 | Allergen information is structured, translated, versioned and shown with approved text plus universal supporting icons; an allergy declaration triggers a prominent confirmation and kitchen alert, not a medical guarantee. Required safety translation gaps block publication in that language. |

| D-030 | Dietary labels | Core | v1.1 | Vegetarian, vegan, fasting, halal and other labels are configurable claims with definitions and approval responsibility per tenant and jurisdiction. |

| D-031 | Recipe versioning | Phase 2/3 | v1.1 | Stock consumption and costing reference the recipe/BOM version effective when the order line was accepted or fulfilled. |

| D-032 | Modifier rules | Core | v1.1 | Modifiers use explicit min/max, included quantity, price delta, preparation effect, allergen effect and recipe effect. |

| D-033 | Menu scheduling | Core | v1.1 | Menus, prices and availability can vary by outlet, channel, daypart, date, service area and fulfillment method. |

| D-034 | Pickup scope | Phase 1 Extension | v1.1 | Pickup supports branch selection, now/scheduled slots, promised-ready time, collection code and customer-arrived notification. |

| D-035 | Delivery scope | Phase 1 Extension/Phase 3 | v1.1 | Phase 1 delivery supports zones/polygons, fees, minimums, GPS pin plus landmarks, manual dispatch, own-driver or manual third-party handoff, COD and proof. |

| D-036 | Advanced delivery | Phase 1 Extension/Phase 3 | v1.1 | Automatic batching, route optimization, marketplace aggregation and fully automated courier APIs are later modules using the same delivery domain. |

| D-037 | Delivery versus kitchen states | Phase 1 Extension/Phase 3 | v1.1 | Preparation status and delivery status remain separate; a ready order may still be awaiting a driver. |

| D-038 | Address doctrine | Phase 1 Extension/Phase 3 | v1.1 | Delivery addresses support map coordinates, landmark, building/floor/unit and free-text instructions; formal street address is not mandatory. |

tory. |

| D-039 | Payment architecture | Core | v1.1 | Payments use provider adapters and tokenized references; raw card data is never stored by the platform. |

| D-040 | Payment methods | Core | v1.1 | Cash, external card-terminal result recording and verified Telebirr/CBE Birr proof confirmation are live pilot paths. Direct online-provider APIs remain adapter/simulator-only until contracted and credentialed. Raw card processing is prohibited and receipts state the actual method. |

| D-041 | Cash and card request | Core | v1.1 | A dine-in customer may request cash/card collection; staff receives a routed, timed payment-service request. |

| D-042 | Split settlement | Core | v1.1 | Checks can be split by item, person, amount or equal share; partial payments and multiple tenders are supported without changing the original order history. |

| D-043 | Tips and service charge | Phase 1 | v1.1 | The bill, optional tip, service charge, tax and every other fee are distinct records and presentation lines. The payment screen contains a separate optional Tip box; no tip is selected by default; total paid equals bill settlement plus separately recorded tip. |

| D-044 | Refund controls | Core | v1.1 | Refunds, voids, discounts and complimentary items require configurable permissions, reason codes and audit; thresholds can require approval. |

| D-045 | Cash-shift control | Core | v1.1 | Cash drawers and cashier shifts have opening float, collections, payouts, expected balance, actual count and variance approval. |

| D-046 | Receipt strategy | Core | v1.1 | The platform supports digital and printed customer receipts in the order language, reprints with attribution, English internal documents and fiscal adapter hooks. |

| D-047 | Stock ledger | Phase 2/3 | v1.1 | Inventory changes are recorded in an append-only stock ledger with lot/location/UOM context; corrections use reversals or adjustment events. |

| D-048 | Negative stock | Phase 2/3 | v1.1 | Negative stock is blocked by default; any permitted override is tenant policy, role restricted, reasoned and visibly exceptional. |

| D-049 | Stock consumption timing | Phase 2/3 | v1.1 | Default recipe consumption occurs at fulfillment completion, with configurable reservation/commit timing and explicit waste/rework adjustments. |

| D-050 | Packaging inventory | Phase 2/3 | v1.1 | Pickup and delivery packaging is modeled as stock and may be costed or charged separately. |

| D-051 | Waste and staff meals | Phase 2/3 | v1.1 | Waste, spoilage, tasting, staff meal, promotion and complimentary consumption use distinct reasoned stock events. |

| D-052 | Procurement neutrality | Phase 2/3 | v1.1 | Procurement supports manual suppliers and a generic supplier integration hub; no supplier owns the restaurant's internal item identity. |

| D-053 | Supplier mapping | Phase 2/3 | v1.1 | Internal ingredients/items map to zero or more supplier SKUs with pack/UOM conversions, validity dates, priority and substitution rules. |

| D-054 | Horeca Addis optionality | Phase 2/3 | v1.1 | The Horeca Addis connector is optional, disabled by default and enabled by tenant/outlet entitlement and configuration. |

| D-055 | Horeca Addis timing | Phase 2/3 | v1.1 | The connector contract and simulator exist in Phase 1; basic transactional integration activates in Phase 2 and intelligent reordering in Phase 3. |

| D-056 | Horeca Addis independence | Phase 2/3 | v1.1 | The restaurant platform continues to operate if Horeca Addis is disconnected or unavailable and can convert drafts to manual supplier orders. |

| D-057 | External order validation | Phase 2/3 | v1.1 | A connected supplier order requires online validation of current price, availability, pack, minimums, account status and delivery slot before final submission. |

| D-058 | Human procurement approval | Phase 2/3 | v1.1 | Automated or forecast-based reorders create recommendations/drafts; a human or configured approval workflow authorizes purchase commitment. |

| D-059 | Receiving chain | Phase 2/3 | v1.1 | Purchase order, supplier acknowledgement, delivery, goods receipt, discrepancy, return/claim, supplier invoice and payment are distinct linked records. |

| D-060 | Financial engine | Phase 2 | v1.1 | The native accounting foundation uses double-entry journals, immutable postings, reversals, periods, cost centers and auditable subledgers. |

| D-061 | Phase 1 finance | Phase 1 | v1.1 | Phase 1 records immutable operational checks, payments,

refunds, tips, cash shifts and settlement summaries. It does not implement a general ledger, AP, AR, inventory accounting, COGS or financial statements. |

| D-062 | Phase 2 finance | Phase 2 | v1.1 | Full AP, AR, purchasing, bank reconciliation, budgets, period close, inventory valuation and outlet/legal-entity reporting activate in Phase 2. |

| D-063 | External accounting | Phase 2 | v1.1 | External accounting/fiscal systems connect through adapters and a system-of-record registry; duplicate authoritative ledgers are prohibited. |

| D-064 | HR foundation | Phase 2 | v1.1 | Phase 1 includes staff identities, outlet memberships, roles and terminal access only. Employee records, rosters, clocking, attendance and workforce management activate in Phase 2. |

| D-065 | Payroll scope | Phase 2/3 | v1.1 | Full payroll is a Phase 3 country package; the core stores payroll inputs and approved results without hard-coding one country's law. |

| D-066 | Loyalty model | Phase 2/3 | v1.1 | Loyalty is optional per tenant and supports phone-based membership, points/tiers/rewards, consent, balances and immutable earn/redeem events. |

| D-067 | Customer privacy | Core | v1.1 | Guest sessions minimize personal data; loyalty and delivery PII are purpose-limited, consented, access-controlled and separately protected. |

| D-068 | Promotions | Phase 2/3 | v1.1 | Promotions are rule-driven, versioned and conflict-resolved; staff cannot invent uncontrolled discounts at checkout. |

| D-069 | Offline customer front | Phase 1 | v1.1 | The customer QR experience is cloud-connected by default but continues over restaurant Wi-Fi during outage using the same per-outlet public hostname, split-horizon DNS and public-trust TLS. |

| D-070 | Offline internal operations | Phase 1 | v1.1 | Customer QR ordering, waiter/POS, KDS/bar/expo, printing, table service, bills, separate tips, cash, external-terminal recording and verified TeleBirr/CBE Birr proof confirmation continue on the outlet network during outage when the underlying provider result can be verified. |

| D-071 | Cloud-only tier | Phase 1 | v1.1 | A production Phase 1 outlet includes the focused local continuity node. Cloud-only mode is allowed only for development, demonstration or explicitly non-production evaluation. |

| D-072 | Edge authority | Core | v1.1 | The outlet node is the authoritative writer for active dine-in operations. Cloud forwarding depends on a valid bidirectional reachability lease and the same durable monotonic signed authority sequence. |

| D-073 | No direct replication | Core | v1.1 | Direct database replication is not the business synchronization mechanism; use transactional outbox/inbox, ordered cursors, idempotency and explicit conflicts. |

| D-074 | Offline restrictions | Core | v1.1 | During outage, direct online-provider payments and other external-authority actions are unavailable or queued. Cash, external-terminal recording and verified TeleBirr/CBE Birr proof confirmation may continue; unverified claims remain pending. |

| D-075 | Print resilience | Phase 1 | v1.1 | Kitchen/bar/receipt printing has a local print agent, printer routing, retry/deduplication and operator-visible failure handling. |

| D-076 | Authentication resilience | Core | v1.1 | Staff uses phone/email identity with secure sessions; trusted local terminals may offer quick PIN only for low-risk actions with step-up for sensitive work. |

| D-077 | Authorization | Core | v1.1 | Permissions are deny-by-default and scoped by tenant, legal entity, outlet, role, resource and action; server-side enforcement is mandatory. |

| D-078 | Segregation of duties | Core | v1.1 | High-risk finance, stock, payroll, refund and configuration actions support maker-checker and prevent self-approval where configured. |

| D-079 | Audit | Core | v1.1 | Security, commercial, stock, finance, menu, permission, sync and administrative events are append-only and attributable to actor, terminal, source and correlation. |

| D-080 | Data retention | Core | v1.1 | Transactional records are retained according to tenant/jurisdiction policy; ordinary users cannot hard-delete completed commercial or financial records. |

| D-081 | File safety | Core | v1.1 | Uploads are private by default, type/size validated, safely named, malware-scannable, hash recorded and served through authorized signed access. |

| D-082 | Observability | Core | v1.1 | Structured logs, traces, metrics, health checks, business alerts and audit are distinct; secrets and sensitive payment/PII data are redacted. |

| D-083 | Deployment | Core | v1.1 | Development, CI/test, staging and production are separate; cont

ainers and adapters support managed cloud and future private/on-prem deployment. |

| D-084 | Backup and recovery | Core | v1.1 | Database and file backups, encrypted off-site copies, restore drills, recovery objectives and edge/cloud reconciliation are part of completion. |

| D-085 | Update safety | Core | v1.1 | Application, database, event and edge protocol versions are checked; incompatible peers pause safely and updates support rollback. |

| D-086 | Testing doctrine | Core | v1.1 | Compilation or page scaffolding is not completion. Real PostgreSQL, production-role, contract, E2E, concurrency, outage, restore and production-image tests are required, with the planted negative-control registry frozen at Package M0. |

| D-087 | Golden journeys | Core | v1.1 | Mandatory end-to-end golden journeys are executable acceptance gates and must use real persistence and integrations/simulators. |

| D-088 | No fake completion | Core | v1.1 | No acceptance-critical TODO, mock-only handler, fake persistence, skipped test, sample secret or manual database fix may remain at handoff. |

| D-089 | Accessibility | Core | v1.1 | Customer and staff experiences support keyboard/touch, semantic structure, sufficient contrast, non-color status indicators and screen-reader essentials. |

| D-090 | Device form factors | Core | v1.1 | Customer web is mobile-first; waiter/POS/KDS are touch-first; back office is responsive desktop-first; all reuse versioned APIs. |

| D-091 | Configuration over forks | Core | v1.1 | Brands, taxes, service rules, roles, numbering, templates, languages, feature flags and connectors are configuration, not tenant-specific code forks. |

| D-092 | Data portability | Phase 2 | v1.1 | Authorized tenants can export their master and transactional data in documented formats; platform lock-in must not rely on withholding customer data. |

| D-093 | AI boundary | Core | v1.1 | AI may draft translations, forecasts, reorder recommendations, menus and analytics, but customer content is never translated live at runtime and AI cannot be sole authority for allergy, payment, accounting, payroll or purchase commitment. |

| D-094 | Reservations and waitlist | Future | v1.1 | Reservations/waitlist are Phase 2 modules connected to the table/session domain; Phase 1 architecture reserves the capability without building a separate booking system. |

| D-095 | Central kitchen and franchise | Future | v1.1 | Central kitchen, commissary, inter-outlet transfer, franchise control and royalty reporting are later modules using existing tenant/outlet/inventory/finance boundaries. |

| D-096 | Hotel integration | Future | v1.1 | Room charge and hotel/PMS integration are optional adapters; core billing never assumes the outlet belongs to a hotel. |

| D-097 | Setup wizard | Phase 1 | v1.1 | The setup workflow configures tenant, brand, outlet, English/Amharic/Arabic customer content, tables and QR codes, stations, taxes, service charges, separate tip settings, payment methods, printers and the local continuity node. |

| D-098 | Demo safety | Core | v1.1 | Demo data is clearly marked, removable and never mixed into production opening balances. |

| D-099 | External blockers | Core | v1.1 | Missing third-party credentials do not justify fake completion. Only explicitly activated Phase 1 providers require adapters and simulators; deferred Phase 2/3 providers must not create dormant production routes or workers. |

| D-100 | Authority and precedence | Core | v2.0.8 amendment AMD-V208-002 | Source of Truth and locked decisions govern conflicts. Independent adverse verdicts pause progression; disputes follow builder response, reviewer rebuttal and founder decision. Overrides remain recorded, name the affected requirement/rule and are re-examined at the next audit. All duplicated narrative, JSON and workbook projections are generated from canonical structured records and parity-checked before pinning. |

| D-101 | Clean repository | Core | v2.0.2 | After Package M0 approval or documented founder adjudication, Phase 1 creates a new empty repository for MOR documentation and conformance only. The PostgreSQL database and executable migration '0001' begin in M1 only after MOR approval or adjudication. The v1.1 repository remains frozen reference. |

| D-102 | Controlled code reuse | Core | v2.0.6 amendment AMD-V206-006B | No code is copied from the frozen repository unless the unit is reviewed in isolation, dependencies and provenance are recorded, and tests written against the exact pinned package prove compliance with that package. |

| D-103 | No dormant future modules | Phase 1 | v2.0.2 | The Phase 1 production artifact and acceptance obligations contain no purchasing, inventory, accounting, workforce, loyalty, pickup, delivery,

operational recipe, costing, supplier or intelligence routes, tables, workers, screens or tests. |

| D-104 | Milestone delivery | Phase 1 | v2.0.2 | The build sequence is Package M0 -> Repository Conformance M0R -> M1 -> M2 -> M3 -> M4 -> M5a -> M5b -> M6. M0R is documentation/conformance only; M1 begins the database and migration '0001'. Every gate closes using only approved predecessors. |

| D-105 | Independent audit timing | Core | v2.0.2 | Codex performs Package M0 before repository creation, Repository Conformance M0R before M1 coding, and audits every completed milestone. An adverse verdict pauses work; final commit/merge authority remains with the founder under documented adjudication. |

| D-106 | Ordinary-command proof | Core | v2.0.2 | Windows and Linux acceptance must use the ordinarily documented commands without CI-only environment injection, hidden PATH changes or skipped tests. |

| D-107 | Production-role proof | Core | v2.0.2 | Readiness, migrations, backup, restore and operational journeys are tested using the exact least-privileged production identities and built production artifacts. |

| D-108 | Same QR continuity | Phase 1 | v2.0.2 | The same table QR uses a per-outlet public hostname. Public DNS routes to cloud and split-horizon outlet DNS routes to the LAN node. A public-CA DNS-01 certificate is centrally issued and securely provisioned. DNS TTL is 30 seconds and supported cloud-to-LAN transition is at most 60 seconds. Raw IP, self-signed bypass and shared cross-outlet private keys are prohibited. |

| D-109 | Split-brain prevention | Phase 1 | v2.0.2 | Cloud forwarding uses signed bidirectional reachability proofs with 5/10/20 timing. Emergency replacement uses a durable monotonic signed authority sequence and cannot become writable until old-node power/network fence evidence and an automated LAN-unreachability probe pass. |

| D-110 | Tip consent | Phase 1 | v2.0.2 | No tip is selected by default. Suggested tips are configurable and clearly optional. Staff-entered tips require the payer's explicit instruction and remain separately auditable. |

| D-111 | Tip settlement model | Phase 1 | v2.0.2 | Each payer may add a separate tip to their payment. Tips do not change order lines, bill allocation, tax/service-charge presentation or sales revenue classification; Phase 2 defines distribution and accounting treatment. |

| D-112 | Phase sequence | Core | v2.0.2 | Phase 1 is customer service and outlet execution; Phase 2 is purchasing/inventory/accounting/workforce; Phase 3 is ingredients/recipes/cost intelligence. Pickup and delivery are deferred customer-channel extensions after the dine-in pilot. |

| D-113 | Requirement gate dependency model | Core | v2.0.3 | Every active requirement carries an introducing gate and, where dependent behavior exists, the later gates at which it is revalidated. A gate closes on its gate-local scope only. No gate may be closed using behavior owned by a later gate, and the dependency graph is regenerated whenever a requirement or gate changes. |

| D-114 | Later-domain obligations are documentation only | Core | v2.0.3 | Excluded domains may be named in Phase 1 only as deferral statements or extension-contract documentation. Phase 1 creates no entity, enumeration value, field, route, screen, registry entry or positive test for an excluded domain, including storage locations, later-domain systems of record, employee/payroll data and operational units of measure. |

| D-115 | Forbidden-surface vocabulary generation | Core | v2.0.3 | The forbidden-surface scanner vocabulary is generated from the phase-boundary exclusion set, so every excluded domain has detection terms and a planted negative control. The scan covers every controlled register, narrative document and workbook, and from Repository Conformance M0R onward every dependency, package, migration, schema, route, job, worker, screen, test, configuration category and generated artifact. |

| D-116 | Per-outlet certificate lifecycle | Phase 1 | v2.0.3 | The per-outlet public-CA certificate has a measurable lifecycle: the outlet node generates and retains the private key, submits only a CSR, central DNS-01 automation obtains the certificate and returns only the certificate chain, and the private key is never exported. Renewal begins 30 days before expiry, uses bounded retry, alerts at 14 and 7 days, records installation evidence verified from the LAN, supports compromise revocation and re-issuance, and fails safely to the cloud-served journey rather than a browser warning. |

| D-117 | Same-QR client resolution acceptance | Phase 1 | v2.0.3 | Same-QR acceptance covers cached public answers, a 30-second DNS TTL, maximum 60-second cloud-to-LAN transition, Android/iOS system-resolver modes, supported and unsupported Private DNS/DoH cases, IPv4-only, IPv6-only and dual-stack

answer matrices, and out-of-band translated guidance delivered through the outlet captive portal and staff signage when local resolution is unavailable. |

| D-118 | Authority replacement activation | Phase 1 | v2.0.3 | The outlet connectivity state machine models one outlet node instance. Emergency replacement can activate only a fenced standby outlet node; the cloud remains a forwarder/control plane and is never a writable dine-in fallback. The replacement becomes writable only after fence evidence, LAN-unreachability proof, readiness and monotonic sequence confirmation, and a replacement-activated event. |

| D-119 | Receipt printing milestone boundary | Phase 1 | v2.0.5 reconciliation | M4 includes the minimum real production printer path required to issue a physical customer receipt. M5a adds durable local queueing, restart recovery, retry, deduplication, printer health, internet-outage continuity and reconciliation. M4 cannot claim M5a resilience, and M5a cannot be the first point at which a real receipt is printed. |

| D-120 | Milestone blocker and bounded P1 adjudication | Core | v2.0.5 reconciliation | Every P0 and every P1 affecting product scope, security, money, authority, milestone executability, acceptance completeness or canonical correctness blocks progression. A P1 limited to publication, projection, wording, identifier or validator coverage may proceed only through recorded founder adjudication when canonical behavior is correct, runtime behavior cannot change, the defect remains visible and repair is scheduled. |

Mandatory Golden Journeys

****16 slices****

- Package: 'Hospitality_OS_Phase_1_Clean_Build_Package_v2.0.9'
- Canonical source root: 'cd6b8cfc2175004cc2057deb7d9de1ad0d6264a23c82f00e15e96faa526c0c96'
- Status: ****Package M0 candidate - M0R and implementation are not authorized****

GJ-01A - English guest QR dine-in order through service

****Milestone:** M3**
****Mandatory:** Yes**
****Predecessors:** M2 approved**
****Personas:** Guest customer, waiter, kitchen**

****Steps:** Scan opaque table QR; open English; continue as guest; browse menu; select modifiers; submit order; the current approved cloud authority persists and no local-authority claim is made before M5b; kitchen accepts and prepares; waiter serves; customer sees served status.**

****Pass:** One table session, one accepted order, linked fulfillment and service timeline exist with no duplicate effect and correct English customer text. The cloud remains the current approved authority at M3. No check, payment, receipt or local-authority capability is required.**

GJ-01B - English bill, cash settlement and receipt

****Milestone:** M4**
****Mandatory:** Yes**
****Predecessors:** GJ-01A**
****Personas:** Guest customer, cashier**

****Steps:** Open the served English table session from GJ-01A; customer requests bill; cashier presents check; customer leaves no tip; cashier settles cash; issue the English digital receipt and print**

the English physical receipt through the minimum M4 production printer path.

****Pass:**** Check, cash payment, no-tip decision, digital receipt, one physical receipt and audit time line are linked to the predecessor order; bill, tip and total paid are shown separately; the physical receipt is printed without relying on M5a outage resilience.

GJ-02 - Amharic customer order and service

****Milestone:**** M3

****Mandatory:**** Yes

****Predecessors:**** M2 approved

****Personas:**** Guest customer, waiter, kitchen

****Steps:**** Select Amharic; browse approved menu and allergen text; place order; receive localized statuses; call waiter; add another order; waiter serves.

****Pass:**** Customer-facing text and snapshots are Amharic; staff screens remain English; cart/session /order identity is preserved; no M4 settlement dependency exists; settlement is proven by GJ-02B at M4.

GJ-02B - Amharic bill, tip, settlement and receipt

****Milestone:**** M4

****Mandatory:**** Yes

****Predecessors:**** GJ-02

****Personas:**** Guest customer, cashier

****Steps:**** Open the served Amharic table session from GJ-02; customer requests the bill; present the Amharic check with the separate optional tip box; customer adds a tip; settle by verified Telebirr proof confirmation; issue the Amharic digital receipt and print the Amharic physical receipt through the minimum M4 production printer path.

****Pass:**** Bill, tip and total paid are separate values and separate visual sections in Amharic; ETB and numerals follow the approved locale policy; packaged Ethiopic fonts render on screen and on the printed receipt with no fallback loss or clipping; the receipt states the actual method, provider and permitted masked/reference identifier; the payment remains proof_pending until staff verify it in the provider app. The M4 print proves a real physical receipt but does not claim M5a durable queue or outage resilience.

GJ-03A - Arabic RTL menu, order and service

****Milestone:**** M3

****Mandatory:**** Yes

****Predecessors:**** M2 approved

****Personas:**** Guest customer, waiter, kitchen

****Steps:**** Select Arabic; browse true RTL menu; use search/modifiers containing Latin SKUs or item codes; view ETB prices/numerals; submit order; view Arabic status timeline; waiter serves.

****Pass:**** Customer PWA mirrors correctly; mixed-direction strings preserve reading order; ETB and numerals follow approved locale policy; no bill or receipt capability is required.

GJ-03B - Arabic RTL bill, tip, payment and receipt

****Milestone:** M4**
****Mandatory:** Yes**
****Predecessors:** GJ-03A**
****Personas:** Guest customer, cashier**

****Steps:**** Open the served Arabic session from GJ-03A; request bill; view separate optional tip box; choose a tip; settle using a permitted live method; view the digital Arabic receipt and print the physical Arabic receipt through the minimum M4 production printer path.

****Pass:**** Bill/tip/payment/receipt layout mirrors correctly; mixed Arabic/Latin provider/reference text, ETB values and numerals preserve reading order; no clipping or string-substitution-only pass is accepted. The physical receipt is real M4 output; durable local print recovery is reserved for M5a.

GJ-04 - Multi-participant table and service requests

****Milestone:** M3**
****Mandatory:** Yes**
****Predecessors:** M2 approved**
****Personas:** Two guest customers, waiter**

****Steps:**** Two devices join one table; keep personal carts; place separate orders; call waiter; waiter acknowledges; one guest adds an order later; authorized staff move the session to another table.

****Pass:**** Participant ownership, service SLA, add-on orders and table move are preserved without expiring another table or duplicating orders.

GJ-05 - Waiter-entered order to KDS and service

****Milestone:** M3**
****Mandatory:** Yes**
****Predecessors:** M2 approved**
****Personas:** Waiter, kitchen/bar, expo, manager**

****Steps:**** Waiter opens table; enters dine-in order; routes lines to kitchen/bar; stations acknowledge; allergy is emphasized; expo marks ready; waiter confirms served; manager handles one authorized amendment.

****Pass:**** Order, tickets and customer-facing statuses remain separate and consistent; unauthorized state jumps fail; no M4 check or settlement dependency exists.

GJ-06 - Split bill with separate tips

****Milestone:** M4**
****Mandatory:** Yes**
****Predecessors:** At least one M3 served order**
****Personas:** Two payers, cashier**

****Steps:**** Split check by item; payer A pays cash and chooses no tip; payer B pays external card terminal and adds a custom tip; print one physical receipt for each payer through the minimum M4 production printer path.

****Pass:**** Bill allocations equal bill total; each payment has independent bill and tip allocations; receipt shows bill, tip and total paid separately; no tip is selected by default. Each physical receipt is produced exactly once.

GJ-07 - Void, refund and tip correction controls

****Milestone:**** M4
****Mandatory:**** Yes
****Predecessors:**** GJ-06
****Personas:**** Cashier, manager

****Steps:**** Attempt cashier self-approval and fail; manager performs purpose-specific step-up; reverse one payment allocation; partially refund bill and tip separately; reconcile the final receipt and print the corrected physical receipt with reprint/correction audit.

****Pass:**** Maker-checker, exact independent bill/tip corrections, reason codes, one-time approval and append-only audit evidence are preserved. The corrected physical receipt is marked and auditable.

GJ-08 - Same QR customer service during internet outage

****Milestone:**** M5b
****Mandatory:**** Yes
****Predecessors:**** M5a approved
****Personas:**** Guest customer, waiter, kitchen, cashier

****Steps:**** Use a real per-outlet public hostname and valid public-CA certificate; disconnect internet; connect customer to restaurant Wi-Fi; scan the same QR; browse cached menu; order; call waiter; prepare/serve; request bill; settle using an available local live method; print receipt; repeat the same-QR entry on a device that joined outlet Wi-Fi holding a cached public DNS answer, on a device with encrypted DNS (Private DNS/DoH) enabled, and on a dual-stack IPv4/IPv6 device.

****Pass:**** Browser shows no certificate warning; local service completes without cloud; IDs and evidence persist for later synchronization; cloud-side new writes are blocked after lease expiry; cached-answer, encrypted-DNS and dual-stack devices either resolve to the trusted local endpoint or fail safely to translated staff guidance, never to a certificate warning or a manual bypass prompt; the served certificate fingerprint and expiry are verified from the LAN.

GJ-09 - Asymmetric partition and emergency authority replacement

****Milestone:**** M5b
****Mandatory:**** Yes
****Predecessors:**** M5a approved
****Personas:**** Outlet operator, platform operator, guest customer

****Steps:**** Cut cloud-to-node and node-to-cloud paths independently; verify challenge/ack failures and 5/10/20 timing; keep LAN node serving; request replacement; record independent approval; power off or network-isolate old node; run LAN-unreachability probe; issue next monotonic signed authority sequence; activate replacement; attempt direct LAN write to old node; reconnect stale node.

****Pass:**** Cloud forwarding expires safely while LAN authority continues; replacement is not writable before fence evidence; direct old-node LAN write fails; every writer rejects rollback; stale events quarantine; recovery requires three valid bidirectional proofs.

GJ-10 - Outlet node durability, synchronization and printing

****Milestone:**** M5a
****Mandatory:**** Yes

****Predecessors:**** M4 approved
****Personas:**** Waiter, kitchen, cashier, outlet operator

****Steps:**** Use staff/POS/KDS endpoints directly on the outlet network; create local session/order/check/payment/tip and print jobs; restart browser, API, worker and node database; replay outbox/inbox with parent-before-child ordering; inject one conflict; reconnect to cloud. Queue a customer receipt, lose internet, restart the local print service, recover the queue, print exactly once, reconnect and reconcile the print-job status.

****Pass:**** Local records and queues survive restart; print retries do not create unmarked duplicates; sync is idempotent and dependency ordered; conflict is visible; no same-QR public hostname, browser TLS or authority replacement claim is required. Receipt printing survives outage and restart through the M5a durable local queue with no lost or duplicate physical output.

GJ-11 - Backup and destructive restore

****Milestone:**** M6
****Mandatory:**** Yes
****Predecessors:**** M5b approved
****Personas:**** Platform operator, outlet operator

****Steps:**** Create cloud and outlet backups from built non-root images using production roles; destroy disposable environments; restore database, configuration, queues, certificates and print evidence; start exact production services.

****Pass:**** Post-restore M3/M4/M5a business slices pass; tenant/outlet isolation, audit, synchronization and recovery time are evidenced; missing grant/script/client makes the drill fail.

GJ-12 - Clean deployment and production-role readiness

****Milestone:**** M6
****Mandatory:**** Yes
****Predecessors:**** M5b approved
****Personas:**** Platform operator, support operator

****Steps:**** Delete generated output and databases; execute ordinary Windows and Linux commands; build production images; start API, workers, outlet node and print agent using least-privileged roles; inspect routes, jobs, users and health.

****Pass:**** All discovered tests run with zero skips; images are non-root and complete; readiness is thorough; no deferred route/table/worker exists; command order does not alter artifacts.

GJ-13 - Second-tenant commercial isolation

****Milestone:**** M6
****Mandatory:**** Yes
****Predecessors:**** M5b approved
****Personas:**** Two tenant administrators, platform support

****Steps:**** Configure two differently branded tenants with sibling outlets, menus, staff, tables, devices and reports; attempt cross-tenant/outlet reads and writes; use time-bound support access; export permitted Phase 1 operational evidence only.

****Pass:**** Branding/configuration are independent; production-role isolation blocks every unauthorized

d CRUD path; support access is time-bound and audited; no Phase 2 portability product is activated.

State Machines

****12 canonical machines****

- Package: 'Hospitality_OS_Phase_1_Clean_Build_Package_v2.0.9'
- Canonical source root: 'cd6b8cfc2175004cc2057deb7d9de1ad0d6264a23c82f00e15e96faa526c0c96'
- Status: ****Package M0 candidate - M0R and implementation are not authorized****

SM-TABLE-SESSION - Table Session

****Phase:** Phase 1**

****States:** pending, open, service_active, bill_requested, settling, closed, cancelled, transferred**

****Transitions****

- pending -> open: valid QR/session creation
- open -> service_active: first accepted order or service request
- service_active -> bill_requested: customer/staff requests bill
- bill_requested -> service_active: bill request withdrawn before presentation
- bill_requested -> settling: check presented
- settling -> closed: all checks settled and service complete
- open/service_active -> transferred: authorized table move
- transferred -> open/service_active: destination confirmed
- pending/open -> cancelled: authorized cancellation without accepted commerce

****Invariants****

- Exactly one tenant and outlet scope.
- Opaque QR never exposes sequential table ID.
- Closing a session does not delete carts, orders, checks or payments.
- Participant tokens cannot cross sessions or outlets.

SM-CART - Cart

****Phase:** Phase 1**

****States:** active, validating, submitted, expired, abandoned**

****Transitions****

- active -> validating: submit requested
- validating -> active: validation error
- validating -> submitted: one order created/idempotent prior outcome
- active -> expired: session/policy expiry
- active -> abandoned: customer intentionally abandons

****Invariants****

- Cart ownership is participant/shared-table explicit.
- Submission revalidates price, availability, translations and modifiers.
- A submitted cart maps to exactly one order outcome.
- Language switching does not lose cart identity.

SM-ORDER - Order

Phase: Phase 1

States: draft, submitted, awaiting_confirmation, accepted, in_fulfillment, partially_ready, ready, partially_served, served, completed, cancelled, voided, on_hold

Transitions

- draft -> submitted: submit
- submitted -> awaiting_confirmation: policy requires staff confirmation
- submitted/awaiting_confirmation -> accepted: accepted
- accepted -> in_fulfillment: tickets released
- in_fulfillment -> partially_ready: some lines ready
- in_fulfillment/partially_ready -> ready: all lines ready
- ready -> partially_served: some lines served
- ready/partially_served -> served: all lines served
- served -> completed: operational and financial conditions satisfied
- submitted/accepted -> cancelled: authorized pre-fulfillment cancellation
- accepted -> on_hold: authorized hold
- on_hold -> accepted: release
- completed -> voided: authorized correction workflow

Invariants

- Accepted price/tax/service snapshots are immutable.
- Order, fulfillment, check, payment and tip states are separate.
- Changes preserve an event timeline.
- A local/cloud retry cannot create a second order.

SM-FULFILLMENT-TICKET - Kitchen/Bar/Expo Ticket

Phase: Phase 1

States: queued, acknowledged, held, preparing, partially_completed, ready, collected, completed, rework, cancelled, exception

Transitions

- queued -> acknowledged: station accepts
- acknowledged -> held: course/capacity hold
- held/acknowledged -> preparing: fire/start
- preparing -> partially_completed: some units ready
- preparing/partially_completed -> ready: all units ready
- ready -> collected: waiter/runner collects
- collected -> completed: served/handoff confirmed
- ready -> rework: quality issue
- rework -> preparing: remake
- queued -> cancelled: upstream cancellation

- preparing -> exception: unavailable/equipment/safety issue

****Invariants****

- Ticket quantities cannot exceed accepted order-line quantities.
- Allergy flags and customer notes persist through routing.
- Rework is reasoned and audited; Phase 1 does not post inventory consumption.
- State transitions are enforced server/database side.

SM-SERVICE-REQUEST - Customer Service Request

****Phase:**** Phase 1

****States:**** new, routed, acknowledged, in_progress, completed, cancelled, expired, escalated, unresolved

****Transitions****

- new -> routed: routing rule
- routed -> acknowledged: staff accepts
- acknowledged -> in_progress: work begins
- in_progress -> completed: outcome recorded
- routed -> escalated: SLA exceeded
- escalated -> acknowledged: alternate accepts
- new -> cancelled: customer withdraws
- routed -> expired: session closes/policy
- in_progress -> unresolved: reason recorded

****Invariants****

- Every active request has an accountable queue/assignee.
- Acknowledgement and completion timestamps are retained.
- Duplicate taps are deduplicated without hiding deliberate repeat requests.
- Customer status uses the session language.

SM-CHECK - Check / Bill

****Phase:**** Phase 1

****States:**** draft, open, presented, partially_paid, paid, issued, voided, credited, written_off

****Transitions****

- draft -> open: line allocation validated
- open -> presented: bill shown/printed
- presented -> partially_paid: bill allocation below balance
- presented/partially_paid -> paid: bill balance fully settled
- paid -> issued: final receipt/fiscal outcome
- issued -> voided: authorized legal/operational void
- issued -> credited: linked credit/refund correction
- presented -> written_off: authorized disposition

****Invariants****

- Order-line quantities cannot be billed twice.
- Bill balance excludes optional tip.
- Issued records are corrected, not overwritten.
- Split/merge preserves original line ownership and audit.

SM-PAYMENT - Payment

****Phase:**** Phase 1

****States:**** created, pending, proof_pending, authorized, captured, partially_allocated, allocated, failed, cancelled, reversed, partially_refunded, refunded, reconciliation_required, reconciled

****Transitions****

- created -> pending: tender initiated
- pending -> proof_pending: Telebirr/CBE Birr proof submitted
- proof_pending -> captured: staff verifies provider receipt and records provider/reference evidence
- proof_pending -> failed: proof rejected, expired or cannot be verified
- pending -> authorized: external terminal authorization recorded
- authorized -> captured: terminal completion recorded
- pending -> captured: cash accepted or terminal returns combined success
- pending -> failed: provider or operational failure
- pending/proof_pending/authorized -> cancelled: authorized cancellation before capture
- captured -> partially_allocated: some bill/tip allocation posted
- captured/partially_allocated -> allocated: all exact allocations posted
- captured/partially_allocated/allocated -> reversed: approved full reversal
- allocated -> partially_refunded: approved refund below captured total
- allocated -> refunded: approved full refund
- partially_refunded -> partially_refunded: additional partial refund below captured total
- partially_refunded -> refunded: cumulative refunds equal captured total
- captured/partially_allocated/allocated/reversed/partially_refunded/refunded/failed/cancelled -> reconciliation_required: provider, cash or allocation mismatch detected
- reconciliation_required -> captured: verified missed capture
- reconciliation_required -> allocated: verified allocation correction
- reconciliation_required -> reversed: verified reversal
- reconciliation_required -> refunded: verified full refund
- reconciliation_required -> failed: verified failure
- captured/allocated/reversed/refunded/failed -> reconciled: evidence matches final commercial outcome

****Invariants****

- Bill and tip allocations are separate exact values.
- Raw card data is never stored.
- A payment retry is idempotent and provider/reference uniqueness is enforced.
- Telebirr/CBE Birr remains proof_pending until staff verifies receipt in the provider application.
- Unverified proof, screenshot or customer claim cannot produce a paid check or live receipt.
- Direct provider simulators cannot be labelled or reported as live pilot payments.
- Offline cash/local-terminal/mobile-proof records cannot be falsely labelled online-provider success.
- Every terminal state records actor, method, provider/reference evidence and reason where applicable.

SM-TIP - Optional Tip

****Phase:** Phase 1**

****States:**** not_selected, proposed, accepted, payment_pending, recorded, settled, failed, cancelled, partially_refunded, refunded, reversed

****Transitions****

- not_selected -> proposed: customer opens tip choice
- proposed -> not_selected: customer declines or clears selection
- proposed -> accepted: explicit amount or percentage confirmation
- accepted -> payment_pending: linked payment intent created
- payment_pending -> recorded: linked payment captured and separate tip allocation created
- payment_pending -> failed: linked payment fails or proof is rejected
- payment_pending -> cancelled: linked payment is cancelled before capture
- recorded -> settled: linked payment allocations are final
- recorded/settled -> partially_refunded: approved tip refund below recorded tip
- recorded/settled -> refunded: approved full tip refund
- partially_refunded -> partially_refunded: additional partial refund below recorded tip
- partially_refunded -> refunded: cumulative refunds equal recorded tip
- recorded/settled/partially_refunded -> reversed: linked payment is fully reversed before completed refund treatment

****Invariants****

- No tip is selected by default.
- Tip is never required to settle the bill.
- Tip does not change order lines, bill allocation, tax, service-charge presentation or sales totals
- .
- Every tip is linked to one payer/payment and separately auditable.
- A failed or cancelled linked payment cannot create a recorded or settled tip.
- Payment reversal/refund creates the corresponding tip reversal/refund path; tip outcome cannot remain silently settled.
- Tip refunds and reversals require purpose-specific approval, reason and exact cumulative limits.

SM-CASH-SHIFT - Cash Shift

****Phase:** Phase 1**

****States:**** planned, open, counting, submitted, approved, variance_review, closed, reopened

****Transitions****

- planned -> open: float accepted
- open -> counting: close initiated
- counting -> submitted: count submitted
- submitted -> approved: within policy/manager approves
- submitted -> variance_review: variance threshold exceeded
- variance_review -> approved: resolved
- approved -> closed: close posted
- closed -> reopened: exceptional approved correction
- reopened -> counting: corrected recount initiated under linked correction evidence
- reopened -> variance_review: correction requires variance adjudication
- reopened -> closed: authorized correction posted with maker-checker evidence and no recount required

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****Invariants****

- Opening float, bill cash, tip cash, payouts, expected cash and actual cash are separate values.
- Cashier cannot self-approve restricted variance.
- Corrections use linked movements/reopen evidence.
- Offline shift survives restart and sync.
- reopened is never terminal; every reopened shift returns to closed through recount and approval or an audited maker-checker correction.

SM-OUTLET-CONNECTIVITY - Outlet Connectivity and Authority

****Phase:** Phase 1**

****States:**** standby, authority_activating, connected, degraded, local_continuity, reconciling, replacement_pending, fence_verified, blocked, maintenance

****Transitions****

- connected -> degraded: one or more bidirectional reachability proofs fail and 10-second threshold is reached
- degraded -> local_continuity: cloud forwarding lease expires at 20 seconds and local readiness passes
- degraded -> connected: three consecutive valid bidirectional proofs before lease expiry
- local_continuity -> reconciling: three consecutive valid proofs plus protocol/cursor compatibility
- reconciling -> connected: queues acknowledged and highest authority sequence confirmed
- connected/degraded/local_continuity -> replacement_pending: step-up and independent replacement approval recorded
- replacement_pending -> fence_verified: old node power-off or network isolation evidence plus LAN-unreachability probe passes
- replacement_pending -> connected/local_continuity: replacement cancelled before new sequence issuance
- connected/degraded/local_continuity/reconciling/replacement_pending/fence_verified -> blocked: node identity, version, storage, security or authority invalid
- connected/local_continuity/blocked -> maintenance: authorized maintenance
- maintenance -> connected: readiness passes under highest accepted authority sequence
- fence_verified -> blocked: the superseded node instance is fenced and its authority sequence revoked
- standby -> authority_activating: fence evidence for the superseded instance passed, independent approval recorded, and the next durable monotonic signed authority sequence is issued to this instance
- authority_activating -> connected: readiness, identity, storage and protocol checks pass, cloud and local writers confirm the highest accepted sequence, and valid bidirectional proofs resume; EVT-AUTHORITY-REPLACEMENT-ACTIVATED is emitted
- authority_activating -> local_continuity: readiness checks pass while the cloud remains unreachable; this instance becomes the writable LAN authority and EVT-AUTHORITY-REPLACEMENT-ACTIVATED is emitted
- authority_activating -> blocked: readiness, identity, storage, signature or sequence validation fails; the instance never becomes writable

****Invariants****

- A valid reachability proof is a signed bidirectional challenge/acknowledgement confirming both directions, authority sequence, protocol compatibility and cursor posture.

- Cloud blocks new dine-in submissions after lease expiry but never revokes the LAN node local authority.
- Emergency replacement cannot become writable until old-node physical or network fence evidence and an automated LAN-unreachability probe pass.
- Authority sequence is durable, monotonic, signed and anti-rollback; every writer persists and compares the highest accepted sequence.
- After replacement, a direct LAN write to the old node fails and stale events quarantine.
- Transitions, approvals, fence evidence, probes and operator overrides are audited.
- A node instance in standby or authority_activating is never writable.
- At most one node instance per outlet is writable at any time; activation of a replacement requires the superseded instance to be fenced and blocked first.
- EVT-AUTHORITY-REPLACEMENT-ACTIVATED is emitted only on a transition out of authority_activating in to a writable state.

SM-SYNC-EVENT - Local/Cloud Synchronization Event

****Phase:**** Phase 1

****States:**** pending, sending, acknowledged, retrying, blocked, conflict, quarantined, superseded

****Transitions****

- pending -> sending: worker sends
- sending -> acknowledged: peer accepts/idempotent prior outcome
- sending -> retrying: transient failure
- retrying -> sending: backoff elapsed
- sending -> blocked: dependency/version/authority unavailable
- sending -> conflict: domain/version conflict
- sending -> quarantined: invalid/security/nonrecoverable
- conflict -> superseded: authorized resolution
- blocked -> pending: dependency resolved

****Invariants****

- Event ID and command idempotency prevent duplicate effect.
- Parent/dependency order is explicit.
- No silent last-write-wins for order, bill, payment, tip, cash or permission data.
- A bad event cannot permanently block unrelated streams.

SM-PRINT-JOB - Local Print Job

****Phase:**** Phase 1

****States:**** queued, claimed, printing, printed, retrying, failed, cancelled, superseded

****Transitions****

- queued -> claimed: print agent claims
- claimed -> printing: device write begins
- printing -> printed: acknowledgement/evidence
- claimed/printing -> retrying: transient failure
- retrying -> claimed: retry due
- retrying -> failed: limit exceeded
- queued -> cancelled: safe upstream cancellation

- failed -> superseded: authorized replacement/reprint

****Invariants****

- A job has a deterministic deduplication key.
- Reprint is attributed and cannot silently duplicate an original receipt.
- Printer failure is visible to staff.
- Jobs persist through node restart and internet outage.

Phase Boundaries and M5 Ownership

- Package: 'Hospitality_OS_Phase_1_Clean_Build_Package_v2.0.9'
- Canonical source root: 'cd6b8cfc2175004cc2057deb7d9de1ad0d6264a23c82f00e15e96faa526c0c96'
- Status: ****Package M0 candidate - M0R and implementation are not authorized****

Phase 1 active scope

- QR dine-in
- English/Amharic/Arabic
- waiter
- KDS
- counter POS
- bill
- separate optional tip
- cash
- external terminal recording
- verified Telebirr/CBE Birr proof confirmation
- receipt
- cash shift
- local Wi-Fi continuity
- printing
- operational reports

Excluded scope

- pickup
- delivery
- loyalty
- CRM
- purchasing
- inventory
- accounting
- HR/workforce
- operational recipes
- costing
- intelligence
- supplier/Horeca runtime
- Phase 2 data portability product

Repository boundary

- ****Package M0:**** documents only
- ****M0R:**** empty repository, docs, plans, CI/scanner design; no database/schema/executable migration/application code
- ****M1:**** create PostgreSQL database and executable migration 0001 after M0R approval/adjudication

M5a ownership

****Requirements:**** FR-CFG-001E, FR-DAT-008C, FR-EDG-001, FR-EDG-002A, FR-EDG-004A, FR-EDG-005, FR-EDG-008, FR-EDG-009, FR-EDG-010, FR-EDG-012, FR-EDG-015A, FR-EDG-016, FR-EDG-017, FR-EDG-018, FR-EDG-025, FR-EDG-027, FR-EDG-029, FR-INT-003, FR-INT-004, FR-OPS-010, FR-OPS-018, FR-POS-008, FR-SEC-002B, FR-TST-008

****Journeys:**** GJ-10

****Services:**** outlet_api, outlet_postgresql, sync_worker, realtime_gateway, print_agent

****Forbidden Claims:**** same_qr_public_hostname, browser_public_tls, cloud_forwarding_lease, emergency_authority_replacement

****Exact service boundary:**** exactly five services; 'local_backup_agent' is excluded. Backup scheduling and destructive restore remain M6 obligations.

M5b ownership

****Requirements:**** FR-EDG-003, FR-EDG-004B, FR-EDG-015B, FR-EDG-021, FR-EDG-022A, FR-EDG-022B, FR-EDG-022C, FR-EDG-023, FR-EDG-024, FR-EDG-026, FR-EDG-028, FR-OPS-017

****Journeys:**** GJ-08, GJ-09

****Services:**** public_dns, outlet_split_horizon_dns, dns01_certificate_service, authority_control_plane, reachability_challenge_service, fence_evidence_service

****Forbidden Claims:**** replacement_writable_without_fence, one_way_heartbeat_as_healthy, self_signed_browser_bypass, shared_cross_outlet_private_key, writable_cloud_fallback

Test, Evidence and Validator Strategy

- Package: 'Hospitality_OS_Phase_1_Clean_Build_Package_v2.0.9'
- Canonical source root: 'cd6b8cfc2175004cc2057deb7d9de1ad0d6264a23c82f00e15e96faa526c0c96'
- Status: ****Package M0 candidate - M0R and implementation are not authorized****

Validator status

- Frozen validator script SHA-256: 'ee65b8ec3292db19798f785e23a2d54969eadd91f00e98888520d16051c9f7b0',
- Validator package SHA-256: '628ae551120497280a82fdcf8fd8bd2a69e4c893b400cd94d1914529788cce9'
- Internal planted mutations: ****10/10 detected****
- Blind Round 1: ****23/24 detected before repair****
- Blind Round 2: ****23/28 detected before repair****
- Disclosed Round 2 misses after repair: ****5/5 detected diagnostically****

The post-repair reruns are diagnostic, not fresh independent evidence. The agreed two-round cap is closed; Codex receives the limitation explicitly.

Negative controls

- **NC-M0-001 (M0):** Package pinning - deliberate break: Change one controlled file after checksum generation - expected: 'CHECKSUM_MISMATCH'
- **NC-M0-002 (M0):** Requirement recount - deliberate break: Remove or duplicate one requirement row - expected: 'REQUIREMENT_COUNT_OR_DUPLICATE'
- **NC-M0-003 (M0):** Phase boundary - deliberate break: Plant one forbidden positive obligation for every excluded domain in the generated vocabulary (pickup, delivery, loyalty, CRM, purchasing/procurement, inventory/storage location, accounting/general ledger, HR/payroll, operational recipes, costing/variance, intelligence/forecasting, supplier runtime and portability product) into a Phase 1 fixture - expected: 'FORBIDDEN_PHASE1_SURFACE_PER_DOMAIN'
- **NC-M0-005 (M0):** Forbidden-surface vocabulary completeness - deliberate break: Remove one excluded domain from the generated forbidden-surface vocabulary while leaving it excluded in phase_boundaries - expected: 'EXCLUSION_SET_NOT_COVERED'
- **NC-M1-001 (M1):** Fail-closed tenant context - deliberate break: Run production role with tenant/outlet context unset - expected: 'VISIBLE_OR_WRITABLE_ROWS_WITHOUT_CONTEXT'
- **NC-M1-002 (M1):** Sibling-outlet isolation - deliberate break: Soften one outlet policy to tenant-only - expected: 'SIBLING_OUTLET_ACCESS'
- **NC-M1-003 (M1):** Future schema protection - deliberate break: Add outlet_id to an existing tenant table without policy upgrade - expected: 'OUTLET_POLICY_NOT_UPGRADED'
- **NC-M1-004 (M1):** Runtime least privilege - deliberate break: Configure API/worker with owner or BYPASSRLS role - expected: 'PRIVILEGED_RUNTIME_ROLE_REJECTED'
- **NC-M2-001 (M2):** Opaque QR - deliberate break: Replace opaque token with enumerable table number - expected: 'ENUMERABLE_QR_REFERENCE'
- **NC-M2-002 (M2):** Session scope - deliberate break: Forge participant token for another outlet/table - expected: 'FOREIGN_SESSION_ACCEPTED'
- **NC-M2-003 (M2):** Safety translation publication - deliberate break: Remove required Arabic allergen translation - expected: 'REQUIRED_SAFETY_TRANSLATION_MISSING'
- **NC-M2-004 (M2):** Arabic RTL depth - deliberate break: Disable dir=rtl/logical CSS or inject mixed Latin SKU string - expected: 'RTL_LAYOUT_OR_READING_ORDER_FAILURE'
- **NC-M3-001 (M3):** Order idempotency - deliberate break: Repeat submit with same idempotency key - expected: 'DUPLICATE_ORDER_EFFECT'
- **NC-M3-002 (M3):** Price snapshot - deliberate break: Change menu price between display and submit without revalidation - expected: 'STALE_PRICE_ACCEPTED'
- **NC-M3-003 (M3):** Allergy propagation - deliberate break: Drop allergy flag before KDS ticket - expected: 'ALLERGY_FLAG_LOST'
- **NC-M3-004 (M3):** State enforcement - deliberate break: Jump accepted order directly to served - expected: 'ILLEGAL_TRANSITION_ACCEPTED'
- **NC-M4-001 (M4):** No tip default - deliberate break: Preselect a suggested tip - expected: 'TIP_PRESELECTED'
- **NC-M4-002 (M4):** Bill/tip separation - deliberate break: Add tip to taxable bill balance - expected: 'TIP_COMMINGLED_WITH_BILL'
- **NC-M4-003 (M4):** Payment truth and live/simulated boundary - deliberate break: Mark unverified Telebirr/CBE Birr proof paid or label a direct-provider simulator as live - expected: 'UNVERIFIED_OR_SIMULATED_PAYMENT_CLAIM'
- **NC-M4-004 (M4):** Maker-checker - deliberate break: Allow cashier to approve own refund/tip reversal - expected: 'SELF_APPROVAL_ACCEPTED'
- **NC-M4-005 (M4):** Amharic settlement and receipt coverage - deliberate break: Remove the packaged Ethiopic font from the receipt print path or fall back to a substitute glyph set - expected: 'ETHIOPIC_FONT_FALLBACK_ON_RECEIPT'

- **NC-M4-006 (M4):** Reopened cash shift closure - deliberate break: Leave a reopened cash shift with no recount, approval or maker-checker correction path and attempt to report it as closed - expected: 'REOPENED_SHIFT_NOT_RESOLVED'
- **NC-M5A-001 (M5a):** Sync dependency order - deliberate break: Upload child before unacknowledged parent - expected: 'CHILD_APPLIED_BEFORE_PARENT'
- **NC-M5A-002 (M5a):** Sync idempotency - deliberate break: Replay order/payment/tip event - expected: 'DUPLICATE_COMMERCIAL_EFFECT'
- **NC-M5A-003 (M5a):** Print idempotency - deliberate break: Retry acknowledged print job - expected: 'DUPLICATE_UNMARKED_PRINT'
- **NC-M5A-004 (M5a):** Durability - deliberate break: Restart browser/node with accepted local order queued - expected: 'LOCAL_RECORD_LOST'
- **NC-M5B-001 (M5b):** Public-trust TLS - deliberate break: Use self-signed, wrong-host or expired certificate - expected: 'TLS_NOT_PUBLICLY_TRUSTED'
- **NC-M5B-002 (M5b):** Bidirectional cloud forwarding lease - deliberate break: Permit cloud write after 20-second lease expiry or accept a one-way reachability proof - expected: 'CLOUD_WRITE_AFTER_LEASE_EXPIRY_OR_ONE_WAY_PROOF'
- **NC-M5B-003 (M5b):** Asymmetric LAN authority - deliberate break: Stop LAN writes merely because the cloud forwarding lease expires - expected: 'LAN_AUTHORITY_REVOKED_BY_CLOUD_OUTAGE'
- **NC-M5B-004 (M5b):** Emergency replacement fencing and monotonic sequence - deliberate break: Activate replacement without recorded power/network fence and LAN-unreachability proof, or allow a direct LAN write to the old node after replacement - expected: 'OLD_NODE_STILL_WRITABLE_OR_REPLACEMENT_UNFENCED'
- **NC-M5B-005 (M5b):** Cloud-to-node reachability direction - deliberate break: Allow node-to-cloud heartbeat traffic but block cloud challenge delivery while treating the lease as valid - expected: 'ONE_WAY_CLOUD_TO_NODE_ACCEPTED'
- **NC-M5B-006 (M5b):** Node-to-cloud reachability direction - deliberate break: Allow cloud challenge delivery but block node acknowledgement while treating the lease as valid - expected: 'ONE_WAY_NODE_TO_CLOUD_ACCEPTED'
- **NC-M5B-007 (M5b):** Certificate lifecycle - deliberate break: Allow the per-outlet certificate to pass its renewal threshold without renewal, escalation or installation evidence - expected: 'CERTIFICATE_RENEWAL_NOT_ENFORCED'
- **NC-M5B-008 (M5b):** Same-QR resolution under real client conditions - deliberate break: Join outlet Wi-Fi holding a cached public DNS answer, and separately enable encrypted DNS (Private DNS/DoH) that bypasses outlet split DNS - expected: 'SAME_QR_RESOLUTION_UNSAFE_OR_UNGUIDED'
- **NC-M5B-009 (M5b):** Replacement activation gate - deliberate break: Attempt to make a standby node writable directly from standby without passing authority activating readiness and sequence confirmation - expected: 'REPLACEMENT_WRITABLE_WITHOUT_ACTIVATION'
- **NC-M6-001 (M6):** Zero-skip CI - deliberate break: Skip one discovered acceptance test - expected: 'SKIPPED_TEST_DETECTED'
- **NC-M6-002 (M6):** Clean build - deliberate break: Leave stale route artifact before build - expected: 'STALE_ARTIFACT_DETECTED'
- **NC-M6-003 (M6):** Ordinary Windows - deliberate break: Require hidden PG_BIN/PATH injection - expected: 'HIDDEN_ENV_DEPENDENCY'
- **NC-M6-004 (M6):** Production image completeness - deliberate break: Remove an advertised runtime script or required PostgreSQL client from the production image - expected: 'REQUIRED_ARTIFACT_MISSING'
- **NC-M6-005 (M6):** Restore realism - deliberate break: Run restore smoke with owner instead of production role - expected: 'NON_PRODUCTION_ROLE_USED'
- **NC-M0-011 (M0):** Accounting alias leakage - deliberate break: Plant a Phase 1 journal entry or ledger-posting obligation using singular/plural aliases. - expected: 'FORBIDDEN_ACCOUNTING_OBLIGATION'
- **NC-M0-012 (M0):** CRM campaign alias leakage - deliberate break: Plant a generic campaign correlation field without the phrase marketing campaign. - expected: 'FORBIDDEN_CRM_OBLIGATION'
- **NC-M0-013 (M0):** Workforce alias leakage - deliberate break: Plant roster, attendance, timekeep

ing or break-record persistence in service presence. - expected: 'FORBIDDEN_WORKFORCE_OBLIGATION'
- **NC-M0-014 (M0):** Canonical projection parity - deliberate break: Change a Source of Truth decision cell without changing the canonical decision record. - expected: 'CANONICAL_PROJECTION_MISMATCH'

Package M0 questions

1. Can every active requirement be traced to a milestone, journey or explicit engineering test?
2. Are any Phase 2/3 or deferred extension capabilities accidentally required by Phase 1?
3. Can the same table QR safely resolve to cloud or outlet node without browser-security warnings?
4. Is the outlet authority model sufficient to prevent split-brain, including emergency replacement?
5. Are local/cloud identifiers, idempotency, authority sequence and dependency ordering specified clearly?
6. Are bill, tip, payment, service charge and tax structurally separate?
7. Does Arabic RTL receive executable mixed-direction, ETB, numeral and receipt coverage rather than string substitution?
8. Are tenant/outlet boundaries enforceable using exact production roles?
9. Can backup/restore and production images be tested without host-only dependencies?
10. Can every CI job fail honestly when tests are skipped, undiscovered, stale or unsupported?
11. Is the build graph executable milestone-by-milestone using only approved predecessors?
12. Is any v1.1 or deferred-module assumption silently retained?
13. Are 5-second proofs, 10-second degradation, 20-second lease expiry, three-proof recovery and asymmetric LAN service independently specified and testable?
14. Are cash, external-terminal recording and verified Telebirr/CBE Birr proof confirmation live pilot paths while direct provider APIs remain simulator-only?

Build Control Plan

- Package: 'Hospitality_OS_Phase_1_Clean_Build_Package_v2.0.9'
- Canonical source root: 'cd6b8cfc2175004cc2057deb7d9de1ad0d6264a23c82f00e15e96faa526c0c96'
- Status: **Package M0 candidate - M0R and implementation are not authorized**

Control principles

1. Build one milestone at a time.
2. Do not start a later milestone before the current gate is approved or adjudicated under FR-GOV-004.
3. Every route, table, worker, screen and test maps to an active requirement and gate.
4. Phase 2/3 surfaces are physically absent.
5. Reused prototype code is reviewed as third-party code.
6. Production-role tests, not owner/superuser tests, prove isolation.
7. Generated projections are rebuilt from canonical sources; manual edits are prohibited.

Repository start rule

M0R creates an empty repository containing only the approved Package M0 documents, conformance plans and CI/scanner design. No database, schema, executable migration, application route, worker or UI is permitted.

Migration start rule

PostgreSQL and migration '0001' begin at M1 after M0R approval or adjudication. No v1.1 migration is

```

imported.

## Requirement ownership by gate

- **M0:** 2 requirements - Platform & Security: 1; Quality Engineering: 1
- **M0R:** 30 requirements - Platform & Security: 11; Service Execution: 6; Quality Engineering: 6;
Customer Experience: 4; Edge & Operations: 1; Commerce & Payments: 1; Platform Engineering: 1
- **M1:** 77 requirements - Platform & Security: 57; Edge & Operations: 9; Quality Engineering: 9; C
ustomer Experience: 2
- **M2:** 49 requirements - Customer Experience: 37; Platform Engineering: 9; Platform & Security: 2
; Quality Engineering: 1
- **M3:** 75 requirements - Service Execution: 41; Customer Experience: 18; Platform Engineering: 9;
Edge & Operations: 3; Platform & Security: 2; Quality Engineering: 2
- **M4:** 55 requirements - Commerce & Payments: 19; Platform Engineering: 16; Quality Engineering:
8; Customer Experience: 4; Platform & Security: 3; Edge & Operations: 2; Service Execution: 2; Billi
ng & Payments: 1
- **M5a:** 24 requirements - Edge & Operations: 18; Platform & Security: 3; Outlet Edge & Operations
: 1; Customer Experience: 1; Quality Engineering: 1
- **M5b:** 12 requirements - Edge & Operations: 11; Edge & Continuity: 1
- **M6:** 12 requirements - Edge & Operations: 7; Quality Engineering: 3; Platform & Security: 2

```

Historical Disposition and Lineage Summary

```

- Package: 'Hospitality_OS_Phase_1_Clean_Build_Package_v2.0.9'
- Canonical source root: 'cd6b8cfc2175004cc2057deb7d9de1ad0d6264a23c82f00e15e96faa526c0c96'
- Status: **Package M0 candidate - M0R and implementation are not authorized**

```

The historical registers preserve all 500 original v1.1 requirements and 100 original decisions at row level. The active reconciled register preserves all 294 imported Phase 1 requirement IDs through retained rows or explicit split-successor lineage.

```

- Original imported Phase 1 requirements: 294
- Mapped originals: 294
- Split originals: 32
- New audit requirements: 2
- Behavior coverage attested: True

```

Split register

Original	Successors	Reason
FR-CFG-001	FR-CFG-001A, FR-CFG-001B, FR-CFG-001C, FR-CFG-001D, FR-CFG-001E	Independent behaviors have different introducing gates, materially different evidence, or combine positive runtime behavior with a separately testable negative phase-boundary control.
FR-CFG-002	FR-CFG-002A, FR-CFG-002B	Independent behaviors have different introducing gates, materially different evidence, or combine positive runtime behavior with a separately testable negative phase-boundary control.
FR-CFG-005	FR-CFG-005A, FR-CFG-005B	Independent behaviors have different introducing gates, materially different evidence, or combine positive runtime behavior with a separately testable negative phase-boundary control.
FR-CFG-007	FR-CFG-007A, FR-CFG-007B	Independent behaviors have different introducing gates, ma

[illegible]

terially different evidence, or combine positive runtime behavior with a separately testable negative phase-boundary control. |
FR-SEC-002	FR-SEC-002A, FR-SEC-002B	Independent behaviors have different introducing gates, materially different evidence, or combine positive runtime behavior with a separately testable negative phase-boundary control.
FR-SEC-010	FR-SEC-010A, FR-SEC-010B	Independent behaviors have different introducing gates, materially different evidence, or combine positive runtime behavior with a separately testable negative phase-boundary control.
FR-SRV-007	FR-SRV-007A, FR-SRV-007B	Independent behaviors have different introducing gates, materially different evidence, or combine positive runtime behavior with a separately testable negative phase-boundary control.
FR-TAB-007	FR-TAB-007A, FR-TAB-007B	Independent behaviors have different introducing gates, materially different evidence, or combine positive runtime behavior with a separately testable negative phase-boundary control.
FR-TEN-002	FR-TEN-002A, FR-TEN-002B	Independent behaviors have different introducing gates, materially different evidence, or combine positive runtime behavior with a separately testable negative phase-boundary control.
FR-TEN-009	FR-TEN-009A, FR-TEN-009B	Independent behaviors have different introducing gates, materially different evidence, or combine positive runtime behavior with a separately testable negative phase-boundary control.
FR-TST-004	FR-TST-004A, FR-TST-004B	Independent behaviors have different introducing gates, materially different evidence, or combine positive runtime behavior with a separately testable negative phase-boundary control.
FR-TST-005	FR-TST-005A, FR-TST-005B	Independent behaviors have different introducing gates, materially different evidence, or combine positive runtime behavior with a separately testable negative phase-boundary control.
FR-TST-007	FR-TST-007A, FR-TST-007B	Independent behaviors have different introducing gates, materially different evidence, or combine positive runtime behavior with a separately testable negative phase-boundary control.
FR-UX-001	FR-UX-001A, FR-UX-001B	Independent behaviors have different introducing gates, materially different evidence, or combine positive runtime behavior with a separately testable negative phase-boundary control.

Amendments and Findings

- Package: 'Hospitality_OS_Phase_1_Clean_Build_Package_v2.0.9'
- Canonical source root: 'cd6b8cfc2175004cc2057deb7d9de1ad0d6264a23c82f00e15e96faa526c0c96'
- Status: **Package M0 candidate - M0R and implementation are not authorized**

Amendments

AMD-REC-001 - Founder Approved Requirement Amendment

Source: reconciliation

Affected: FR-GOV-004, D-120

Old value: Do not begin a later milestone until the current gate has clean evidence and no unresolved P0/P1 defect. An adverse independent verdict pauses progression; disputes follow builder response, reviewer rebuttal and founder decision. Overrides remain recorded, name the affected requirement/rule and are re-examined at the next audit.

****New value:**** Do not begin a later milestone while any P0 remains unresolved, or while any P1 affects product scope, security, money, authority, milestone executability, acceptance completeness or canonical correctness. A P1 limited to publication, projection, wording, identifier or validator coverage may proceed only through recorded founder adjudication when canonical behavior is correct, runtime behavior cannot change, the defect remains visible and repair is scheduled. An adverse independent verdict pauses progression until the builder response, reviewer rebuttal and founder decision are recorded. Overrides name the affected requirement or rule and are re-examined at the next audit.

****Change:**** Old: Do not begin a later milestone until the current gate has clean evidence and no unresolved P0/P1 defect. An adverse independent verdict pauses progression; disputes follow builder response, reviewer rebuttal and founder decision. Overrides remain recorded, name the affected requirement/rule and are re-examined at the next audit.

New: Do not begin a later milestone while any P0 remains unresolved, or while any P1 affects product scope, security, money, authority, milestone executability, acceptance completeness or canonical correctness. A P1 limited to publication, projection, wording, identifier or validator coverage may proceed only through recorded founder adjudication when canonical behavior is correct, runtime behavior cannot change, the defect remains visible and repair is scheduled. An adverse independent verdict pauses progression until the builder response, reviewer rebuttal and founder decision are recorded. Overrides name the affected requirement or rule and are re-examined at the next audit.

****Reason:**** Align the requirement with the approved stopping rule: substantive P1 remains blocking, bounded publication/projection debt may proceed only through recorded adjudication.

****Verification:**** Exact wording is asserted by the standard-library validator.

****Residual risk:**** None.

AMD-REC-002 - Requirement Split

****Source:**** reconciliation

****Affected:**** FR-CFG-001, FR-CFG-002, FR-CFG-005, FR-CFG-007, FR-DAT-008, FR-EDG-002, FR-EDG-004, FR-EDG-015, FR-EDG-022, FR-FUL-016, FR-GOV-001, FR-I18N-001, FR-MNU-002, FR-ORD-001, FR-ORD-007, FR-ORD-012, FR-ORD-016, FR-ORD-019, FR-PAY-010, FR-POS-003, FR-POS-010, FR-RCP-008, FR-SEC-002, FR-SEC-010, FR-SRV-007, FR-TAB-007, FR-TEN-002, FR-TEN-009, FR-TST-004, FR-TST-005, FR-TST-007, FR-UX-001

****Old value:**** 32 original mixed-behavior requirements.

****New value:**** 72 successor requirements with full original lineage.

****Change:**** Old: 32 original mixed-behavior requirements.

New: 72 successor requirements with full original lineage.

****Reason:**** Different introducing gates, evidence or positive/negative phase-boundary behavior.

****Verification:**** Every original is mapped to all successors; active IDs are unique; no successor depends on a later gate.

****Residual risk:**** Human semantic coverage remains subject to Codex audit.

AMD-REC-003 - New Requirements

****Source:**** reconciliation

****Affected:**** FR-BIL-017, FR-EDG-029, D-119

****Old value:**** No requirement separated minimum M4 physical receipt printing from M5a print resilience.

****New value:**** M4 minimum physical receipt path and M5a durable local print resilience are separate requirements.

****Change:**** Old: No requirement separated minimum M4 physical receipt printing from M5a print resilience.

New: M4 minimum physical receipt path and M5a durable local print resilience are separate requirements.

****Reason:**** Resolve the v2.0.4 P0 milestone dependency.

****Verification:**** M4 journeys require real print; GJ-10 proves M5a recovery and deduplication.

****Residual risk:**** Implementation audit required.

AMD-REC-004 - Gate Reconciliation

****Source:**** reconciliation

****Affected:**** FR-EDG-003

****Old value:**** M5a local authority with M5b revalidation.

****New value:**** M5b introduction with M6 revalidation.

****Change:**** Old: M5a local authority with M5b revalidation.

New: M5b introduction with M6 revalidation.

****Reason:**** Authoritative write ownership depends on the M5b lease and fencing model; M5a provides local execution only.

****Verification:**** No M5a requirement depends on FR-EDG-003.

****Residual risk:**** None.

AMD-REC-005 - Gate Reconciliation

****Source:**** reconciliation

****Affected:**** FR-RPT-001, FR-RPT-002, FR-RPT-003, FR-RPT-004, FR-RPT-005, FR-RPT-013, FR-RPT-015

****Old value:**** M6 introduction.

****New value:**** M4 introduction, with concrete M5a or M6 revalidation where required.

****Change:**** Old: M6 introduction.

New: M4 introduction, with concrete M5a or M6 revalidation where required.

****Reason:**** Operational reports are a Phase 1 product capability and become executable when M4 completes settlement data; M6 is hardening, not first implementation.

****Verification:**** Report prerequisites exist by M4 and no later-gate prerequisite remains.

****Residual risk:**** None.

AMD-REC-006 - Security Contract Amendment

Source: reconciliation

Affected: FR-EDG-022A, D-116

Old value: Per-outlet key generation and documented custody.

New value: Node-generated and retained private key, CSR-only submission, certificate-chain return and no private-key export.

Change: Old: Per-outlet key generation and documented custody.

New: Node-generated and retained private key, CSR-only submission, certificate-chain return and no private-key export.

Reason: Resolve the v2.0.4 key-custody ambiguity.

Verification: Exact phrases are asserted by the validator.

Residual risk: Implementation keystore selection remains open.

AMD-REC-007 - Journey Amendment

Source: reconciliation

Affected: GJ-01A, GJ-01B, GJ-02B, GJ-03B, GJ-06, GJ-07, GJ-10

Old value: Ambiguous M3 authority wording and incomplete M4/M5a print ownership.

New value: Cloud/current authority at M3; real physical print at M4; durable print recovery at M5a.

Change: Old: Ambiguous M3 authority wording and incomplete M4/M5a print ownership.

New: Cloud/current authority at M3; real physical print at M4; durable print recovery at M5a.

Reason: Align journeys to the reconciled milestone contracts.

Verification: Journey text is asserted by the validator.

Residual risk: None.

AMD-008 - Journey and revalidation evidence alignment

Source: validator-build pre-freeze audit

Affected: FR-SEC-002A, FR-TST-006, FR-AUTH-003, FR-MNU-012, FR-RPT-001, FR-RPT-004, FR-RPT-005

Old value: Not separately recorded.

New value: Added missing later-gate revalidation/test evidence where a linked journey occurs after introduction, and removed journey links whose milestone precedes the reporting requirement's introducing gate.

Change: Added missing later-gate revalidation/test evidence where a linked journey occurs after introduction, and removed journey links whose milestone precedes the reporting requirement's introducing gate.

****Reason:**** A journey may prove a clause only at its introducing or an explicitly listed revalidation gate.

****Verification:**** Validator rule JOURNEY_GATE_ALIGNMENT and workbook projection parity.

****Residual risk:**** None.

AMD-009 - Blind Mutation Round 2 validator and projection hardening

****Source:**** Blind Mutation Pack Round 2

****Affected:**** canonical/reconciled_requirements.json, canonical/reconciled_decisions.json, Hospitality_OS_Reconciled_Register_v0.1.2.xlsx, validator/validate_hospitality_os.py

****Old value:**** Not separately recorded.

****New value:**** Added declared-count validation, supersession consistency validation, full decision-register workbook projection, component-path/classification workbook projection, and decision-level tip-separation validation.

****Change:**** Added declared-count validation, supersession consistency validation, full decision-register workbook projection, component-path/classification workbook projection, and decision-level tip-separation validation.

****Reason:**** Round 2 exposed five already-approved rules or projection contracts that the validator did not enforce.

****Verification:**** Baseline validation plus post-repair diagnostics for R2-BMP-S-12, R2-BMP-S-18, R2-BMP-P-14, R2-BMP-P-16 and R2-BMP-M-24.

****Residual risk:**** None.

AMD-V206-001 - M5a service inventory correction

****Source:**** Codex v2.0.5 Package M0 review

****Affected:**** FR-EDG-002A, m5_ownership.json, D-113

****Old value:**** M5a listed six services including local_backup_agent.

****New value:**** M5a lists exactly five required services; backup scheduling and destructive restore remain M6.

****Change:**** M5a lists exactly five required services; backup scheduling and destructive restore remain M6.

****Reason:**** Resolve Codex P1-01 and restore the exact five-service edge boundary.

****Verification:**** M5 ownership, requirements and publications agree; validator asserts exact service set.

****Residual risk:**** None.

AMD-V206-002 - Lifecycle-state reconciliation

****Source:**** Codex v2.0.5 Package M0 review
****Affected:**** residual_issues.json, finding_register.json

****Old value:**** Canonical records said blind mutations and validator freeze had not occurred.

****New value:**** Canonical lifecycle records show both blind rounds, frozen validator, v2.0.5 Codex review and v2.0.6 repair state.

****Change:**** Canonical lifecycle records show both blind rounds, frozen validator, v2.0.5 Codex review and v2.0.6 repair state.

****Reason:**** Resolve Codex P1-02.

****Verification:**** Lifecycle validator rejects stale next-step wording and unresolved historical states.

****Residual risk:**** None.

AMD-V206-003 - Uniform clause canonical contract

****Source:**** Codex v2.0.5 Package M0 review
****Affected:**** reconciled_requirements.json, reconciled_canonical_register.schema.json

****Old value:**** Ten clauses used substitute fields instead of exact_clause_text.

****New value:**** All 337 clauses use the same required exact_clause_text contract; substitute fields are removed.

****Change:**** All 337 clauses use the same required exact_clause_text contract; substitute fields are removed.

****Reason:**** Resolve Codex P1-03.

****Verification:**** Schema and validator require the uniform clause fields for every clause.

****Residual risk:**** None.

AMD-V206-004 - Lossless generated projections

****Source:**** Codex v2.0.5 Package M0 review
****Affected:**** generator, workbooks, human publications

****Old value:**** Amendments, findings, split reasons and 96 non-regression rules were incompletely projected.

****New value:**** Every canonical field is mapped; all split reasons and all 96 identified rules are published in Markdown and Excel.

****Change:**** Every canonical field is mapped; all split reasons and all 96 identified rules are published in Markdown and Excel.

****Reason:**** Resolve Codex P1-04.

****Verification:**** Field-level projection validator compares all rows and required fields.

****Residual risk:**** None.

AMD-V206-005 - Executable forbidden-surface rules

****Source:**** Codex v2.0.5 Package M0 review

****Affected:**** forbidden_surface_rules.json, validator

****Old value:**** Validator used hard-coded incomplete vocabulary and sense handling.

****New value:**** Validator loads the normative vocabulary, negation markers and sense exclusions; every excluded domain and permitted sense has planted tests.

****Change:**** Validator loads the normative vocabulary, negation markers and sense exclusions; every excluded domain and permitted sense has planted tests.

****Reason:**** Resolve Codex P1-05.

****Verification:**** Internal mutation suite exercises every forbidden domain and every permitted sense .

****Residual risk:**** None.

AMD-V206-006A - D-100 projection-authority provenance

****Source:**** Codex v2.0.5 Package M0 review

****Affected:**** D-100, original_decision_dispositions.json

****Old value:**** D-100 contained a generated-projection parity clause absent from the v2.0.3 disposition without amendment evidence.

****New value:**** The clause is retained and explicitly recorded as AMD-V206-006A.

****Change:**** The clause is retained and explicitly recorded as AMD-V206-006A.

****Reason:**** Resolve Codex P1-06.

****Verification:**** Decision register and amendment register cross-reference D-100.

****Residual risk:**** None.

AMD-V206-006B - D-102 exact pinned-package reuse evidence

****Source:**** Codex v2.0.5 Package M0 review

****Affected:**** D-102, FR-GOV-003

****Old value:**** D-102 named v2.0.4 tests as evidence for a later package.

****New value:**** Reused code must pass tests written against the exact pinned package.

****Change:**** Reused code must pass tests written against the exact pinned package.

****Reason:**** Resolve Codex P1-06.

****Verification:**** Validator rejects stale package-version test wording in D-102.

****Residual risk:**** None.

AMD-V206-007 - Canonical test-ID namespace

****Source:**** Codex v2.0.5 Package M0 review

****Affected:**** 336 requirements, 337 clauses, dependency_graph.json, workbooks

****Old value:**** Top-level acceptance IDs and clause engineering-test IDs diverged on 18 requirements.

****New value:**** Top-level test IDs equal the ordered clause union; single- and multi-clause formats are explicit.

****Change:**** Top-level test IDs equal the ordered clause union; single- and multi-clause formats are explicit.

****Reason:**** Resolve Codex P2-01.

****Verification:**** Validator enforces exact equality, gate coverage and active-ID prefixes.

****Residual risk:**** None.

AMD-V208-001 - Occurrence-linked forbidden-surface semantics

****Source:**** Codex v2.0.7 Package M0 review

****Affected:**** FR-GOV-002, FR-TST-017, D-103, D-115, forbidden_surface_rules.json, validator

****Old value:**** Negation and sense exclusions could authorize unrelated forbidden occurrences within a sentence or requirement.

****New value:**** Each forbidden occurrence is evaluated in its own adversative clause; negation must govern that occurrence, double-negation fails, and sense exclusions require approved phrase patterns.

****Change:**** Replaced sentence-wide and requirement-wide authorization with occurrence-linked negation and exact phrase/sense matching.

****Reason:**** Resolve Codex v2.0.7 P1-02.

****Verification:**** Mandatory probes include all four Codex bypasses, cross-sentence/adversative variants, double negation, and permitted senses.

****Residual risk:**** Automated scanning does not replace independent human semantic review.

AMD-V208-002 - D-100 JSON projection wording

****Source:**** Codex v2.0.7 Package M0 review

****Affected:**** D-100, decision projections

****Old value:**** D-100 referred to narrative, YAML and workbook projections although controlled machine-readable projections are JSON.

****New value:**** D-100 refers to narrative, JSON and workbook projections.

****Change:**** Corrected the projection-format name and regenerated all decision publications.

****Reason:**** Resolve Codex v2.0.7 P2-01.

****Verification:**** Decision JSON, workbook, Markdown, DOCX and PDF contain identical corrected wording.

****Residual risk:**** None.

AMD-V208-003 - Mandatory outer ZIP pin delivery

****Source:**** Codex v2.0.7 Package M0 review

****Affected:**** FR-GOV-004, D-100, D-102, Package M0 delivery

****Old value:**** The v2.0.7 review delivery omitted the separate .zip.sha256 publisher pin.

****New value:**** The v2.0.8 delivery contains the ZIP and matching .zip.sha256 as two separate artifacts.

****Change:**** Locked the final handoff to a two-file delivery and explicit expected hash in the review prompt.

****Reason:**** Resolve Codex v2.0.7 P1-01.

****Verification:**** Final response exposes both artifacts and the sidecar contains the exact ZIP digest.

****Residual risk:**** The sender must attach both files.

Findings

V204-P0-01 - M4 mandatory journeys require physical receipts while the first printer capability was owned by M5a.

****Severity:**** P0

****Disposition:**** resolved

****Affected:**** FR-CFG-001D, FR-BIL-017, FR-EDG-029, D-119, GJ-01B, GJ-02B, GJ-03B, GJ-06, GJ-07, GJ-10

****Finding:**** M4 mandatory journeys require physical receipts while the first printer capability was owned by M5a.

****Verification:**** M4 has a minimum real physical-receipt path; M5a separately owns durable queue, retry, restart recovery, deduplication, health, outage continuity and reconciliation.

****Residual risk:**** None; v2.0.5 Codex verified M4/M5a print separation.

V204-P1-02 - Decision D-113 provenance differed between narrative and machine-readable sources.

****Severity:**** P1

****Disposition:**** resolved

****Affected:**** D-113

****Finding:**** Decision D-113 provenance differed between narrative and machine-readable sources.

****Verification:**** Canonical decision source is v2.0.3 in the reconciled decision catalog.

****Residual risk:**** None.

V204-P1-03 - Original package assessment counts were stale.

****Severity:**** P1
****Disposition:**** resolved
****Affected:**** original_requirement_lineage.json, original_decision_dispositions.json

****Finding:**** Original package assessment counts were stale.

****Verification:**** The v2.0.6 generator publishes current row-level counts and all projections from canonical records.

****Residual risk:**** None.

V204-P1-04 - Row-level 500 original requirement and 100 original decision dispositions were missing.

****Severity:**** P1
****Disposition:**** resolved
****Affected:**** original_requirement_lineage.json

****Finding:**** Row-level 500 original requirement and 100 original decision dispositions were missing.

****Verification:**** The frozen canonical baseline contains the 500/100 row-level evidence; this package retains the exact source hashes and maps all 294 active originals.

****Residual risk:**** None for reconciliation.

V204-P1-05 - The dependency graph was generic and milestone-wide rather than requirement-specific.

****Severity:**** P1
****Disposition:**** resolved
****Affected:**** reconciled_requirements.json

****Finding:**** The dependency graph was generic and milestone-wide rather than requirement-specific.

****Verification:**** Two independent clause-level reviews were reconciled and the final graph has requirement-specific gates, prerequisites, journeys and tests.

****Residual risk:**** Codex remains the independent semantic reviewer.

V204-P1-06 - The package validator relied on PyYAML and checked structure rather than full semantics.

****Severity:**** P1
****Disposition:**** resolved
****Affected:**** tools/validate_reconciled_stdlib.py

****Finding:**** The package validator relied on PyYAML and checked structure rather than full semantics

.

****Verification:**** The standard-library validator was frozen and challenged through two blind mutation rounds; final limitations are disclosed.

****Residual risk:**** Post-repair diagnostics are not fresh blind evidence.

V204-P1-07 - GJ-01A said outlet authority persists at M3, implying local authority before M5b.

****Severity:**** P1
****Disposition:**** resolved
****Affected:**** GJ-01A

****Finding:**** GJ-01A said outlet authority persists at M3, implying local authority before M5b.

****Verification:**** GJ-01A now states that the current approved cloud authority persists and makes no local-authority claim before M5b.

****Residual risk:**** None.

V204-P2-08 - Private-key custody was inconsistent.

****Severity:**** P2
****Disposition:**** resolved
****Affected:**** FR-EDG-022A, D-116

****Finding:**** Private-key custody was inconsistent.

****Verification:**** The node generates and retains the private key, submits only a CSR, receives only the certificate chain, and never exports the private key.

****Residual risk:**** Operational HSM/keystore choice remains an implementation decision under this contract.

V204-P2-09 - FR-EDG-028 test ID used inconsistent M5b casing.

****Severity:**** P2
****Disposition:**** resolved
****Affected:**** FR-EDG-028

****Finding:**** FR-EDG-028 test ID used inconsistent M5b casing.

****Verification:**** Engineering test ID is normalized to TST-M5b-FR-EDG-028.

****Residual risk:**** None.

V204-P2-10 - Amendment and finding logs were not exact enough.

****Severity:**** P2
****Disposition:**** resolved
****Affected:**** finding_register.json, amendment_register.json

****Finding:**** Amendment and finding logs were not exact enough.

****Verification:**** Each prior finding has an ID, severity, disposition, affected IDs, verification and residual risk.

****Residual risk:**** Future findings must use the same structure.

VAL-P1-001 - Journey links were not consistently aligned to introduction/revalidation gates

****Severity:**** P1

****Disposition:**** REPAIRED

****Affected:**** FR-SEC-002A, FR-TST-006, FR-AUTH-003, FR-MNU-012, FR-RPT-001, FR-RPT-004, FR-RPT-005

****Finding:**** Journey links were not consistently aligned to introduction/revalidation gates

****Verification:**** Every journey milestone now equals the requirement introducing gate or a declared revalidation gate.

****Residual risk:**** none

VAL-P1-002 - Round 2 exposed five validator coverage gaps

****Severity:**** P1

****Disposition:**** REPAIRED_AND_REFROZEN

****Affected:**** FR-CFG-002B, FR-DAT-003, D-043, canonical count declarations, canonical decision projection

****Finding:**** Round 2 exposed five validator coverage gaps

****Verification:**** All five disclosed Round 2 misses are detected after repair. No Round 3 is required under the agreed two-round cap unless Codex later identifies a P0 validator defect.

****Residual risk:**** The post-repair rerun is diagnostic rather than fresh independent evidence. This limitation is disclosed to Codex.

V205-P1-01 - M5a service boundary contradiction

****Severity:**** P1

****Disposition:**** resolved

****Affected:**** FR-EDG-002A, FR-OPS-006, FR-OPS-007, FR-SEC-019, FR-TST-009, D-113, GJ-10, GJ-11

****Finding:**** m5_ownership.json listed local_backup_agent although FR-EDG-002A requires exactly five services.

****Verification:**** M5a now lists exactly five services; backup scheduling and restore remain M6.

****Residual risk:**** None.

V205-P1-02 - Stale canonical governance state

****Severity:**** P1

****Disposition:**** resolved

****Affected:**** residual_issues.json, finding_register.json

****Finding:**** Residual/finding records described validator and blind rounds as future work.

****Verification:**** Lifecycle state is current and projections are regenerated.

****Residual risk:**** None.

V205-P1-03 - Non-uniform clause schema

****Severity:**** P1

****Disposition:**** resolved

****Affected:**** reconciled_requirements.json, reconciled_canonical_register.schema.json

****Finding:**** Ten clauses omitted exact_clause_text and used substitute fields.

****Verification:**** All 337 clauses share one enforced contract.

****Residual risk:**** None.

V205-P1-04 - Projection content loss

****Severity:**** P1

****Disposition:**** resolved

****Affected:**** amendment_register.json, finding_register.json, original_requirement_lineage.json, non_regression_rules.json

****Finding:**** Amendments, findings, split reasons and non-regression rules were incompletely published.

****Verification:**** All canonical fields and all 96 rules are projected and field-compared.

****Residual risk:**** None.

V205-P1-05 - Forbidden-surface rules not executable

****Severity:**** P1

****Disposition:**** resolved

****Affected:**** forbidden_surface_rules.json, validator

****Finding:**** Validator did not load the normative vocabulary and sense exclusions.

****Verification:**** Validator executes the normative rule file and planted tests cover every domain and permitted sense.

****Residual risk:**** None.

V205-P1-06 - Decision provenance and stale reuse evidence

****Severity:**** P1

****Disposition:**** resolved

****Affected:**** D-100, D-102, FR-GOV-003

****Finding:**** D-100 lacked amendment evidence and D-102 named v2.0.4 tests.

****Verification:**** D-100 amendment is explicit and D-102 requires exact pinned-package tests.

****Residual risk:**** None.

V205-P2-01 - Divergent test-ID namespaces

****Severity:**** P2
****Disposition:**** resolved
****Affected:**** requirements.json, dependency_graph.json

****Finding:**** Top-level and clause-level test IDs diverged on 18 requirements.

****Verification:**** One canonical test-ID policy is enforced across all projections.

****Residual risk:**** None.

V207-P1-01 - Required outer artifact pin was not delivered

****Severity:**** P1
****Disposition:**** resolved_by_v2.0.8_delivery
****Affected:**** FR-GOV-004, D-100, D-102

****Finding:**** The Package M0 review received the ZIP without the separately supplied .zip.sha256 publisher pin.

****Verification:**** The v2.0.8 ZIP and matching sidecar are delivered together and the prompt states the expected digest.

****Residual risk:**** External delivery must include both files.

V207-P1-02 - Forbidden-surface validator admitted coherent deferred-domain obligations

****Severity:**** P1
****Disposition:**** resolved
****Affected:**** FR-GOV-002, FR-TST-017, D-103, D-115

****Finding:**** Sentence-wide negation and requirement-wide sense allowances could permit unrelated positive payroll, inventory or variance obligations.

****Verification:**** Occurrence-linked parser rejects all four Codex examples and targeted variants while preserving approved senses.

****Residual risk:**** Human semantic review remains required for novel wording.

V207-P2-01 - D-100 referred to nonexistent YAML projections

****Severity:**** P2
****Disposition:**** resolved
****Affected:**** D-100

****Finding:**** The package uses JSON as its controlled machine-readable projection, not YAML.

****Verification:**** All decision projections use the corrected JSON wording.

****Residual risk:**** None.

```

---

# Known Validator Limits

- Package: 'Hospitality_OS_Phase_1_Clean_Build_Package_v2.0.9'
- Canonical source root: 'cd6b8cfc2175004cc2057deb7d9de1ad0d6264a23c82f00e15e96faa526c0c96'
- Status: **Package M0 candidate - M0R and implementation are not authorized**

# Known Validator Limits

- Occurrence-linked scanning is deliberately conservative and cannot replace independent human semantic review.
- Blind mutation testing stopped after two rounds as approved. Focused post-review probes are disclosed implementation verification, not a third blind round.
- The publisher checksum sidecar is external to the ZIP and must be delivered separately.

---

# Non-Regression Rules

**96 canonical rules**

- Package: 'Hospitality_OS_Phase_1_Clean_Build_Package_v2.0.9'
- Canonical source root: 'cd6b8cfc2175004cc2057deb7d9de1ad0d6264a23c82f00e15e96faa526c0c96'
- Status: **Package M0 candidate - M0R and implementation are not authorized**

Every rule is identified and published. The machine-readable register remains authoritative.

## NR-001

Do not implement future modules as hidden routes, tables, workers or feature-flagged pages.

## NR-002

A clean phase boundary is physical as well as visual.

## NR-003

Build one vertical milestone at a time.

## NR-004

Do not clone the frozen repository.

## NR-005

Reused code is treated as third-party code and reviewed in isolation.

## NR-006

The local node contains only customer service and outlet execution.

```

NR-007

Customer-visible ingredients are not operational recipes.

NR-008

Bill, payment and tip are separate business records.

NR-009

Missing tenant/outlet context must return zero rows and permit zero writes.

NR-010

Use populated owner-side fixtures so zero-row tests cannot pass vacuously.

NR-011

Test sibling outlets inside the same tenant, not only different tenants.

NR-012

Test SELECT, INSERT, UPDATE and DELETE.

NR-013

Policies must contain both 'USING' and 'WITH CHECK' where applicable.

NR-014

Adding 'outlet_id' later must automatically strengthen existing policies.

NR-015

API, jobs, files, caches, reports and sync paths are part of the isolation boundary.

NR-016

Production services reject owner, superuser, BYPASSRLS and maintenance database roles.

NR-017

Do not use binary floating-point for money, percentages or quantity outcomes.

NR-018

Accepted orders, issued bills, payments, tips and cash movements are immutable or reversal-based.

NR-019

No tip is selected by default.

NR-020

Bill allocation and tip allocation are separate.

NR-021

A tip cannot hide an unpaid bill balance.

NR-022

Refunds and reversals require purpose-specific step-up, permission, reason and audit.

NR-023

Quick PIN cannot authorize sensitive financial actions.

NR-024

Split bill participants may choose different tips.

NR-025

A protocol document is not a deployable outlet service.

NR-026

Test the real outlet node, worker, database, print agent and restart behavior.

NR-027

Cloud must not accept new dine-in writes when it cannot reach the outlet authority.

NR-028

Use an authority epoch/lease to prevent split brain.

NR-029

Persist IDs, cursors, dependencies and queues durably.

NR-030

A bad event must not block unrelated streams forever.

NR-031

Retries must be idempotent.

NR-032

Conflicts are explicit; silent last-write-wins is prohibited.

NR-033

The same QR must work locally with valid TLS, not a raw IP and ignored warning.

NR-034

Create the empty repository only after Package M0; create PostgreSQL and executable migration '0001' only in M1 after M0R approval or adjudication.

NR-035

Never edit an applied migration.

NR-036

Lock migration checksums.

NR-037

Test clean build and supported upgrade.

NR-038

Test historical/edge-case fixtures before declaring upgrade safety.

NR-039

Use real PostgreSQL and exact production roles.

NR-040

Cross-platform line endings and tool invocation are acceptance concerns.

NR-041

A source-tree test does not prove the production container.

NR-042

Production images contain every advertised script, helper and database client.

NR-043

Containers run as non-root.

NR-044

Required paths remain usable by the non-root runtime user.

NR-045

Health/readiness uses least-privileged application roles.

NR-046

Readiness is unhealthy when an advertised job cannot do real work.

NR-047

Backup tests use the same binaries, roles and image as production.

NR-048

Restore is destructive into a clean environment and runs post-restore business journeys.

NR-049

Missing grants, scripts, policies or binaries must make the drill fail.

NR-050

Build output is deterministic and independent of command order.

NR-051

Zero discovered tests is failure.

NR-052

Any skipped acceptance test is failure.

NR-053

Empty, unsupported or unknown security-scan coverage is failure.

NR-054

Every critical gate has a planted negative control.

NR-055

Tests cannot rely on another test's artifact or environment mutation.

NR-056

Clear 'dist', '.next', generated output and test databases before canonical validation.

NR-057

Ordinary Windows and Linux commands must work without CI-only PATH or 'PG_BIN' injection.

NR-058

Pin the exact commit audited.

NR-059

Store commands, logs, evidence and verdict on an isolated audit branch.

NR-060

Do not weaken assertions merely to make CI green.

NR-061

When an inherited gate becomes logically invalid after a repair, replace it with a stronger audit-only control and prove both repaired and deliberately broken cases.

NR-062

Compilation, page scaffolding and green unit tests are not completion.

NR-063

Production-path evidence takes precedence over claims.

NR-064

Every active requirement maps to code, tests and evidence.

NR-065

Every deferred requirement is absent from the production artifact.

NR-066

A milestone or repository gate cannot proceed while its P0/P1 defects remain unresolved or unadjudicated under Source of Truth Section 9.1.

NR-067

Final merge requires an independent staging verdict plus either reviewer approval or documented founder adjudication under Source of Truth Section 9.1; founder adjudication is never represented as reviewer approval.

NR-068

An adverse reviewer verdict pauses progression but does not create an unrecorded permanent veto; disputes follow written founder adjudication.

NR-069

Every founder risk acceptance names the affected requirement or non-regression rule and is re-examined at the next milestone.

NR-070

M5a and M5b have separate executable exit journeys, ownership registers and audit gates.

NR-071

Same-QR local service uses a per-outlet public hostname and public-trust certificate; self-signed by pass is prohibited.

NR-072

Cloud lease expiry blocks cloud writes but never revokes LAN authority; reachability must be proved bidirectionally.

NR-073

Only cash, external-terminal recording and verified Telebirr/CBE Birr proof confirmation are live pilot payment paths until a provider API is contracted.

NR-074

A simulator cannot be represented as a live payment result or printed as a real provider receipt.

NR-075

Negative controls are enumerated, identical across narrative/YAML/workbook and frozen at Package M0, not improvised after implementation.

NR-076

Every milestone journey uses only capabilities owned by that milestone and approved predecessors; M3 cannot require M4 settlement or receipt behavior.

NR-077

Deferred loyalty, pickup, delivery, supplier, inventory, accounting and data-portability behavior cannot remain as active Phase 1 requirements, tests, screens or prompt obligations.

NR-078

Package M0 occurs before repository creation; Repository Conformance M0R occurs after the empty repository plan and before any M1 application code.

NR-079

Emergency authority replacement cannot become writable until documented old-node power-off or network isolation and an automated LAN-unreachability probe are recorded.

NR-080

Authority sequence is durable, monotonic, signed and anti-rollback; every writer persists the highest accepted sequence.

NR-081

Reachability is a signed bidirectional challenge/acknowledgement; failure of either direction expires the cloud forwarding lease.

NR-082

The canonical Package M0 review register contains exactly 14 questions and every review response answers all 14.

NR-083

Normative M5 ownership is generated from one source and must match requirements, journeys, screens, events, risks and negative controls.

NR-084

No gate is closed using behavior owned by a later gate; every requirement declares its introducing gate, gate-local scope and revalidation gates.

NR-085

Excluded-domain capability appears in Phase 1 only as a deferral statement or extension-contract documentation, never as an entity, field, enumeration value, route, screen, registry entry or positive test.

NR-086

The forbidden-surface vocabulary is generated from the phase-boundary exclusion set and every excluded domain carries a planted negative control.

NR-087

A per-outlet certificate renewal is complete only when installation evidence is verified from the LAN; failed renewal fails safe to the cloud journey, never to a browser warning or bypass prompt.

NR-088

Same-QR acceptance includes cached-DNS, encrypted-DNS and dual-stack client conditions with a safe documented failure mode.

NR-089

A replacement node becomes writable only through the explicit activation state after the superseded instance is fenced and blocked; at most one node instance per outlet is writable.

NR-090

A reopened cash shift is never terminal and returns to closed through recount and approval or an audited maker-checker correction.

NR-091

Source of Truth, YAML, workbooks and combined publications are generated from canonical structured records and must pass exact parity checks.

NR-092

Every active requirement receives a human-semantic introducing-gate and revalidation-gate disposition.

NR-093

A generic journal, campaign, roster, attendance, timekeeping or break-record obligation is a forbidden future-domain leak unless explicitly negated or historical.

NR-094

GJ-02B is mandatory at M4 and reverse-linked to localization, bill, payment, tip, receipt, print and testing requirements.

NR-095

The cloud is never a writable emergency dine-in authority in Phase 1; only a physically or network-fenced standby outlet node may replace authority.

NR-096

Same-QR acceptance uses a 30-second DNS TTL and maximum 60-second cloud-to-LAN transition, with captive-portal and staff guidance for unsupported encrypted DNS.